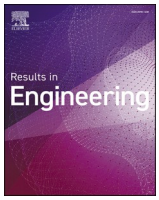
Results in Engineering 27 (2025) 106010



Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering



Review article



Occupant-centric indoor assessments and control solutions in healthcare facilities: a critical review of restorative perception

Jiake Yang, Yidan Wu, S. Thomas Ng *

Department of Architecture and Civil Engineering, City University of Hong Kong, Hong Kong

ARTICLE INFO

Keywords:
Healthcare facilities
indoor environmental quality (IEQ)
Control
design
evaluation
individual

ABSTRACT

The primary function of healthcare facility environments is to support the recovery of its occupants. Numerous studies have examined how indoor environmental quality (IEQ) and physical design factors in healthcare facilities affect occupants and the effectiveness of control methods, respectively. However, results from independent studies on influential elements may conflict, lack transferability, and recoverability. Therefore, a holistic review on indoor environments and related control methods for healthcare facilities from the restorative perspective is needed. This study analyzes 249 articles to examine the indoor environment's impact on residents' restorative perceptions and related control behaviors. It also explores automation modeling methods across various healthcare settings. The results show that the setting of different environments and occupant roles in healthcare buildings can have an impact on the evaluation of the indoor environment. Indoor environmental factors are diverse but lack generalization, and existing standards and models fail to adequately address practical needs, highlighting the need for expanded cross-disciplinary research. Although qualitative evaluation and results on IEQ and physical design factors could be applied to satisfy some groups' needs, there is a lack of quantitative, personalized and dynamic evaluation frameworks in current healthcare environments. Existing control methods are still predominantly manual, relying on personal literacy and perception, and lack integration and application with automated control technologies. To bridge this gap, this paper outlines future directions and designs a conceptual framework, offering insights for stakeholders on design standards and advanced technology applications to better align healthcare interiors with occupants' needs.

1. Introduction

Healthcare facilities, among the most complex buildings in modern society, aim to improve patient safety, enhance operational efficiency, and reduce environmental impact [49,330]. The primary mission of these facilities extends beyond medical treatment to creating environments that actively support healing, recovery, and the well-being of all occupants—patients, healthcare staff, and visitors [141,152]. With rising physician burnout attributed to workplace stressors and chaotic work environments [218], there is an urgent need to optimize healthcare indoor environments through occupant-centric approaches.

The concept of "healing environments" has gained significant momentum in recent years [141,190], emphasizing spaces that psychologically comfort and empower both patients and healthcare providers [31]. Research has consistently demonstrated that indoor environmental quality (IEQ) profoundly impacts occupant well-being [18], patient recovery rates, and staff performance [152,324]. Current research hotspots include smart healthcare facilities leveraging adaptive environmental control technologies [13,258] and investigations into the psychological impacts of environmental factors [191].

Despite these advances, several critical gaps persist in the field. First, there is no universal standard for IEQ in healthcare settings that adequately considers individual needs and perceptions [298,305]. Second, the impact of key environmental factors—lighting [43,174], acoustics [187,286], thermal conditions [324], and indoor air quality (IAQ) [137,252]—remain underexplored, particularly their interactive effects [210]. Third, existing frameworks lack comprehensive integration of environmental assessment and control strategies specifically tailored for healthcare contexts [69,209].

Current theoretical frameworks, while valuable, present significant limitations. The environment–behavior (E–B) model focuses primarily on Alzheimer's patients [328], while the environment–human-performance framework has been validated mainly in educational settings [40–42]. The environment–occupant–health

* Corresponding author.
E-mail addresses: J.K.Yang@my.cityu.edu.hk (J. Yang), yidanwu4-c@my.cityu.edu.hk (Y. Wu), thomasng@cityu.edu.hk (S.T. Ng).

https://doi.org/10.1016/j.rineng.2025.106010
Received 23 March 2025; Received in revised form 16 June 2025; Accepted 27 June 2025
Available online 3 July 2025
2590-1230/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

framework [340] represents progress but lacks quantitative evaluation methods and differentiated impacts for various occupant groups. Existing standards like WELL Building Standard [301] and Leadership in Energy and Environmental Design (LEED) for Healthcare [121], while comprehensive, fall short in addressing specific healthcare needs and individual differences [121,305].

1.1. Research objectives

The primary objective of this research is to develop a comprehensive, occupant-centric framework for assessing and controlling indoor environments in healthcare facilities. Specific objectives include:

1. Systematically review literature on indoor environmental assessment and control in healthcare facilities
2. Evaluate current research status across different subjects, scenarios, and contexts
3. Comprehensively assess the impact of healthcare indoor environments on occupants' restorative perceptions
4. Synthesize existing occupant control behaviors and automation models
5. Identify occupant-centric operational challenges and future development directions
6. Develop an individual-based environmental assessment system and comprehensive control framework

1.2. Research methodology and structure

This study employs a systematic literature review methodology combined with scientometric analysis to examine peer-reviewed publications from multiple databases. The review follows established protocols for healthcare environment research, incorporating both quantitative analysis of research trends and qualitative assessment of theoretical frameworks.

The paper is structured into seven sections: Section 1 introduces the research context and objectives; Section 2 presents the methodology and review procedures; Section 3 provides quantitative analysis results; Section 4 assesses environmental factors influencing restorative perceptions; Section 5 examines control strategies and automation models; Section 6 discusses findings and proposes the new framework; and Section 7 concludes with implications and future directions.

1.3. Strengths and limitations

As getting a clear understanding of healthcare environments is a multidisciplinary field, analyzing previous reviews is necessary. Based on the literature review method and related results in the second section, 30 review papers related to this research topic were identified in the Appendix, covering physical design, IEQ, and control methods (e.g., behavioral adjustments, HVAC controls) [149]. Recent reviews, like Shen et al. [254] and Eijkelenboom and Bluyssen [88], have a comprehensive understanding but fail to integrate physical design as well as occupant behavior with environmental factors and control methods, revealing a gap in understanding their interactions in healthcare facilities.

Based on the analysis above, strengths of this study include: (1) Comprehensive scope covering multiple environmental factors and occupant types; (2) Integration of assessment and control methodologies; (3) Focus on healthcare-specific requirements; (4) Development of individual evaluation framework; (5) Synthesis of interdisciplinary research findings.

Limitations include: (1) Reliance on existing literature may limit findings to published research; (2) Framework validation requires future empirical testing; (3) Geographic and cultural variations in healthcare practices may affect generalizability; (4) Rapid technological advances may require framework updates.

The findings of this research will provide valuable insights for healthcare facility designers, facility managers, and policymakers seeking to create optimal healing environments that support both patient recovery and staff well-being.

2. Methodology and procedures

This study employs a systematic literature review (SLR) complemented by critical literature review (CLR) elements to synthesize literature. This current article used SLR method [54] first, which followed by Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) criteria [219], to identify, assess, and synthesize all the studies relevant to the research scopes and then applied CLR method to further analyze the findings. SLR is a scientific approach that involves systematically evaluating and interpreting literature in a specific field, aiming to identify, analyze, and synthesize key findings, methodologies, and gaps in existing research [280]. CLR method emphasizes critical thinking and objective assessment to understand the current state of knowledge and contributes to ongoing scholarly discourse. This dual approach aligns with prior IEQ reviews that combine systematic synthesis with critical evaluation to address both technical and interpretive dimensions (E.g., [210]). This review adopted the SLR methodology based on previous research [126,333] and comprised four consecutive steps:

Step 1: Define the conceptual boundaries of the research.
Step 2: Identify key questions on the research.
Step 3: Literature search and selection.
Step 4: Scientometric analysis of collected bibliographies.

2.1. Description of perceptions of restorative on occupants

Identifying the specific research area is critical before doing the literature review as the definitions of perceptions of restorative on occupants in healthcare facilities are different in the multidisciplinary presented in current research [3,22,80,190,205,294]. Current discourse on how occupants react to indoor environments encompasses various concepts, such as comfort [233], well-being [122,303], satisfaction [19], and attention [263]. Some studies have examined these aspects from multiple angles to evaluate how occupants perceive their environments, highlighting the intersections among these concepts [17, 152]. Within the healthcare architecture, the design of indoor environments is pivotal to promoting healing [221]. For instance, Mahmood and Tayib [191] identify four critical design factors—interior appearance, privacy, comfort and control—as essential in creating a healing environment. This notion is expanded by incorporating factors like natural settings, IAQ, lighting, noise, thermal comfort, and aesthetics [66,80], which are recognized for their contribution to enhancing recovery, health, and work efficiency in healthcare settings [85,207,303].

Environmental psychology introduces the term "restorative perception" to assess the degree to which an environment supports an occupant's recovery, health, and stress alleviation [3,216]. Such environments are expected to promote positive health outcomes, improve mood, reduce anxiety, and empower individuals with a sense of control, covering both psychological and physiological aspects [97,190, 193]. Traditionally, the quality of therapeutic environments was evaluated using satisfaction and acceptance as key metrics [216], with assessments incorporating both subjective and objective measurements [109].

Considering the diverse needs of healthcare facility occupants—patients focusing on recovery [141], visitors on comfort [233], and staffs on efficiency [116,288] — this paper proposes using the concept of restorative perception to elucidate the specific requirements of different individuals. This strategy enables a nuanced assessment of healthcare indoor environmental quality and the development of

appropriate control measures, thereby aligning the indoor environment with the varied priorities of its occupants.

2.2. Identifying key questions

Based on the above analyses, three key questions could be proposed for the research:

RQ 1: What is the current state of research on the indoor environment of hospitals?
RQ 2: What are the indoor environmental and design factors that influence an individual's sense of restorative perception?
RQ 3: What are the existing regulatory behavior and related control models that meet the regulatory requirements of restorative perception?

2.3. Literature search and selection criteria

This review process is followed by PRISMA criteria [219]. This section defines the search criteria and screening procedure in the literature review process. As shown in Fig. 1, from the main scientific databases, the Scopus, and Web of Science (WOS) databases were selected for the peer-reviewed literature search on the restorative perceptions of occupants in healthcare facilities and their relationships by searching in title, abstract, and keywords. The PubMed database was not chosen because it mainly collects research aimed at the life sciences and biomedical sciences, which is not compatible with the theme of this paper. The final search was conducted on March 3, 2024, and a search for updated literature was conducted on August 11, 2024.

The first step is to define the scope of the provisions. According to the objective of this review, the search keywords should cover people, environment, perception, IEQ or design, and control five areas. The types and terms used by the authors searched for the titles, abstracts, and keywords of articles published in English between 1977 and 2024 are listed in Table 1.

In the second step, we filtered the types of articles, and due to the incompleteness of some conference papers, "articles" and "reviews" of journal literature were selected, while other types of literature were excluded. The main restriction applied to three primary categories of "environmental science", "engineering" and "energy", where the subcategories in the WOS included (Engineering Environmental or Psychology or Engineering Civil or Construction Building Technology or Management or Environmental Sciences or Multidisciplinary Sciences or Computer Science Information Systems or Ergonomics or Environmental Studies or Green Sustainable Science Technology or Architecture).

Table 1
Categories and keywords.

Category	Keyword
Environment	"hospital" OR "healthcare facility" OR "ward" OR "nursing unit"
People	"occupant" OR "human" OR "personal" OR "individual" OR "patient" OR "doctor" OR "nurse"
Design OR IEQ	"design" OR "spatial" OR "layout" OR "indoor environmental quality" OR "IEQ" OR "thermal" OR "visual" OR "acoustic" OR "indoor air quality" OR "IAQ" OR "air" OR "lighting" OR "sound"
Control	" control" OR "optimiz*" OR "regulat*"
Perception	"cognitive" OR "percept*" OR "comfort" OR "mood" OR "satisfaction" OR "recovery" OR "attention" OR "restorative"

In the final phase, the relevance of the articles was evaluated. Two reviewers screened the titles and abstracts of articles independently. The first reviewer extracted the data into an Excel spreadsheet and the data were verified by second reviewer. For discrepancies that occurred during data extraction, a third reviewer was consulted to reach consensus. In detail, an abstract-based screening was conducted to filter out literature primarily focused on nursing services. After this initial assessment, the remaining articles were reviewed to determine their alignment with the review's research objectives and questions; those that did not align were excluded. Ultimately, this process resulted in a curated selection of 249 articles that were included in the final classification for this review (Tables A and B).

2.4. Scientometric analysis of collected bibliographies

In this section, a further analysis of the keywords was done by Citespace [67], which is a widely useful tool can help researchers analyze collected literatures. The time distribution of the collected bibliographic records was studied and analyzed over the last ten years to identify the research area in Fig. 2.

As shown in Fig. 2A, the most frequent keyword analyses in the past twelve years are thermal comfort; hospital; indoor environmental quality; healthcare facilities; and IAQ, this is also in line with the research priorities of the indoor environment proposed by [254]. In the

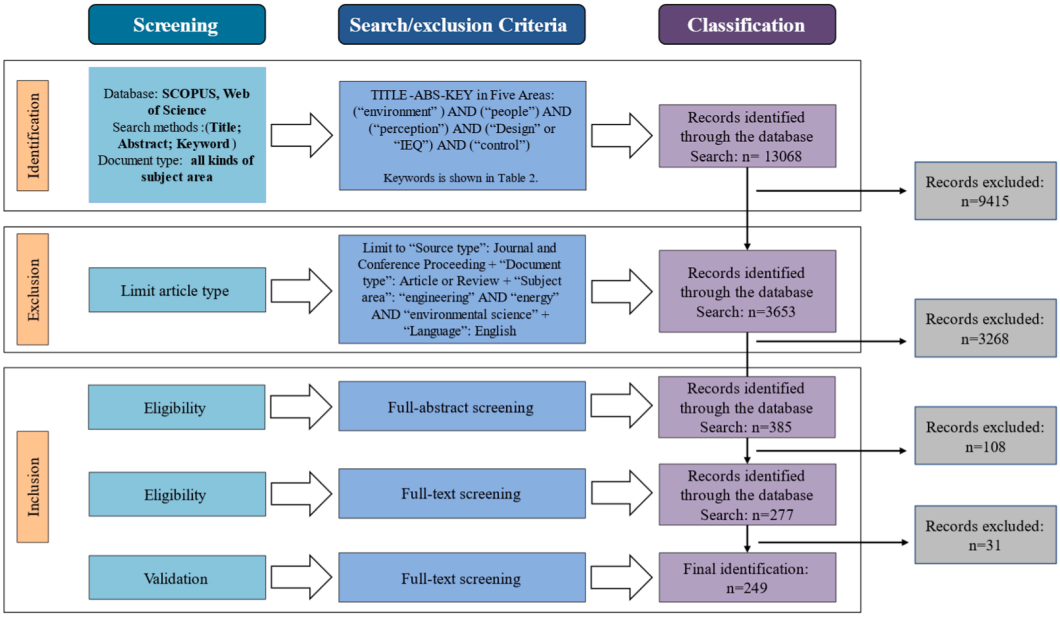


Fig. 1. The review screening procedure framework.

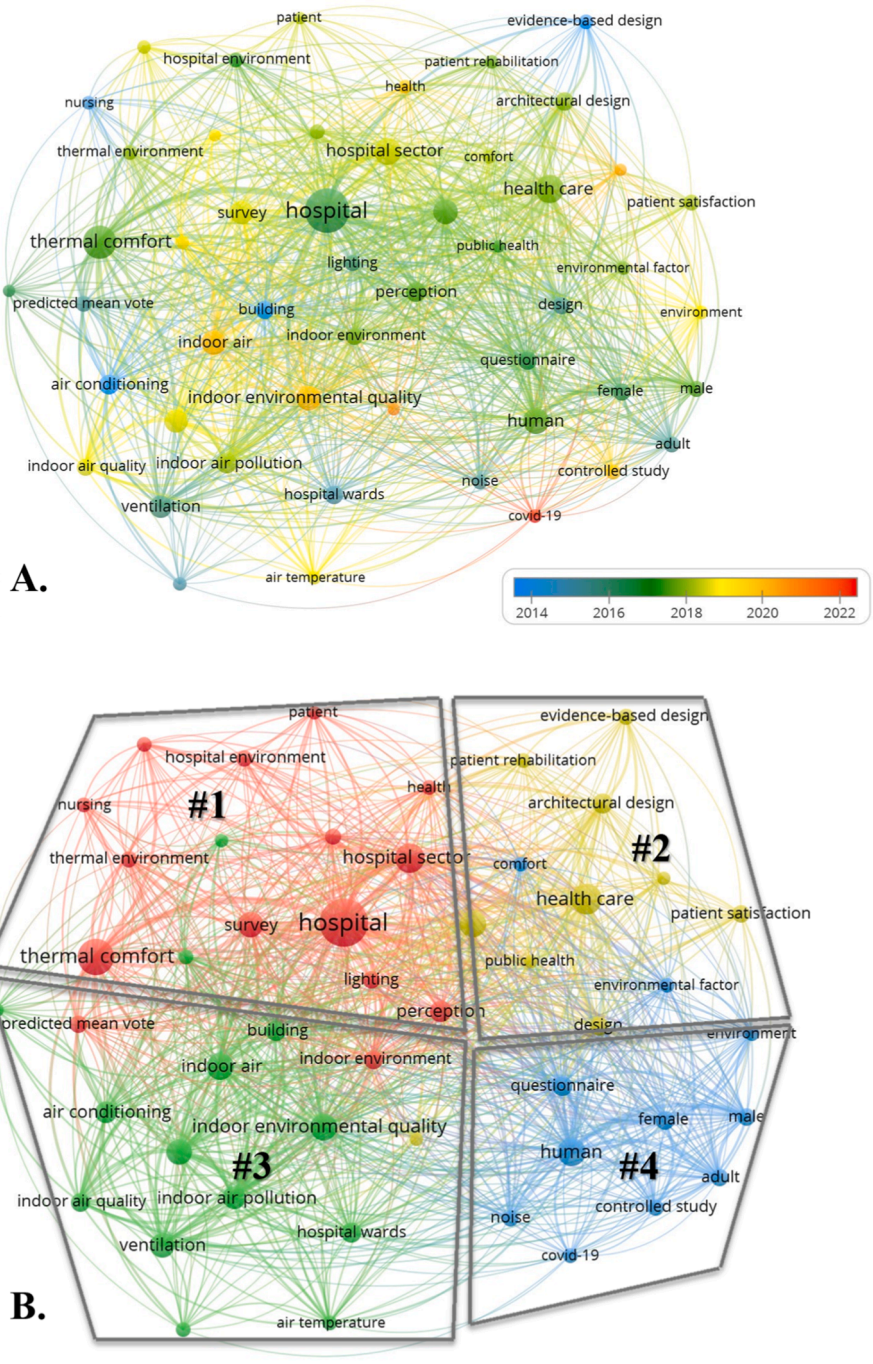


Fig. 2. Keywords analysis.
Fig. 2A shows the 2012–2024 frequent keyword analysis.
Fig. 2B shows clusters of data from 249 articles.

past three years, the keywords that have gradually attracted attention include naturally ventilated; COVID-19; infection control; energy efficiency; patient ward; and well-being. In recent years, the research trend has shifted from a single study of thermal environment and thermal comfort to a comprehensive evaluation of IEQ, and at the same time, with the outbreak of the COVID-19 epidemic, the focus of researchers on COVID-19 and indoor ventilation control strategies has been concerned. The rest of the key themes are energy efficiency, and there is a trend in the scene from the large space research of the hospital to the concentration of small spaces such as single wards and nursing units.

To identify the research status, a cluster analysis on nearly 12 years of research was conducted in Fig. 2B The four clusters identified in this article are #1 Hospital environment, #2 design factors, #3 indoor environmental factors, and #4 human-related factors, as shown in Fig. 2B For cluster #1 and cluster #3, containing the major research topics, focusing on the hospital sectors and related environment. The IEQ-related factors attract researchers' interest and related control facilities with indexes are selected. Cluster #2 shows design-related factors in healthcare factors, the evidence-based design method is used to improve patient satisfaction and public health. Cluster #4 identifies human-related factors in research, demonstrating investigation methods and accessible control studies. Those factors are close to each other and could reflect the status of concentrated research.

Based on the clusters mentioned in the above analysis, the focus of

this study includes four themes: 1) surveys related to restorative perception; 2) factors related to restorative perception (including physical environment design and indoor environmental quality); 3) measures to improve restorative perception, and 4) state-of-the-art environmental control methods. These four themes are also within the scope of the three questions raised in Section 2.2.

3. Quantitative analysis

To answer the first question (RQ1), the quantitative analysis is addressed to introduce the current state of research on the healthcare facilities' indoor environment. The quantitative analysis provides a brief history of publications and the research scenarios, subjects, methods, individual statistical information, and relationships between different research factors. An analysis of basic information on selected papers is conducted in Fig. 3.

Publication trends for the eight hottest research sites over the last decade are illustrated in Fig. 3A, China has the highest number of publications, followed by the USA and Italy. Research interest is on the rise in countries. Influenced by major health events, the number of publications in 2020 has surged. In addition to the COVID-19 pandemic bearing part of the blame, the significance of the whole indoor environment in healthcare facilities is attracting increasing attention worldwide [207,305].

Considering the differences in perception, the extraction of individual information is important. Existing studies are highly variable in terms of study groups, with as few as eight or >10,000 subjects participating in questionnaires. As shown in Fig. 3B, the majority of studies have focused on the order of magnitude of hundreds. For the different subject groups, studies on staff had a slightly higher level of interest than patients, which may be related to the level of data availability and patient specificity [34]. As to visitors, it is the minimal proportion in studies. Overall, the proportion of the investigated population may be related to the length of residence.

As shown in Fig. 3C, 42 % of the non-review articles used both subjective and objective methods to study the indoor environment, and objective methods appeared more frequently than subjective methods. Review articles have been analyzed separately in the previous section. In subjective research, questionnaires [30,89], including standardized scales, non-standardized scales, and open-ended items, and interviews [194] are used to collect data on subjective experiences and perceptions of the occupants. While objective studies focus on analyzing indoor environmental factors using simulation models and field measurements [8,179]. The percentage of studies using a hybrid method has remained high in recent years due to validating empirical data and obtaining more rational solutions [1].

Fig. 3D shows the frequencies of six important factors in the indoor environment. The amount of the type of IEQ-related research includes more than one main factor in IEQ. The largest percentage of research is related to healthcare facilities' design, which goes through a long research cycle, with a small number of these deeply integrated with IEQ [304,330]. In addition, thermal comfort also gained a great deal of attention, followed by papers on IEQ, IAQ, visual and acoustic.

As multiple research plans were conducted in selected literature, identifying the basic information on research factors and their relationship is crucial to determining the gaps and directions. Fig. 4 is designed to present basic factors and illustrate the correlations between the main personal information, research subjects, research scenarios and research topics.

For the research scenarios, healthcare facilities include many places and the hospital is seen as the common and representative scenario in actual life, which is composed of clinical areas, public areas, administrative areas, and other components [88,192].

In addition to the general hospital buildings [146,275], there are also

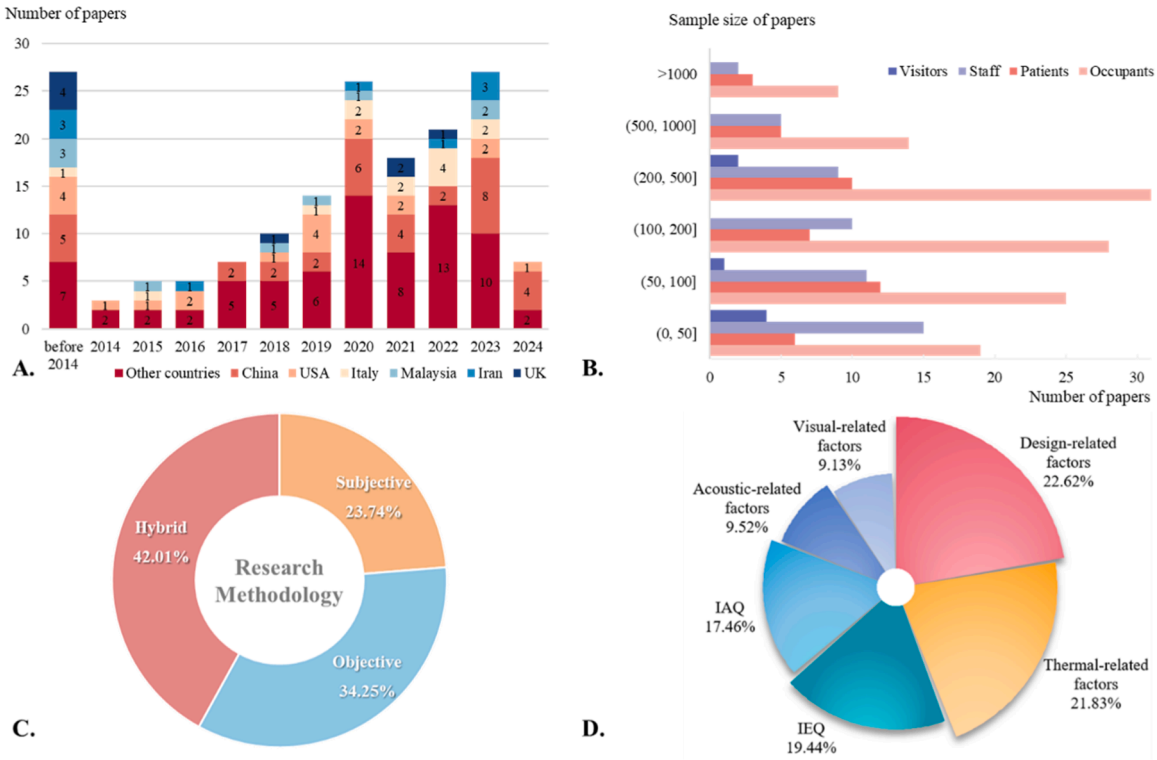


Fig. 3. Basic information about the selected paper.
Fig. 3A shows the trend in publications over time for the eight most popular research regions.
Fig. 3B shows the distribution of occupants across categories and totals.
Fig. 3C shows the proportions of three research methods for non-review articles.
Fig. 3D shows the frequencies of six important factors in the indoor environment.

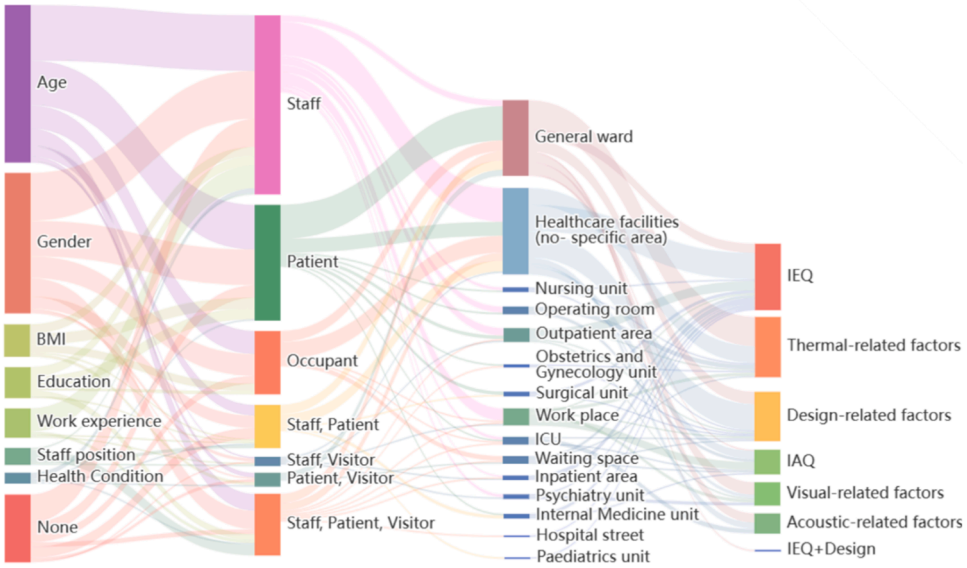


Fig. 4. Relationship map between personal information, research subjects, research scenarios and research topics in the chosen paper.

cancer centers [86], children's hospital [277] and dental hospital [264]. However, nearly 30 % of studies focused on healthcare facilities with no specific area. In specific research scenarios, studies have the most interest in focusing on the general physical space setup [336], with a lack of studies on specific departments.

As to the component studies, general wards are studied mostly, followed by workplace, outpatient areas [16.249] and operating rooms, intensive care units (ICU) [289,296]. While fewer relevant studies have been conducted in obstetrics and gynecology units, pediatrics units, as well as public places. Improving the IEQ of those places is important because pregnant women and children have lower immunity [134] as well as visitors often congregate in public places [87].

The general ward, housing the longest and largest number of patients, is a key scenario that influences patient satisfaction and recovery. Particularly when patients were studied individually, correlations with specific wards cannot be ignored [22,222,295]. However, the type of wards in some studies is not clear and rarely analyzed. To fill this gap, a summarization of ward information includes the maternity ward [60, 149,224,234,235,251], psychiatric ward [11,152,195], medical ward [78], surgical ward [70,230], pediatric wards [78,231], geriatric ward [45,52] and modular wards [105].

For the collection of research subjects' information, four groups were classified including staff, patient, visitor, and occupant (without introductions on research subjects in articles). The main personal information included age [236], gender [79], BMI [237], as well as individual physiological parameters determined by questionnaires [331], health condition [158], disease information [74], metabolic rate [236] or survey on patients with disabilities [55], as well as the possible effects of medications (opioids) [36].

From the results in Fig. 4, gender and age continue to be the most collected information when confronted with different research subjects [178,217]. Apart from this, other types of information were more balanced when there were only patients. Work experience was emphasized when the staff was studied alone. Self-rating of health conditions appeared in different types of groups, but the overall distribution was small [167]. In ward-related studies, patients received more attention than staff. The most varied scenarios were associated with the inclusion of three types of occupants, reflecting that residential satisfaction is diversifying [231,290]. More IEQ studies have been conducted in general wards and outpatient areas [91]. While thermal studies focused on wards and operating rooms [293]. IAQ was in the spotlight in workplaces such as administrative offices [269]. In ICUs, acoustics are the most frequent consideration [28].

Overall, integrating different research interests provides additional insights into research directions. Through the combination, it can be found that different personal information may have an impact on research factors. However, there are conflicting and personal conclusions in research. Huang et al. [139] showed that the characteristics of managers did not affect their perception. While Patrick Amoatey et al. [29] and Lam et al. [167] illustrated that staff are more sensitive to sensory disturbances such as noise. Patients tend to have a wider range of acceptability for IEQ [249.272], but age and education level influence patients' overall satisfaction [190]. Gender makes a slight difference in temperature satisfaction [310], females prefer warmer environments [287,306]. Younger patients are more likely to accept indoor environments in healthcare facilities due to better regulatory mechanisms [236, 336]. Zhao and Mourshed [342] found that female outpatients attended to sensory design factors more than males. Residents' perceptions of indoor environmental factors, such as lighting and ventilation, varied based on their education and conditions like hyperlipidemia, osteoporosis, or stroke [140].

4. Critical review of factors influencing occupants' perceptions of restorative in healthcare facilities

As shown in Fig. 4, former publications researching IEQ-related factors and physical environment factors are complex and isolated. Besides, it could be noted that question 2 (RQ2) and question 3 (RQ3) are related from the logistic perspective, so in this stage and the next stage, we focus on the influencing factors assessment in healthcare facilities and control strategies conclusion. The CLR methodology was used to do the analysis, and the structure is shown in Fig. 5.

Physical design factors and IEQ are distinct yet complementary components of healthcare facility environments, each contributing uniquely to restorative perception.

From an architectural perspective, physical design encompasses structural and aesthetic elements, such as spatial layouts, material selections, and circulation patterns, which define the environment's form and initial occupant experience [283]. In contrast, IEQ, as defined in environmental engineering, includes dynamic, measurable conditions like air quality, lighting, acoustics, and thermal comfort, which directly influence occupant health and well-being [32]. Psychologically, physical design facilitates restorative settings through features like natural views, while IEQ supports immediate sensory comfort, as per Attention

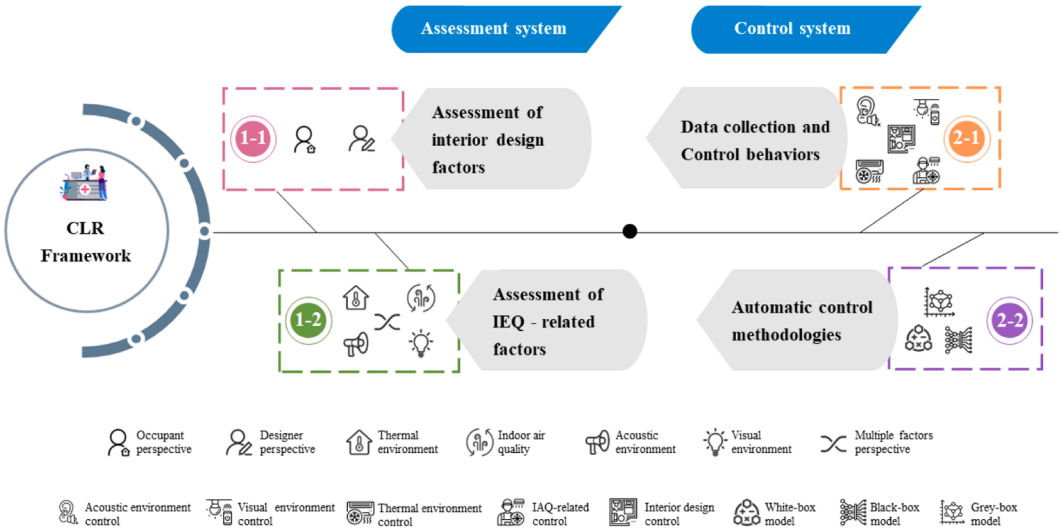


Fig. 5. The research framework for the critical review of factors influencing occupants' perceptions of restorative and related control strategies in healthcare facilities.

Restoration Theory. This multidisciplinary distinction, grounded in architecture, engineering, and psychology, underpins the proposed framework's integrative approach, ensuring a comprehensive assessment of restorative perception [141].

To answer RQ2, a whole analysis and assessments of the indoor environment for healthcare facilities are shown in this section, which consists of two parts: interior design factors and IEQ-related factors.

4.1. Assessment of design factors

Ulrich's theory [283] of supportive design conceptualizes how the healthcare physical-social environment affects patients' well-being and was tested by Andrade and Devlin [30] showing that design features had a positive effect on human perceptions. Proper layout design decreases healthcare facilities' risks and increases customer satisfaction and service quality [74,170,221]. In healthcare, designing is complex [133] and design factors have a great impact on occupants, apart from the efficiency and safety requirements [245], identifying a satisfactory environment is a second priority for designers, especially since the actual impact of each design factor has not been clarified. So, determining interior design is related to two key stakeholders: the designer and the user. In this part, we would like to evaluate the interior design from two perspectives.

4.1.1. Perspective of occupants

From the perspective of users, identifying key design factors that influence human perception is of paramount importance. Research has utilized various methods, including questionnaires like the H-PTPS [17], to isolate significant design elements.

However, the classification methods are not identified in published research, some researchers have proposed frameworks to evaluate interior design or rank the design factors. Han et al. [130] proposed a framework for Intelligent Healthscope Quality (IHQ), identifying five dimensions and 17 factors as reliable and valid measurements of health scope evaluation to cater to diverse needs and preferences. Another study found that furniture and spatial connections are crucial dimensions affecting the restorative quality of the ward environment, identifying four principal components (the space, furniture, safety feature, and privacy) and seven sub-dimensions [19]. Mahmood and Tayib [190] demonstrated a strong relationship between the quality of the healing environment and patient satisfaction, identifying four key factors: Interior appearance, Comfort and control, Views, and Privacy, which was similar to the results by Mahmood and Tayib [191]. In the research by Zhao and Mourshed [341], design factors ranked in terms of importance, with cleanliness, environment, and safety design emerging as top priorities while entertainment facilities and pleasant exterior views were deemed less critical.

Despite IEQ-related factors and control behaviors in literature, based on previous experience and literature collection results, the design factors in this paper are set into three parts: Decoration [104,114,134,205], nature [46,65,66,80,143,154,189,205,215,294,316,321], and spatiality [71,113,116,138,343].

For the decoration of healthcare facilities' interior environment, the decoration concluded with artwork and furniture layout. Environmental factors affecting patient outcomes include form, unit layout, floor material, room features, medical equipment visibility, nature, lighting, and music [257]. A study by Gao and Zhang [114] identified 19 design characteristics that significantly correlate with the restrictiveness of ward environments, including nature, artwork, doorway position, furniture layout, room size, and wall color. Ferrante and Villani [104] introduced the photometric characteristics of finishing materials and furnishings that would affect the daylight. The wall color may influence users' emotions [259]. Song et al. [264] showed that the color perception by eye tracking and the color preference by survey did not tend to match, as pink attracted a high level of attention, regardless of personal preference. Based on virtual reality, Amirbeiki Tafti et al. [27] demonstrated that incorporating natural blue elements in PET/CT scan room design elicits positive physiological responses in patients, enhancing the vitality of medical imaging environments. Facility design features can also promote healthy work environments to motivate nurses to continue working [266]. As to artworks, Nejati et al. [205] compared many photos with different elements and identified landscape artworks that had greater restorative value than indoor plants. It is similar to Hill et al. [134], who determined that the addition of music and natural landscape images could release patients undergoing pelvic reconstructive surgery.

Nature attracted many researchers' attention and they used biophilia design [3,86,129,143,321] to improve the positive affect on occupants. Patients specify a strong need for connection with nature [66]. Incorporating plant walls and digital elements into healthcare environments can synergistically enhance emotional well-being. This combination not only diminishes negative emotions but also fosters positive emotional changes, thereby having a complementary effect on the overall mood of individuals [154]. Berg et al. [45] discovered that the greening of a geriatric ward may support patient well-being and reduce functional decline. Nejati et al. [205] illustrated that physical access to the outdoors had greater value than window views. People often incorporate

gardens in hospitals, which could improve occupants' perceived restorative qualities [65], especially for hospitalized children [241, 316]. Six design-relevant factors in the garden assessment system, including mood, well-being, nature, mysticism, tranquility, and sense of touch, may be helpful to the understanding of gardens [65].

Spatiality is the traditional design factor which designers consider. It impacts patient satisfaction like the centralized unit transformed decentralized unit [138]. In general, the focus of spatiality is to protect occupant privacy [64] and safety [113]. For special groups such as the visually impaired, they can understand the principles of spatial organization normally, but vision and the length of time spent in the work environment can affect the spatial cognition of the visually impaired, so it is necessary to enhance the safety of the space to enhance their familiarity with the specific space [343]. Cho [71] revealed spatial quality, social program, and space for the successful delivery of therapeutic healthcare design. A study by Gao and Zhang [114] identified that doorway position and room size would influence patients' recovery. Relatively spacious rooms make patients feel free by providing enough space for their families. For doorway locations, the different relative positions between the door and the bed may lead to different perceptions of acoustic and visual privacy, as the door has the function of partitioning and connecting. Faghiholislam et al. [97] suggested that safe spaces, such as suitable positions of furniture, could help reduce depression in therapeutic spaces. Moreover, patients with high sensory and hygienic design requirements prefer single rooms [341].

4.1.2. Perspective of designers

Healthcare facilities are one of the most complicated buildings [49]. The complex spatial environment may impact cognitive mapping [203] and the layout design assures the quality of healthcare facilities services [221]. As to the division methods, based on previous research, it could be divided into three categories: Materials, layout, and circulation.

Selecting suitable materials is crucial for designers. In interior design, materials often affect 1) flooring, 2) walls, 3) ceilings, and 4) furniture. Flooring is often related to sound characteristics [223,284], as the materials affect the vibration. Walls are not only responsible for carrying artwork but also affect human cognition, especially the wall color is seen as a main factor in restorative perception [114]. As warm colors and cold colors are opposites of the color temperature, they have arousal-inducing and calming properties, respectively, which should be considered in suitable scenarios [264]. Ceilings are similar to floorings as they also affect the sound [223], heating [105,262] and pollutants [292]. Furniture is a broad concept, as it could be related to green, artwork, and decorations. Incorporating natural materials or nature-inspired elements can significantly boost patient recovery and reduce stress [104,244]. Using safe and easily cleaned surface materials [244] is also significant for furniture usage.

Layout design is a crucial perspective for physical environment design. For example, visual factors, physical environmental factors, and human–environment interactive factors significantly impact the spatial cognition of visually impaired people [343]. The layout of nursing units significantly impacts healthcare delivery and patient experience [88]. Decentralized nursing units are considered best practice, as centralized layouts restrict visual access to patient rooms [244], but Hua et al. [138] considered centralized or decentralized unit designs to analyze effective team member communication and illustrated decentralized unit design may decrease communications. Besides, inpatient unit design and patient experience are also connected, which reflects that the hybrid decentralized design supplies a more optimal experience than centralized design [58]. Single-bed rooms enhance infection control, privacy, and confidentiality, benefiting patients but decreasing the satisfaction of healthcare staff [88]. Additionally, incorporating a family zone or waiting zone within these settings provides a dedicated space for patients, their families, and visitors to gather, supporting a more inclusive and comforting environment [228].

Circulation refers to the way people move through and interact with a building, and it is often related to spatiality. Evaluating the circulation in healthcare facilities should involve many perspectives [172], including efficiency, restoration, connectivity [117] and so on. A study proposed combining design factors in therapeutic spaces, which could extract effective cycles to reduce patients' depression [97]. Geng et al. [117] used space syntax to test spatial arrangement, including wayfinding, spatial integration, visibility from the nursing station, connectivity for public space and visual integration, suggesting the nursing station should be set to enhance visibility and operational efficiency. As to the room design, designers should consider the room size and doorway position [113]. Cai et al. [58] also concluded that room design should align with and support care flow, operational model, and call button communication technology, as well as improve visibility and proximity between patient rooms and nurse stations, to improve patient experience. Cho [71] emphasized that safety is one of the most important parameters in healthcare facilities, which needs to be assured in the entry layout design, airflow circulation and flow of occupants. As to the waiting area, Qi et al. [228] proposed the distance from the waiting area to the consulting room should be $<$20 m, with no more than one turning point, which could help to keep a suitable flow organization and improve satisfaction. For the visually impaired [343], keeping safe, colorful especially green spaces and stylized traditional architectural forms could increase their behaviors in space recognition.

4.2. Assessment of IEQ factors

In recent years, some reviews summarized the various factors influencing IEQ within healthcare facilities. However, a critical gap in the existing literature is the infrequent integration of these factors into possible assessment metrics that can be correlated with individual feedback. This omission limits the ability to provide a comprehensive and nuanced understanding of how specific IEQ components affect occupants' restorative perceptions. Besides, previous research focused on single factor individual assessment, the whole assessment of IEQ is lacking.

4.2.1. Assessment of visual environment

Firstly, the visual environment is summarized. As interior design factors are collected in the 4.1 section, the lighting environment is the main research object in this section. The primary research on lighting environments can be categorized into two types: natural and artificial light, and their impacts on occupants [196].

In lighting assessments, brightness, illuminance, and color temperature are the main components for evaluating light sources [24]. Natural light, which offers higher brightness and better color rendering, is more susceptible to variations due to time and weather conditions. In practice, natural light is often associated with views and windows, and most of the time, both patients and staff showed a preference for natural light or conditions near windows, despite potential glare related to the reflectivity of interior materials [104]. Choi, Beltran, and Kim [73] showed that exposure to natural light can reduce stress, increase satisfaction, and work efficiency for staff, and alleviate pain and depression for patients, thereby enhancing patient satisfaction. Overall, ample cumulative photic flux leads to higher satisfaction, with employees exposed to longer periods of daylight reporting greater job satisfaction [50]. Additionally, the impact of lighting, natural and artificial, on humans is influenced by location [93]. The influence of natural light necessitates considering factors like bed position, curtains, and windows, combined with patients' actual needs [104], while the impact of artificial light sources also requires assessment of illuminance, spectral composition, control strategies, and actual environmental needs.

The illuminance and spectrum of light sources are associated with staff satisfaction, performance, and visual fatigue [145], as well as patients' sleep and perception. Environmental deficiencies such as thermal conditions and light intensity, along with the lack of proper management plans for night shift nurses, are detrimental and negative factors

for the health and well-being of this workforce [36]. Regarding lighting requirements in various hospital settings, many respondents reported no significant differences between professions. Concerning control strategies, these relate to the type of fixtures and reflectivity [232], with glare, flicker, and shadows to be avoided as they can affect performance, alertness, and sleep patterns [250]. Meanwhile, hospital lighting control strategies generally aim to support circadian rhythms [197], benefiting overall satisfaction for both patients and healthcare staff [106], although further consideration of patients' psychological needs and avoiding blue wavelengths at night is necessary to prevent disrupting patients' sleep patterns by reducing melatonin [14].

Long-term occupants, or workers (healthcare staff), have higher and more diverse demands for indoor lighting, potentially conflicting with the needs of short-term occupants (patients and visitors). However, overall, their needs are similar, desiring higher daylight flux and more color during the day, with satisfaction increasing under sufficient control [196].

4.2.2. Assessment of acoustic environment

When it comes to the acoustic environment, the noisy environment in hospitals is often problematic [44,185,226]. Regardless in general or ICU wards, noise often exceeds the standard value [29]. However, the sources of noise are different, and its effects are different. Previous studies have often categorized sounds into two categories: noise and music [254]. Given the differences in individual perceptions, music is often thought to help heal patients [70,134,281] but it may also be considered noise at times [175]. In this research, we discussed acoustic environment from the source, the complex composition and whether it is conscious.

Firstly, the sources of indoor sound in healthcare facilities were analyzed, and the influencing factors mainly included four parts: internal sound of buildings (ventilation and air conditioning equipment sound, cleaning equipment sound, cart, door, etc. [25,251,284]); building exterior sounds (external sounds transmitted through building openings, envelopes, pipes, etc.); human activity sounds (staff conversations, visitor voices, snoring, footsteps, etc. [153,251]; actively creating a sound environment (Music [70,231], Television, Alarm [217]).

The second is the composition of sounds, i.e. the proportion of different sounds, which is often expressed in terms of the acoustic complexity index [102]. Overall, complex sound environments can contribute to cardiovascular disease, sleep disturbances, and hearing loss among residents. For patients who are asleep, noise impacts brain and cardiovascular functions by influencing heart rate and blood pressure, subsequently deteriorating their sleep quality [29]. For patients in certain settings, such as the ICU, the complex composition of sound can cause fear and distress [29]. At the same time, healthcare workers suffer and even respond more strongly than patients [168,273]. An irregular and loud sound environment can reduce the satisfaction of medical staff, and even the sound of alarms has a similar effect to overall noise [217]. In certain situations, such as the operating room, even regular music designed to create a relaxing environment can be considered by some medical staff (anesthesiologists) to be a factor affecting their work [175]. However, music can also be considered helpful for ICU patients requiring treatment [231]. For patients, different sound interventions (natural sound, steady-state sound and written sound source) may improve people's satisfaction [186]. People tend to have a lower tolerance for these background noises. In certain scenarios or under specific needs that drive activities, the sounds actively produced by individuals become the dominant composite sound, like playing music with headphones [70], overshadowing other background noises. This can enhance occupants' satisfaction, although this point remains contentious.

Acoustic elements can influence additional domains as well. For instance, Dalirnaghadeh et al. [76] demonstrated that sound-related landmarks could offset the absence of visual cues, potentially aiding in the development of navigational systems for visually impaired users. Meanwhile, in a randomized trial, Hill et al. [134] discovered that although music and landscape imagery did not significantly alter postoperative pain scores, they did enhance the overall postoperative experience. Besides, thermal stimulation can create a comfortable thermal environment, which can improve the patient's assessment of acoustics, while negative stimulation has the opposite effect [307].

4.2.3. Assessment of thermal environment

The thermal environment is generally considered to be the most important factor affecting people's comfort [161,324], and it is also the factor with the most research content. The PMV model is one of the basic models established by many standards [35]. However, it is often found to be inconsistent with the actual human comfort in the actual environment [22,227,249,285,318]. Standards including ASHRAE 55 and ISO 7730 are present, but they are difficult to respond to the real-time needs of different individuals, and it is difficult to bring the entire hospital environment into compliance [293].

Thermal comfort emerges as a subjective perception influenced by various factors, including gender [61,227], age [62], participant type [249] and room orientation [62], yet consensus remains elusive.

Gender differences in thermal perception and preference are noted [287], potentially linked to health status [156]. Women are more sensitive to extremes of heat and cold [79] and prefer warmer room temperatures [307], generally expressing greater satisfaction with thermal conditions and air quality than men [227,268,335]. In contrast, men report more comfort with healthcare facility temperatures [310].

Age influences thermal comfort sensitivity, with older individuals being more sensitive than younger ones [62], showing less acceptance toward hot and dry conditions [307]. Young people (<25 years) as well as those with a normal BMI demonstrate better environmental adaptation [236,237]. Interestingly, older people rate indoor air quality lower than younger counterparts [336], and weight gain is associated with increased thermal sensation in patients [237].

Evaluations of thermal conditions and preferences among patients and visitors have been conducted [158], emphasizing the need for focused thermal comfort considerations for healthcare workers and the development of adaptive models [184]. These workers prefer cooler conditions, indicative of an adaptation to less warm environments. Moreover, a patient's thermal needs are further modulated by physical health [144,156], highlighting the complexity of achieving optimal thermal comfort in healthcare settings that accommodate the varied needs of all occupants.

In general, six main factors directly affect thermal perception [35] including four environmental factors: air temperature, radiation temperature, air velocity, and humidity [62] as well as two individual factors: the thermal insulation of the clothing and metabolic rate [233, 331].

Temperature generally has the highest correlation with thermal sensation [75] and low temperature may be an important risk factor for hospitalization [247]. Existing thermoregulation strategies do not take into account the patient's temperature needs often affect the patient's perceived level of attention [21]. Air velocity and humidity not only affect the sensation of heat but also cause other sensations, such as the sensation of blowing wind or stuffiness. Therefore, even if the combination of these six parameters reaches a neutral state of thermal perception, it may still cause discomfort due to the presence of a feeling of wind, etc. Managing the microclimate through behavioral controls of evenly distributed air temperature, relative humidity, and airflow is essential, as they are the three most important thermal environmental aspects in healthcare facilities [265]. These may also be influenced by the season and geographic location on thermal perception [236,260, 318]. Rahman et al. [2] determined that information on the climatic characteristics of the building's surroundings and the current state of the indoor thermal environment will help the building owner to develop an appropriate bioclimatic or passive design approach.

Relative humidity directly or indirectly affects the comfort of

occupants. Low humidity can cause discomfort [287] and impact health, such as dryness of the nose, throat, eyes, and skin, particularly when the dew point falls below 0 °C, leading to increased discomfort [292]. Conversely, high relative humidity levels above 65 % frequently foster microbial growth on indoor surfaces within inpatient settings [272]. Interestingly, patient perceptions of air dryness may not always align with actual humidity measurements, often feeling more pronounced due to factors like age and health status [287]. Managing indoor humidity, therefore, involves either reducing or increasing moisture levels through specialized equipment to maintain a comfortable environment [260]. Additionally, air humidification during the heating season in colder climates can alleviate symptoms of Sick Building Syndrome (SBS) and the sensation of dry air, enhancing overall comfort [211].

Ventilation, or airflow, is also pivotal for thermal comfort within buildings [322]. A properly maintained ventilation system enhances employee satisfaction and alleviates symptoms associated with poor air quality, such as dry skin and eyes. Natural ventilation generally provides better thermal comfort compared to mechanical ventilation systems due to its direct impact on occupants [325]. Factors such as the size and placement of windows, building orientation [291], and the size of heating elements like infrared ceiling plates [262] also play critical roles in shaping thermal comfort. In specialized settings like operating rooms, the type of ventilation system is particularly important for ensuring the restorative perception of staff [38,127]. Common problems include insufficient indoor air exchange, and insufficient control of indoor thermal conditions. This can be obtained by behavioral control of microclimate parameters (e.g., temperature, relative humidity, and wind speed) [63] and evaluation of control methods can be evaluated according to Fan et al. [98].

4.2.4. Assessment of indoor air quality

Farhadi et al. [101] concluded several standards and the acceptable limit for IAQ parameters were proposed. However, the application in studies remains diverse and controversial.

For the study of IAQ, there are two main parts: the first part is the control of particulate, pollutants and aerosol, including PM2.5, PM10, microbial counts, dust particles and the concentrations of carbon dioxide (CO_2), carbon monoxide (CO), formaldehyde (HCHO) and total volatile organic compounds (TVOC) [33,68,188,252] and so on. Those pollutants affect the mental health of occupants [125]. At the same time, they also affect the health of the occupants and lead to the occurrence of diseases. For example, sick building syndrome (SBS) [12,212,270] is a common disease closely related to the contaminants of IAQ, even asthma [188] and chronic obstructive pulmonary disease (AECOPD) [99] may be caused. Sanna et al. [246] mentioned that Rhodotorula spp could increase the risk to hospitalized patients, highlighting the importance of microbiological monitoring of hospital air and surfaces, especially in high-risk wards.

Besides, the COVID-19 pandemic is related to IAQ [90,147,180]. Hospital work during the COVID-19 pandemic has had important personal, professional, and psychological impacts on healthcare workers [136]. Respondents identified opportunities to better address information, training, and support needs. Personal behaviors, such as smoking [72,334], or speaking [177], may affect air quality by producing pollutants and altering aerosol distribution, respectively. Liu et al. [177] studied the effects of different respiratory behaviors such as normal breathing, prolonged speech, phrases, and alternating speech on the spatiotemporal distribution of aerosols in negative pressure wards, which had implications for both the risk of infection and IAQ in healthcare workers.

The other research part is to identify residents' satisfaction with IAQ. Zhang et al. [336] investigated perceptions of the air environment in hospital waiting areas, emphasizing the critical role of IAQ in shaping patient and visitor experiences. Uścinowicz et al. [287] focused on patient perceptions of the indoor environment in Polish hospitals, identifying a connection between demographic characteristics such as gender and age, and their assessments of the indoor environment's impact on recovery processes. This underscores the personalized nature of environmental sensitivity impacts on patient care. Yau and Phuah [320] revealed that many hospital areas exceeded recommended IAQ thresholds, potentially leading to SBS. This discrepancy points to the complex relationship between different aspects of IAQ and the subjective nature of environmental satisfaction. Kagan et al. [155] linked healthcare staff's job satisfaction with their awareness and experiences related to air quality issues, particularly concerning carbapenemase-producing Klebsiella pneumoniae in isolation rooms. This association between knowledge, environmental conditions, and job satisfaction highlights the importance of air quality not only for patient care but also for the health and satisfaction of healthcare providers. Besides, there are differences in the negative impacts of air pollution on occupants, with women and low-income groups feeling it more acutely [125].

4.2.5. Assessment of multiple perspectives

For the evaluation and research of the whole IEQ in healthcare facilities, the first controversial topic to be solved is whether the existing standards and personnel satisfaction can be harmonized. In the U.S., the LEED rating system is the most widely recognized certification system for green buildings [122]. Thiel et al. [277] and Xuan [309] showed that LEED standards were useful. Kim et al. [163] proved that staff in controlled IEQ scenarios would get higher satisfaction than non-LEED-certified hospitals. Xuan [310] did further research in the quantitative comparison of LEED-certified and non-LEED-certified hospitals and found detailed differences in IEQ factors. Liu et al. [176] investigated a one-year post-occupant evaluation and compared a 3A and a 2A hospital in different climate zones, they found that the insignificant correlation between the objective environment and subjective satisfaction because of their self-perceived health and individual differences, and seasons would affect occupants' perceptions of different factors in IEQ. Liu et al. [176] suggested that hospital types cannot be the same as the occupant satisfaction as the over-crowdedness was the major impact factor.

Standards still need to be improved to satisfy the needs of occupants and evaluating methods is a point for researchers. For the evaluation of IEQ in healthcare facilities, Nimlyat [207] collected data in three hospitals and built a structural regression model to represent occupants' satisfaction. It gave a standardized protocol for IEQ assessment and supplied the building index rating standards. Teoh et al. [275] used multilevel structural equation models to draw on the job demands-resource model to test the associations between hospital-level care with hospital-level demands and self-reported doctors' job demands, job resources and wellbeing.

IEQ have impacts on different occupants' restorative perceptions, and individuals show different feedback on it. Huang et al. [140] did a questionnaire on occupants in long-term care facilities and got a comprehensive understanding of the characteristics (profession, education level, age, illness) of building users in LTC facilities were associated with differences in the importance given to indoor environmental indicators.

In Madagascar, Nematchoua et al. [206] carried out a complex survey of 5 hospitals showing that there were not very great differences between males and females in their thermal sensations, humidity sensations, and air movement sensations. Ma et al. [183] revealed significant correlations between IEQ parameters (temperature, TVOC, and HCHO levels) and the spatial distribution of patient behavioral changes. Eijkelenboom et al. [92] presented a study on health and comfort of staff in outpatient areas to evaluate the impact of room types on human perceptions. It showed a larger prevalence of building-related symptoms and dissatisfaction with comfort aspects in outpatient areas than in offices or inpatient area. In the next research, Eijkelenboom and Bluyssen [91] made six clusters according to the preference and comfort level of the IEQ and three clusters were made for preferences and comfort in the social aspect. These clusters suggested that IEQ preferences were related

to health and comfort.

However, due to the limited overlap between the characteristics of IEQ clusters and the characteristics of social clusters, it was necessary to consider both at the same time. Based on the research results, it can be concluded that the cross-modal effects of IEQ have multiple influences on human cognition, perception and behavior, but the conclusion may be conflicted and be influenced by individual factors and groups [305], even objective factors. Sun et al. [268] showed that occupant satisfaction was influenced by gender, working hours, and length of stay, which may affect the energy efficiency regarding their hospital length. Therefore, the integration of all factors of IEQ at the building design stage was needed, which provided a hospital environmental setting that reflects occupants' requirements and expectations, promoting patient healing processes [208].

Some studies also emphasized the evaluation and enhancement of indoor environmental conditions in hospitals. The earliest case study was conducted in the Padua Hospital [78] and proposed to pay attention to the building envelope in an optic of building renovation, especially for corner rooms to avoid deficiency in building or system design and maintenance, which could help improve IEQ assessment. Tang et al. [272] conducted a three-day survey in inpatient buildings in a humid and hot region in China, and they revealed that HVAC systems adjustment could be connected to IEQ conditions. Khan et al. [160] identified that factors such as the occupancy rate, ambient thermal conditions, the type of HVAC system, and building orientation significantly influence IEQ across various hospital settings, including emergency rooms, operating theaters, ICUs, and medical wards. Tang et al. [273] focused on the hospital-street layout impact on a hospital and proposed that air quality and uneven illumination in the hospital street with hospital conditions (roof materials, IAQ) would cause discomfort.

Besides, emergencies like the COVID-19 pandemic could affect the preferences for IEQ and the social comfort of outpatient workers [90]. Tang et al. [274] demonstrated POE could identify deficiencies in the indoor environment and provided a reference for assessment of IEQ in nonresidential buildings. Jain et al. [146] believed that compliance with strict IEQ requirements in hospitals comes at the cost of energy (e.g., filtering and conditioning large amounts of fresh air), which is critical to striking the right balance between reducing energy use and greenhouse gas emissions. This requires the development of an appropriate environmental control strategy.

5. Critical review of control strategies on healthcare facilities considering occupants' perceptions

To address RQ3, occupants take many actions to control interior equipment, and much guidance is proposed to accomplish the proper actions. Nowadays, many standards are generally considered best practices and are often incorporated into national regulations and guidelines [101]. ISO Standards provide clear standards for environments with high requirements, such as operating theatres, ASHRAE Standards [204] are technically detailed, particularly suitable for temperature and air quality control, LEED Certification [310] is designed for eco-friendly and energy-efficient healthcare buildings, like using natural light and sustainable materials, and lastly, JCI Standards mainly focus on processes and management, may lack specificity in environmental details.

These standards each have their unique features and considerations. But in actual applications, considering their specific environmental contexts, patient care needs, and local regulations, healthcare facilities often need to integrate various standards. However, there is a large gap in the performance capacity of medical institutions [181] and the standards set by the hospital are not consistent with the actual satisfaction of the occupants [176], and adaptation strategies seem to be influenced in a psychological way [302]. Besides, people's life behaviors have different degrees of influence on the elements of indoor physical environments [338].

Based on analysis above, this section will be divided into two parts: the collection and description of existing control behaviors (Section 5.1), and the analysis and summary of the latest automated control and related methods (Section 5.2).

5.1. Data collection and control behavior

Based on the results of the previous study and referring to the previous study [312], types of data that control behaviors and influencing factors can be divided into two categories: external factors and internal factors. External factors are environmental factors, physical information factors. This data is mainly based on environmental parameters, mainly objective data and statistics collected by sensors. The internal factors are mainly based on human information data. This includes physiological and psychological information, as well as data from surveys of an individual's subjective willingness to control. In this paper, the occupant control behavior and the design and adjustment of the indoor environment are collected and analyzed.

5.1.1. Interior design control

There are a variety of factors that need to be considered to control the design of a hospital, and it needs to consider the requirements of efficiency, capacity, zoning, etc. According to the classification in Section 4.1, it could be divided into three perspectives, including spatiality: choosing the right space size [113], increasing the protection of privacy and safety [64,97], balancing the efficiency and comfort of the streamlined design [276]; green: the arrangement of green plants in the interior, the choice of the natural environment outside the window and increasing the window-to-wall ratio [300]; and decoration: the selection of floor, artwork [205], decorations [114], and wall materials and colors [264] and so on. Those factors are related to designers, as the confirmation of materials, layout, and circulation need to be based on the actual conditions.

In the past, based on the selected factors from occupants, designers often used the Evidence-Based Design (EBD) method to create physical environments that positively impact patients, staff, and visitors [59]. Recently, space syntax [117,276] and even AI methods [172] were proposed to model layout and circulation designs as well as critical qualities in healthcare settings. These methods help designers do interior design.

Many designers welcome EBD and focus on creating positive physical environments and building standards like the LEED [277]. It is traceable. Hospitals are among the most expensive facilities to build and their design should be guided by evidence-based research [152]. This study provides evidence that family rooms off wards and mixed-sex communal areas can potentially increase treatment satisfaction. Selecting key factors is important. Bingham et al. [50] documents the implementation of EBD elements relating to patient privacy and comfort in hospital renovation projects. Qi et al. [228] identified four first-level indicators in four dimensions: functional layout flow organization, supporting facilities and environmental details in the waiting space environment of pediatric clinics. Based on EBD methods, many researchers developed tools to assess buildings. Huisman et al. [141] selected the impact of physical environmental factors on users in EBD and designed a tool to evaluate building performance. Similarly, Joseph et al. [151] built a POE toolkit to help workers to evaluate building performance. Alotaibi et al. [20] proposed a new design tool considering inpatients' various medical and complex conditions to fill the gap. For special patients, Kirch and Marquardt [164] developed a validated assessment tool on design for dementia available to improve the functional ability of patients at discharge from the hospital.

Space Syntax is a traditional design method, usually used to optimize the streamlined design to connect different components in hospitals efficiently [58]. Two Space Syntax-based metrics [276] are introduced to model proximity and ease of access in layout designs to control infection. Brambilla et al. [56] provided meta-design indications for the

improvement of healthcare facilities, emphasizing the role of spatial design in the management of epidemic health emergencies. Space syntax also could assist global researchers in setting up future studies with a comparative nature. Geng et al. [117] compared spatial analysis and observation in four hospitals by space syntax, which provided designers with both the spatial qualities and the practical use of spaces.

In addition, maintaining feedback between designers and users, and increasing the importance of residents' suggestions, can help to develop suitable solutions [116]. Nowadays, the use of more advanced technology can lead to a better understanding of the exact needs of different occupants and can help designers refine the design process. Zou and Zhou [343] collected physiological monitoring signal data from visually impaired people walking along prescribed routes, including electrical skin activity, heart rate variability, and electroencephalograms, and then compared the physiological monitoring signal data of each route to help designers design and transform the space. The most advanced technology currently is the brain-computer interface (BCI), Yang et al. [314] have developed a BCI tool to help designers manipulate the position and size of windows in a virtual space, helping to refine the design process. Similarly, Jung et al. [154] facilitated the widespread adoption of biophilic design by measuring emotional changes associated with biophilic design in an experimental setting through a combination of electroencephalography (EEG) and VR methods.

5.1.2. Visual environment-related control

Lighting is considered one of the key architectural factors affecting individuals [106]. Early designs for lighting control schemes have been proposed [26]. But literature surveys indicate a divergence in perceptions and needs regarding lighting control between healthcare staff and patients, with both groups showing a high preference for natural light [73]. Nurses desire increased brightness to aid their work in wards [196], whereas patients often seek greater autonomy, possibly preferring a dimmer environment, leading to potential conflicts [64] However, enhancing their control over lighting can improve satisfaction with the light environment.

Lighting control is categorized into natural light control and artificial lighting control. Control of natural light primarily involves modifying the window-to-wall ratio [23,279] adjusting blinds [93], and utilizing shades to regulate natural light exposure [297], which helps avoid glare [24]. Besides, roofs also impact on humans. As to roof design, applying green roof could improve physical health and emotional well-being [215]. Selecting suitable furniture surface materials can also improve daylighting conditions [104]. Besides, roof materials are important, the transparent ethylene-tetra-fluoro-ethylene (ETFE) membrane roof may cause visual discomfort [273]. Artificial lighting control is more complex, encompassing relocating light sources to improve the hospital's lighting environment [174], altering spectral composition to allow variable color temperatures in different areas, implementing automatic lighting control technologies to reduce blue light and increase red light [107] for alertness and circadian rhythm-aligned lighting schemes [250]. However, success requires close collaboration with occupants and continual adjustments.

5.1.3. Acoustic environment-related control

For the control of noise, providing patients with a sound environment that is conducive to recovery and satisfaction, and reducing the stress of healthcare workers, is the core of current work [187]. The World Health Organization (WHO) recommend that hospital noise levels should not exceed 35 A-weighted decibels (dBA) during the day and 30 dBA at night [47]. They are still difficult to achieve for many hospitals, even indoors and outdoors [226]. Many studies do not provide recommendations for the control of sounds, and there is a lack of multiple scenario validations [173]

Sound control can be implemented from three perspectives: the sound source, the propagation route, and the human ear. Source control includes control of human voices and device sounds [142]. For the control of human voices, it is necessary to carry out noise reduction education and training for employees, set up conversation areas, or install noise-sensitive lights to remind employees of activities and set quiet times [132]. On the other hand, for patients or visitors, it is necessary to set up reminder signs and limit the patient's visit during quiet moments. However, for the ICU environment, behavioral modification alone is not adequate to control excessive noise [166]. For environments with special purposes, patients can make a lot of noise and need to set up sound locks to meet their needs [223]. For equipment sounds, it is possible to change the wheels that make noise from the trolley, change the position of the equipment, and keep maintenance of the equipment or doors that make noise [142].

The control of sound propagation pathways is mainly carried out by setting up materials, such as the use of vegetation [294], installing high-performance sound-absorbing materials for ceilings and walls, and noise-reducing bed curtains. Another approach is to close windows and doors. The control of the human ear, such as the provision of earplugs or noise-cancelling headphones [220], or the playing of soft music to mask noisy environments and calm the patient. In addition to this, there may be a combination of other factors, such as visible vegetation could improve human perception of environmental noise [294], and the comfort thermal environment may also improve the evaluation of the acoustic environment [294].

5.1.4. Thermal environment-related control

Enhancing occupant satisfaction with the thermal environment is challenging due to the complexity of thermal comfort. This comfort is influenced by several factors including the building's size, the facilities and equipment available, and the number of people present. Additionally, individual perceptions of heat play a significant role, varying with each person's body type, age, clothing, activity level, and even gender [158]. Considering the differences between individuals and the environment, Zaniboni et al. [326] proposed to develop a database to meet the thermal needs of residents of different identities through a combination of subjective and objective research. Similarly, Alotaibi et al. [20] proposed a comprehensive framework, and Dovjak et al. [83] also suggested related control methods, but there are still many restrictions on how these can be applied. Because climatic conditions vary from country to country, the necessary temperature range and relative humidity levels differ. To address this, centralized air conditioning systems are shifting toward personalized comfort solutions tailored to individual needs [7, 200]. This is also in line with the patient's need for relevant control autonomy [64].

The control scheme can be divided into two schemes: active and passive heating [282]. Active heating is the setting of products that can be heated, such as personal items, heated blankets, and warm or heated clothing, which are often used in the Operation Room [157]. Passive heating solutions are closely linked to ventilation installations, and the effects of natural and mechanical ventilation need to be taken into account during the set-up process [120]. The advantage of natural ventilation is the ability to achieve a high ventilation rate. This emphasizes the impact of the sensation of the outside, which gives them the impression that they have a higher degree of control over their environment [325]. Buildings' thermal environment and energy can be significantly affected by their natural ventilation strategies [7]. Besides, Using the radiant ceiling panel (RCP) systems [292] to heat air is also useful. The main advantage of mechanical ventilation is the ability to precisely set indoor conditions. This is the most common method for maintaining a hospital indoor thermal environment, but due to individual differences, it is difficult for control strategies to meet the needs of all individuals [293]. The use of a personalized ventilation system may improve their thermal comfort [200] but substantial research on this new solutions is needed. At the same time, it is also necessary to meet the control strategy for IAQ. Combining natural and mechanical ventilation also could help improve thermal comfort in the wards [105, 201].

Healthcare professionals, varying in their activity levels and dress codes due to different occupational roles, require distinct thermal environments. To address this, creating multiple thermal zones has been suggested as an effective solution [21,81]. These zones cater to diverse thermal comfort needs by adjusting wind speed and temperature [81,80]. At the same time, the thermal environment is also closely related to other conditions, such as sound, materials, etc. Saikia et al. [242] discovered that deploying polystyrene as insulation on the ward envelope could be a retrofit measure that could minimize indoor heat gain.

5.1.5. IAQ-related control

The ventilation system not only has an impact on thermal comfort but also plays a major role in controlling IAQ [238], which is the focus of current research. Zhang et al. [332] proposed that the traditional methods of improving the aerosol removal capacity of ventilation systems could be divided into three main categories: increasing air changes per hour (ACH) [108], changing the location of inlets and outlets [248], and adding inlets to existing overall ventilation [271]. Therefore, the control of the ventilation system is divided into three parts in this paper: device setting, policy control, and position control. Besides, applying air quality monitoring systems could enhance indoor comfort for patients [77].

When it comes to device setting, it mainly refers to replacing equipment or optimizing equipment design. Increasing upgrading filters is a common method. At present, it mainly refers to the use of high-efficiency particulate air (HEPA) filtration systems [329]. Improving the ventilation system can also have the same effect. Zhai and Osborne [329] showed that maximizing the area of the laminar flow diffuser allows optimal air sterility to be fully achieved and provides very low levels of contamination. Modifying the design of air-conditioning and mechanical ventilation (ACMV) systems has been proven to be effective [320]. IAQ audits in hospitals have shown that occupants tend to feel more discomfort under centralized ACMV systems compared to non-centralized systems. Nonetheless, hospitals with centralized ACMV systems demonstrate superior control over chemical and particulate contaminants in their IAQ [320]. Wang et al. [300] used CFD to simulate the load distribution of air conditioning design in large spaces and suggested a layered air conditioning approach. In negative pressure wards, ventilation system designs include ceiling-supplied floor return (CSFR) and ceiling-supplied ceiling return (CSCR) as well as additional curtain ventilation (ACV) systems [332]. Zhang et al. [332] proposed to reduce ACH and improve energy efficiency by evaluating the accuracy of ventilation systems and using ACV as a single method.

Policy control strategies can be divided into active and passive strategies [1]. Natural ventilation strategies often use CFD to model. Adamu et al. [7] evaluated four buoyancy-driven natural ventilation strategies (single window, double openings on the same side, low intake and exhaust chimneys, and ceiling-based natural ventilation) for single rooms using a regional dynamic thermal model (DTM) and a CFD model. Verma and Sinha [296] used CFD analysis to simulate and compare the removal of contaminating particles (viruses, fungi, bacteria, etc.) using several different ventilation systems. Adamu and Price [6] set a new system called natural personal ventilation (NPV) is feasible to decouple airflow delivery from windows. Berlanga et al. [48] evaluated ventilation performance and contaminants released by health workers exposed to closed patients, evaluating the ease of using four different mixed ventilation configurations in individual wards. The local exhaust strategy could give the best performance for proper airborne infection control [95]. As a supplement to the traditional evaluation indicators introduced before, Lu et al. [180] introduced two indicators to help researchers improve the evaluation of risk control strategies for airborne infections.

Position control, which compensates for the lack of policy control strategies. For example, the conventional approach is to increase the fresh air supply rate (or ACH), but this does not always ensure a reduction in the build-up of stale air in all areas of the ward. In fact, under-planned designs may be a tendency to do the opposite [242]. CFD simulations [4,110–112,239,240] and experiments are normal methods to design hospital contamination. Many studies have been tested through them to design and model HVAC systems [313,315], helping designers to design accurately and avoid the spread of infections such as the COVID-19 virus [147]. Erdogan and Yilmazoglu [95] simulated the exclusion effect of air aerosols from three different ventilation layouts in the ICU and judged that local exhaust can effectively improve the exclusion rate. Aganovic et al. [9] tested a new ventilation in a single-bed isolated room to verify that a concept called protected occupied zone ventilation (POV) could improve the quality of inhaled air and satisfy occupants' comfort needs. Ho et al. [135] did a three-dimensional analysis of thermal comfort and contaminant removal in hospital operating rooms and found that better overall performance could be achieved if the supply grille was placed closer to the vertical centerline of the wall, while the location of the exhaust grille was insignificant. Stale air tends to accumulate near the central roof area [332], and placing the outlet grille near the roof along the roof centerline is an effective measure to expel the stale air that accumulates. Exhaust locations should be positioned as close as possible to the source [95]. It is also a good practice to place the patient's head towards the posterior wall with an exit grille because due to the presence of the exit grille on the posterior wall, stale air tends to accumulate in the patient's leg area, but not directly above the patient's head [332].

Many other aspects other than the ventilation system can also affect IAQ. Decorative materials can also affect IAQ because some pollutants may remain in fixed locations [177]. Besides, clothing materials and types, such as disposable spun-bond polyester gowns and reusable woven polyester gowns, may impact IAQ [238]. Specific shapes in the environment can also affect airflow, for example, in the operating theatre, surgical lights, surgical equipment and the surgeon's posture can seriously interfere with ventilation airflow [238]. Frequent opening of doors can also have a disturbing effect on airflow, so in operating theatres equipped with hybrids, preoperative planning and communication should be improved to minimize unnecessary opening of doors [238]. Rooms or spaces with specific requirements should be controlled with differential pressure control and directional airflow [332]. Nursing staff should open windows and ventilate rooms depending on actual conditions, and clean and sterilize rooms on time to avoid pollutants [254].

5.2. Automatic control methods considering occupant perception of restorative

Beyond the initial, hard-to-alter setup of a space (like equipment, natural scenery, and decorations), most behaviors are under individual control [339]. With technological advancements, building automation systems have found applications across various scenarios, yet their integration into hospitals is still at a preliminary stage. A critical gap identified is the lack of methods that combine individual perception models with control systems, which benefits users [82]. For instance, a user-centered system developed for burn wards [85] was tested in a lab setting. Effective indoor environment control requires a comprehensive strategy that recognizes different occupant needs [339]. This paper identifies three main control methods based on literature: adjustments of automatic window openers or blinds [255], HVAC adjustments, and lighting automation [192,250], covering the majority of key personal actions in managing the indoor environment.

For a variety of control behaviors, it is important to establish a suitable model for the control strategy. Current models, such as PMV, are often thought to be influenced by individual differences [79]. The development of the Personal Comfort System (PCS) has made it possible to provide more refined control schemes to address individual differences, which have been studied mainly by occupant behavior [156] and occupancy. Due to the uncertainty and unpredictability of occupant behavior, as well as the complexity of the interaction between occupant

behavior and the building's physical system, there are many ways to classify suitable models [311] but there is no consensus on them. According to the topic of personalized control, the classification is now mainly focused on human modeling and human-environment modeling. In this paper, we divide the modeling methods into three categories according to the data source and processing method: white-box models, black-box models, and grey-box models, and summarize their applications in the context of healthcare facilities.

5.2.1. The application of white-box models

White-box models, characterized by their transparency, play a crucial role in understanding the intricate dynamics between individuals and their environments and are widely used in actual projects. For human perceptions, studies like those by Zaniboni et al. [327] have explored the impacts of metabolic rates and subjective parameters on thermal comfort, employing models like the Fanger [100] and adaptive models [249] alongside statistical analyses to assess neutral temperature preferences and the interaction between thermal comfort and other domains, highlighting the challenges and adaptability in estimating thermal satisfaction across different populations. Uścinowicz and Bogdan [285] also proposed ways to modify the thermal perception model. In recent years, the consideration of physiological signals [123], including EEG signals, has also become a new trend in modeling, which can be combined with predictive models to propose more specific control schemes for relevant building systems.

On the environmental front, CFD simulations epitomize the application of white box models in predicting airflow within architectural spaces, thereby facilitating the design of buildings optimized for natural ventilation and energy efficiency. This approach relies on solving Navier-Stokes equations to simulate air movement and distribution, crucial for effective ventilation strategies. It could be used to test a variety of hybrid ventilation strategies [201]. Research by Kong et al. [165] and works by Dovjak et al. [84] have extended these models to virus control strategies and energy-efficient thermal comfort solutions, respectively. Furthermore, in the lighting control scenario, the integration of lighting control based on circadian rhythms into white box modeling, as discussed by Barroso and Den Brinker [43], underscores the significance of aligning environmental controls with human biological rhythms for optimal comfort and energy use.

What is more, from a whole perspective, Maleetipwan-Mattsson et al. [192] integrated theories like the Theory of Affordances, Theory of Planned Behavior (TPB) and Planned Behavior Control (PBC) to link lighting design with user interaction, aiming for optimal usage that supports occupant comfort and energy conservation. These instances illustrate the breadth of white box modeling in bridging the gap between theoretical understanding and practical applications for enhancing human-environment interactions, highlighting the potential for more targeted and efficient interventions in building design and environmental control.

5.2.2. The application of black-box models

In contrast, black-box models are essential tools in many fields, particularly in scenarios where the underlying mechanisms of a system are complex, unknown, or deliberately hidden. A key aspect of black box modeling is its reliance on empirical data to establish mathematical relationships between inputs and outputs. For instance, Fattahi and Sharbatdar [103] highlight the importance of system performance measurement data in constructing these relationships, often utilizing real-time data components like seasonal variations and setpoints for training models. It utilized data from the Green Buildings Innovation Cluster (GBIC) project. By considering a variety of factors, including physiological, environmental, and personal, the study designed several feature sets to develop predictive thermal models of individual comfort responses.

Moreover, black-box models have been employed to assess the effects of physical characteristics and configurations of building components, like windows and shading devices, on energy usage, lighting, and visual comfort. In the healthcare domain, preliminary evidence suggests benefits from daylight in hospital bedrooms, yet the effects of using automated dynamic lighting systems remain inconclusive [60]. Eisazadeh et al. [93] found that the right combination of windows and blinds can significantly reduce cooling loads while maintaining adequate lighting levels, emphasizing the importance of selecting appropriate configurations for overall performance. Research by Jia et al. [149] and Shi et al. [255] utilizes logistic regression to model human behaviors, such as window opening and closing in hospitals, based on seasonal variations, demonstrating the adaptability of these models with accuracies exceeding 70 %.

In summary, black-box models offer a pragmatic approach to dealing with complex systems by focusing on observable inputs and outputs. Their utility across various applications—from building environmental control to healthcare—demonstrates their versatility and effectiveness in scenarios where a detailed understanding of internal processes is either unavailable or unnecessary.

5.2.3. The application of grey-box models

Grey-box models meld the detailed insight of white-box models with the efficiency of black-box models, making them ideal for systems where complete information is not available. These models use a combination of empirical data and theoretical principles to predict system behavior, which is particularly beneficial in complex environments like buildings or for applications such as Model Predictive Control (MPC) [37]. Grey-box models can provide a more feasible and reliable basis for MPC.

They also can adeptly manage the intricacies of thermal dynamics in buildings, leading to more optimized and energy-efficient control strategies. The research conducted by Nimlyat [207] illustrates the application of grey-box modeling through structural equation modeling (SEM) to establish relationships between IEQ parameters, patient satisfaction, and health outcomes. Lan et al. [169] combined thermodynamics, CFD, and thermal comfort models for effective analysis and comparison of thermal comfort impacts stemming from various low-energy building retrofit concepts. Khan et al. [159] explored a grey-box modeling approach to enhance infection control in hospital wards by combining response surface methodology (RSM) and genetic algorithm (GA). This work aimed at optimizing ward design to reduce airborne transmission risk while maintaining patient comfort and minimizing environmental impact.

These examples highlight the strengths of grey-box models in various applications, from building energy management to healthcare environment optimization. By judiciously combining theoretical knowledge with empirical data, grey-box models enable more nuanced and effective solutions to complex problems where neither purely theoretical nor empirical approaches alone would suffice.

6. Discussion

6.1. Summary of main findings

There are many literature reviews on the indoor environment design of healthcare facilities and some factors of IEQ, but there is no relevant review on the impact of the overall indoor environment of healthcare facilities on human restorative perception. This paper summarizes the factors of interior design and environmental quality that affect the residents' sense of restorative perception in healthcare facilities and evaluates the related control behaviors and the advanced automatic control methods.

This paper first explores and defines the perception of recoverability in healthcare facilities, determines the scope of the search, and finally selects 249 related articles for literature review. This paper then analyzes these articles quantitatively and identifies the current research status and basic information in this field, including the changes in the research situation in various countries, the research methods and sample

size, and the objects and proportions of attention. The content of the studies, including personal information, research subjects, research scenarios and research topics was analyzed, and the relevance was determined. The analysis reveals that hybrid methodologies are the most prevalent (42.01 %), followed by objective (34.25 %) and subjective (23.74 %) approaches, indicating a trend toward integrating diverse research methods to capture the complexity of restorative perception. Among the factors studied, design-related factors are the most researched (22.62 %), followed by thermal-related factors (21.83 %), IEQ (19.44 %), and IAQ (17.46 %). Acoustic-related (9.52 %) and visual-related (9.13 %) factors receive less attention, suggesting opportunities for future research to address these underrepresented areas.

Wards, the most studied hospital settings, encompass specific research areas involving various departments, followed by workplaces, outpatient areas, ICU, operating rooms, waiting areas and other scenarios, among which the studies that consider the needs of different places are still a minority. At present, there are three main types of hospital residents: patients, staff, and visitors, among which the research on staff accounts for a higher proportion, followed by patients, and there are many surveys of various types of groups, but many studies do not indicate specific research objects. At the same time, personal physiological information (age, gender, height, weight, and BMI) is considered more, and social attributes are less considered. Personal information can have an impact on the conclusions of relevant research, which may be relevant to specific research methods and research scenarios.

Regarding the review of the overall environment of the hospital, this paper examines two aspects: interior design and IEQ. For interior design, there are two perspectives: occupants and designers. For occupants, the design of the indoor environment will have an impact on their perception, comfort, health, privacy, and more. There are many ways in which research has been categorized in this area, and many papers have focused on the setting of physical space in general. This article elaborates on spatiality, decoration, and natural objects, and recommends meeting the spatial requirements of occupants for safety and privacy as much as possible, adding appropriate decorations and helping them to get in touch with nature as much as possible. As for the designers, they care about materials, layout, and circulation, which could relate to occupants' needs. Balancing efficiency, safety and comfort are crucial in designing. The design approach they adopt is mainly focused on evidence-based design research, and the development of new technologies, such as spatial syntax and AI methods, also helps them. The use of new technologies to improve feedback between designers and occupants can be more effective in helping them, but this needs to be further studied.

For the research related to IEQ, this paper summarizes the visual environment, acoustic environment, IAQ, thermal environment, and overall evaluation scheme. Diverse factors on occupants and environment may impact the restorative assessments. The demands for staff, patients and visitors are different, causing some conflict, but the needs for a clear and healthy environment are consistent. At present, the research on IEQ in healthcare facilities is still mainly based on the evaluation of a single factor, and the overall evaluation mainly focuses on the establishment of evaluation standards and systems for the medical environment, but there is still a lack of clear analysis of the composite impact of multiple factors. As to the related control behavior, circadian rhythm lighting and more natural lighting, personalized sound control but without disturbing others, increasing natural ventilation and pollution control, as well as proper heating methods control for occupants, could improve occupants' restorative perceptions. Increasing control over residents, setting suitable materials [308], and regulating simple control behaviors could effectively improve evaluations. Applying appropriate and automatic control methods may be helpful for the overall control system in the immediacy, personalize, and comprehensiveness. But the application is still limited, and developing suitable control models (black, white, and grey models) in specific scenarios is a challenge, especially for individuals. Considering all factors to build and maintain a suitable indoor environment can help residents recover faster, relieve stress and anxiety, increase productivity, improve sleep, improve comfort, and promote healing, which can reduce hospital stay for patients and increase efficiency and strengthen links for workers.

6.2. *Underexplored areas and future directions*

Based on the aforementioned analyses, future suggestions are proposed below compared to underexplored areas in the literature, which are shown in Fig. 6.

- **Applying advanced technologies to increase information form occupants' feedback collection.**

  Getting precise feedback from occupants is a fundamental step; however, comprehensive and real-time collection methods of personal data are still lacking in application. There are many studies on satisfaction with IEQ, and most of them still rely on questionnaires and monitoring data on the environment. Whether from the HCAHPS (Healthcare Provider and Systems Assessment) survey [171] or the researchers' questionnaires, the results are subjective, and there is a lack of uniform criteria. With the advancement of technology and the application of various new wearable devices, it is possible to test people's EEG, ECG (electrocardiogram) [343] and GSR (electrical skin response) [113] anytime, anywhere. With these tools, more accurate personal information can be obtained, and future satisfaction studies will have more consistent measurements and more scientific indicators and results.

- **Utilizing human-environment interaction methods and advanced design techniques to achieve fast and accurate interior design.**

  The initial layout and environmental design of healthcare buildings are influenced by personal perception, making traditional layout designs inadequate for swiftly meeting individual needs. The knowledge gap between designers and occupants presents a challenge and the connection between the environment and personal perception warrants further investigation. This indicates a need for rapid design changes and advanced methods of integration and experience to meet practical demands [150].

  Simulation for architectural design is a common strategy, with researchers employing artificial intelligence to generate layouts, such as improved Genetic Algorithms (GA) or Particle Swarm Optimization (PSO), aiming to fulfil objectives centered around staff efficiency and patient-centric goals [172]. However, achieving high precision in simulations is challenging due to the nonlinear nature of human behavior. Therefore, creating interactions with users in yet-to-be-realized environments might offer a more effective solution. Utilizing technologies such as (VR and 3D printing to facilitate layperson participation in design can lead to more precise and targeted outcomes. It also aids in understanding the impact of the architectural environment on individual perception [303].

- **Developing dynamic evaluation criteria based on objective evaluation of scenarios and personal characteristic factors.**

  Lack of comprehensive impact assessments of specific scenarios and individual differences in healthcare facilities. Research predominantly focuses on the objective impact on the human body, yet the subjective will and specific demands significantly influence evaluations of environmental factors [64]. Autonomy in environmental control can markedly improve individuals' evaluations of their surroundings. The combined effects of different factors vary significantly, influenced by specific environments and populations, making it crucial to identify individual preferences and impacts. Meanwhile, most studies lack a categorized investigation into research scenarios and occupants, failing to explore the impact of various settings, individual differences, and the relationship between

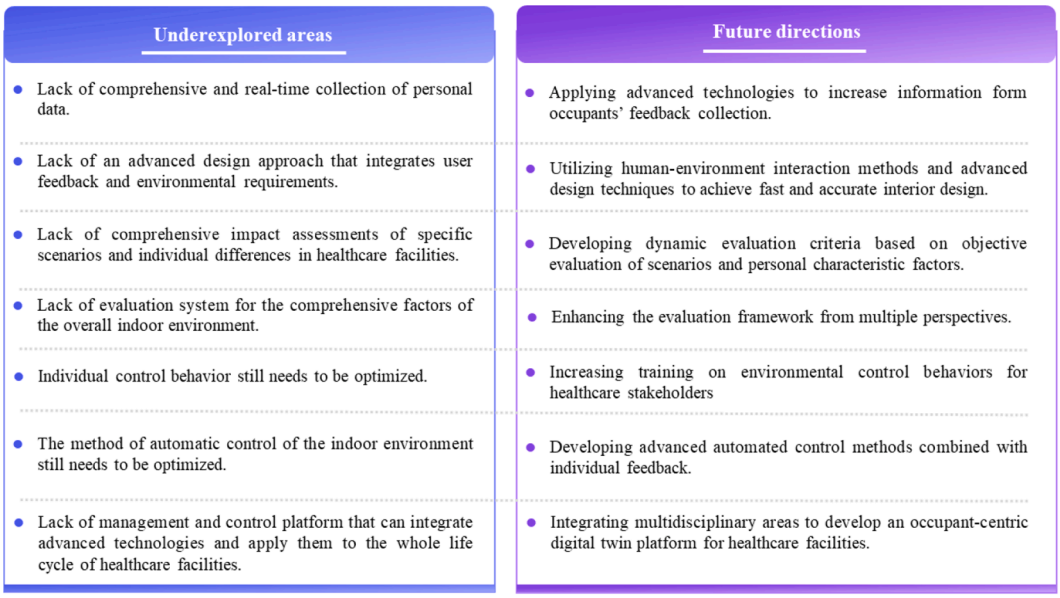


Fig. 6. Research gaps and future research suggestions.

these and ward classifications. Proposing an adaptation model like [303] may help to connect the well-being theory with the latest technologies in IEQ and therapeutic environment research, clarifying the relationship between the architectural environment, perceived control, adaptation strategies, and the continuum of pleasure to happiness.

- **Enhancing the evaluation framework from multiple perspectives**

Although existing single-factor evaluations encompass numerous indicators, there's a need for further application and development of evaluation systems that consider environmental indicators in a unified manner. Similarly, Ghaffari et al. [118] focused on identifying perceptual environmental properties in hospital public spaces that influence salutogenic components and overall patient satisfaction, proposing a conceptual framework for assessment. Moreover, standardizing the assessment of green hospital performance is an effective approach to developing an evaluation system. Zhan et al. [330] constructed a BIM-based pre-assessment system for green hospital performance. The advanced framework should be flexible and changeable based on the needs of different scenarios and occupants.

- **Increasing training on environmental control behaviors for healthcare stakeholders.**

In the actual environments, occupants' behaviors are one of the most important factors affecting healthcare facilities' environment and energy. Individual control behavior still needs to be optimized. Training on environmental control behaviors should encompass several key areas:

IAQ: Training should focus on techniques for controlling sources of indoor pollution [339], understanding and maintaining HVAC systems with proper filtration, and the use of ultraviolet germicidal irradiation (UVGI) [182] to ensure optimal air quality.

Noise Reduction: Training programs should include strategies for installing sound-absorbing materials [256], maintaining quiet operation of medical equipment, and managing noise levels through staff practices such as minimizing loud activities and using noise alert systems.

Lighting and Visual Environment: Including training on the use of natural light, the installation of appropriate artificial lighting systems that minimize glare and shadow [174], and how to create visually restful environments that contribute to patient recovery and staff well-being.

Thermal Comfort: Educating staff on maintaining comfortable temperature and humidity levels [337] that suit both patient and staff needs, which involves the proper use of thermostats and understanding airflow patterns in patient areas.

- **Developing advanced automated control methods combined with individual feedback.**

The automatic control of indoor environments in healthcare requires optimization to better align with individual needs [267] and enhance system efficiency [51]. Current methods, including automated windows, shading devices, and HVAC systems, demand modular yet comprehensive strategies. However, the adoption of such automated systems remains less common in healthcare compared to traditional methods due to challenges in integrating multiple control capabilities to satisfy diverse objectives. These should be addressed using different models of control methods, especially the grey-box model that combines individual behavior [96] and automated control without sufficient data, but this still needs to be discussed.

To bridge this gap, future improvements should focus on advanced, occupant-centric controls [198,199,202]. This includes developing sophisticated human detection technologies, multisource data modeling, and collaborative control behaviors within a central control system. Specifically, implementing a Building Management System (BMS) [148] equipped with AI [131] capabilities may revolutionize patient care environments. This system may utilize real-time data on room occupancy and individual preferences to adjust environmental conditions automatically, thus ensuring both energy efficiency and enhanced patient comfort. Such a personalized approach, supported by AI's ability to learn from historical data, allows for dynamic adjustments across different facility zones, ensuring optimal environmental conditions tailored to individual needs and activities.

- **Integrating multidisciplinary areas to develop an occupant-centric digital twin platform for healthcare facilities.**

As highlighted above, achieving occupant-centric environmental design and control is influenced by various factors [160] and requires

the collection, integration, and processing of vast amounts of data through multiple means [323]. Additionally, this involves interdisciplinary research across several stages, engaging multiple stakeholders [15]. Consequently, the application process often faces conflicts of information and interest, leading to significant time and economic barriers. There is a lack of management and control platforms that can integrate advanced technologies and apply them to the whole life cycle of healthcare facilities. As an enhancement to technologies like BIM [119] and BPS, digital twin technology offers twin control of the real world, enabling real-time and precise control schemes [53]. This provides occupants with the opportunity for prior experience. Coupled with previously mentioned advanced technologies, such as physiological signal monitoring and virtual displays, it collects various data in real-time [299]. This supplies designers and operators with accurate information and assists stakeholders in exchanging information and operating on a unified platform, facilitating real-time monitoring and control [128].

In addition, emergencies such as the COVID-19 pandemic have changed the way in which people access healthcare, placing new demands and challenges on the indoor environments of healthcare facilities [10]. Responding to challenges like this requires preparation for change, and for the COVID-19 pandemic, researchers should chase the changing needs of occupants for multiple environmental requirements.

6.3. Advanced assessment system and control framework

Based on the review results and analysis before, similar to those assessment systems [19,222,254] but different, a dynamic evaluation system and comprehensive control framework is designed for the personalized indoor environment in healthcare facilities, as shown in Fig. 7. This system first considers the underlying individual information and environmental settings and makes connections about possible control behaviors. The system can help the occupants evaluate the indoor environment and make all aspects of the indoor environment have a positive impact on the occupants and improve the resilience of the occupants. The system applies to different countries or regions and different populations, but it needs to be evaluated more accurately based on actual and individual data.

The proposed restorative evaluation framework explicitly accounts for individual differences in restorative perception by integrating personalized data collection and adaptive control systems. For instance, occupant-specific data, such as age, health status, and occupational roles, are collected via surveys and wearable sensors to tailor environmental conditions (e.g., lighting, temperature) to individual needs. Adaptive control systems, leveraging AI and MPC, enable real-time adjustments, ensuring environments are responsive to diverse occupant profiles, such as patients with chronic conditions or staff with high cognitive demands [92]. This approach enhances the framework's applicability across varied healthcare settings.

It is worth noting that this paper analyzes and summarizes all the searched literature to illustrate the impact of indoor environment-related factors on individuals and related interventions, which can effectively help the impact of the healthcare environment on occupants. However, when used in practice, accuracy needs to be considered, and the evaluation and adjustment of specific parameters for different individuals in the actual environment should be based on prior assessment of the individual. At the same time, we recommend that we should maintain flexibility for specific measures, as a single measure often proves to be effective, while the combined application effect of multiple measures needs to be verified and evaluated after the fact to determine whether to proceed. Moreover, the measures are not limited to the systematic approach, which can be continuously improved, which are mentioned before. In addition, increasing occupant control can also significantly increase their satisfaction [82].

7. Conclusion

Improving the overall indoor environmental factors is crucial for occupants (patients, staff, visitors, etc.) in healthcare facilities, as it affects their restorative perceptions. This impacts their evaluation of the living environment, determining their willingness to stay and thereby influencing hospital revenue and costs. Compared to former similar research [5,209,222,305,340], this paper presents the first comprehensive review and evaluation of occupant restorative perceived quality on the overall indoor environmental factors (IEQ and physical design factors) of healthcare facilities, alongside occupant-centric control methods. It identifies and analyzes the current state for research and growing interest in healthcare facility indoor environments, proposing recommendations for an occupant-centric evaluation and related potential manual-automatic comprehensive control schemes. This study

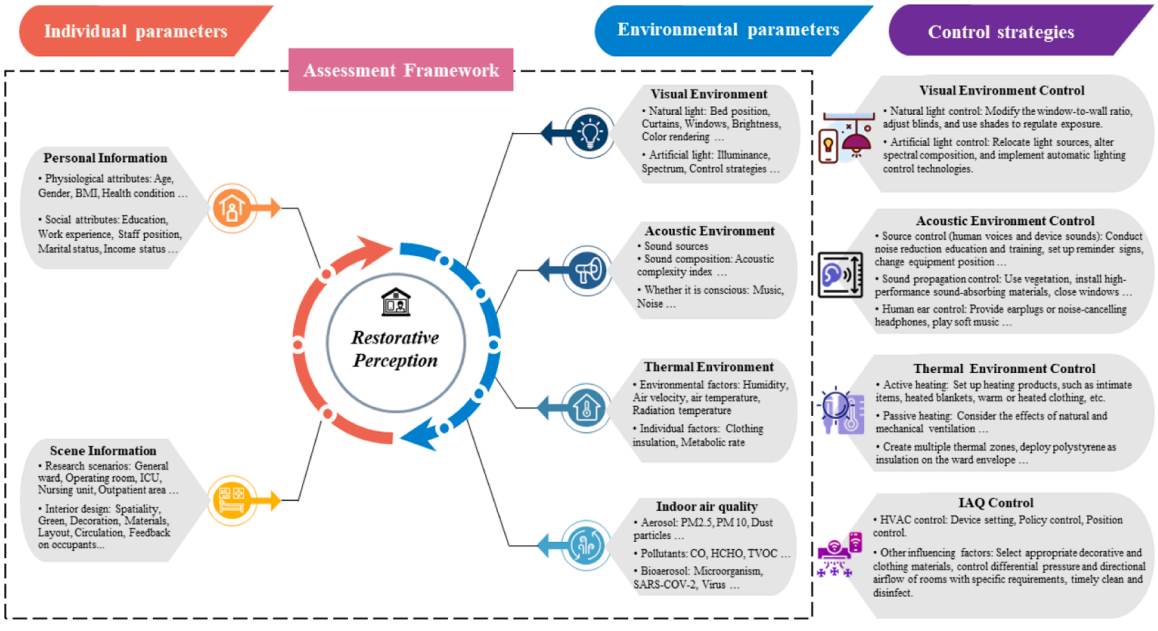


Fig. 7. Individual-based environmental assessment system and comprehensive control framework.

moves the body of scientific knowledge forward by introducing a novel restorative perception framework that integrates physical design, IEQ, and adaptive control solutions, addressing critical gaps in healthcare facility assessment models and providing a foundation for occupant-centric research and practice [210,254]. Specifically, the detailed points are summarized as follows, which reflects the inconsistencies in former research implementations:

- Research in healthcare settings reveals inconsistencies in sample sizes and methodologies, leading to poor transferability of findings across different studies. To address this, future research must focus on more detailed categorization, such as differentiating between hospital wards, public spaces, and patient treatment areas, and employing advanced monitoring techniques like real-time sensors and physiological data collection. These approaches could help standardize environmental assessment systems. Currently, studies tend to focus on hospital wards, neglecting multi-occupant public spaces. Given the importance of shared environments, future research should prioritize settings that accommodate multiple users. Furthermore, disease types, treatment protocols, and professional roles influence outcomes, adding complexity to comparisons across studies between studies. The predominance of cross-sectional designs limits the ability to infer causal relationships or assess long-term impacts, highlighting the need for longitudinal studies.
- Assessing environmental factors within healthcare facilities reveals that individual differences and actual needs play a crucial role. This paper categorizes previous studies into three primary areas: improving greenery, optimizing space while maintaining privacy and accessibility, and enhancing decor. These elements can promote recovery, but differences in task demands between healthcare staff and patients require further investigation. For instance, healthcare staff prioritize convenience, while patients prioritize privacy. Additionally, even for the same individual, needs may vary over time. Therefore, hospital interior environments should be optimized based on the specific tasks conducted in each area. Implementing pre-assessment protocols, automating design generation, and utilizing virtual reality for scenario planning can improve design efficiency and reduce costs.
- The evaluation of indoor environments typically falls into two categories: single-modality studies, and cross-disciplinary research. Current studies often rely on surveys to explore the relationship between group demographics (age, gender, BMI and so on) and environmental factors, providing qualitative insights but lacking quantitative assessments. Personal evaluation models like PMV are frequently debated for their validity. Single-modality studies, primarily based on simulations, yield effective conclusions but have limited real-world application. Overall, reducing noise, enhancing natural light, maintaining circadian rhythms, reducing blue light at night, increasing natural ventilation, adjusting temperature and humidity according to demand, and adopting personalized ventilation can effectively improve occupant satisfaction. The next step is to identify the acute and personal needs and enhance the rapid adjustment. Cross-disciplinary research, though more relevant to real-world settings, the way the factors are combined are different, and the conclusions need to be subject to quantitative studies. Moreover, even for the same individual, environmental needs may change over time. Existing standards, such as LEED, do not sufficiently account for the complexity of individual experiences. Based on these findings, optimization of different IEQ areas within hospital interiors should be task-specific. Hence, more advanced measurement techniques and adaptive systems are required to capture the dynamic and diverse impacts of environmental factors across different contexts.
- The exploration of control behaviors and methods in healthcare settings indicates that occupants should receive training on the importance of IEQ and its impact on health outcomes. Such awareness can empower them to implement simple measures effectively enhance their comfort. Additionally, there is a need for more sophisticated automated control methods in hospitals. Advanced control methods should be adjusted to the personal assessment framework. Current methods predominantly focus on zone-level controls (e.g., thermostats, light switches) rather than system controls (e.g., Air Handling Units, chillers, boilers). Integrating these approaches into the future is essential. Notably, providing occupants with control over their environment can enhance their satisfaction, making it worthwhile exploring a balance between automated and manual control methods, while also considering occupant privacy and data protection.

This review offers significant contributions to knowledge and valuable implications for stakeholders involved in healthcare and architectural design, as outlined below:

- Comprehensive review and policy formulation: This paper guides researchers in identifying evolving focuses and methodological gaps and assists policymakers in enhancing regulatory frameworks based on proven effectiveness. It serves as a foundational tool for shaping informed, evidence-based policies and research directions.
- Establishment of design standards: The findings aid architectural designers in creating environments that support occupant well-being, providing healthcare administrators with actionable insights for facility improvements that enhance recovery and performance. This promotes a holistic approach to design that prioritizes human health and comfort.
- Training and advanced control technology implementation: Effective management awareness is enhanced through targeted training programs, which are recommended for staff and extended through educational materials such as videos and posters for patients and visitors to improve environmental quality at minimal costs. The introduction of automated technologies enhances operational efficiency and occupant comfort, contributing to smarter building management.
- Development of a dynamic framework: This paper presents a comprehensive control system that integrates person-environment-behavior dynamics, combining individual evaluation systems with existing environmental control strategies. It empowers healthcare administrators and facility managers to prioritize modifications that enhance patient recovery and staff performance, laying the groundwork for developing responsive healthcare environments through advanced technology.
- Application of advanced technologies: In addition to implementing automated control technologies, this review advocates for enhancing the digital sophistication of entire medical facilities, establishing a robust digital twin control platform. This approach provides a strategic direction for future research and practical applications, encouraging the integration of cutting-edge technologies and data-driven designs to align healthcare interiors with occupant needs, facilitating personalized and efficient healthcare environments.

The findings of this review have significant implications for healthcare facility design and operation, offering practical applications and future research directions. The proposed restorative perception framework can guide architects and facility managers in creating occupant-centric environments that enhance patient recovery and staff well-being, for instance, through adaptive lighting and improved ventilation systems. We recommend integrating these findings into healthcare architectural policies, aligning with standards like WELL Building Certification, to promote health-focused design. Future research should focus on validating the framework through empirical studies in diverse healthcare settings, exploring cultural and temporal influences, and developing simulation models to optimize control solutions [254]. These efforts will further advance occupant-centric approaches in

healthcare environments.

However, there are some limitations in the current paper as follows:

- Cultural, social, and economic influences may also play a significant role in shaping restorative perception in healthcare facilities, but are not fully addressed in this study. For instance, cultural preferences for natural elements, social expectations for privacy, and economic constraints on advanced control systems can influence occupant experiences and framework applicability [162]. While the proposed framework focuses on technical integration, future research should incorporate these factors to ensure broader relevance across diverse healthcare settings, particularly in low-resource environments or culturally distinct regions.
- While this study summarizes control behaviors and automated systems, it does not propose specific models to simulate human controller behaviors in hospital settings. Simulation models, such as agent-based modeling, offer significant potential to predict how occupants interact with environmental controls, such as adjusting lighting or ventilation based on personal needs. These models can enhance the design of occupant-centric control solutions by simulating dynamic interactions and optimizing system performance [214]. Future research should prioritize developing such models to validate the proposed framework and improve its applicability in healthcare facilities
- Temporal variables, such as day/night cycles and length of stay, are critical for understanding restorative perception in healthcare facilities but are under-explored in the reviewed literature. For instance, circadian lighting that mimics natural day/night cycles can enhance patient recovery, while prolonged hospital stays may alter occupant sensitivity to IEQ factors like thermal [229]. The absence of temporal considerations in current models limits their applicability to dynamic healthcare environments. This study highlights the need for future research to incorporate these variables into assessment frameworks to better capture occupant experiences.
- The proposed restorative evaluation framework is designed for practical implementation in real-world healthcare facilities through strategic integration with existing BMS and pilot testing. By leveraging BMS, the framework can utilize real-time data from sensors (e.g., air quality, lighting) to adjust environmental conditions, ensuring scalability and cost-effectiveness. Pilot studies in diverse healthcare settings, such as intensive care units and outpatient clinics, can validate the framework's effectiveness and identify context-specific adaptations [91]. Stakeholder engagement, including facility managers and healthcare providers, is essential to address implementation challenges, such as retrofitting costs and staff training, ensuring the framework's applicability across varied healthcare environments.
- This study does not analyze the geographical diversity or regional comparisons of the reviewed studies. Future research could explore regional variations to enhance the framework's applicability across diverse healthcare contexts.

CRediT authorship contribution statement

Jiake Yang: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Yidan Wu:** Writing – original draft, Investigation, Formal analysis, Data curation. **S. Thomas Ng:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors would like to thank the Research Grants Council of the HKSAR Government for financially supporting this project through the Collaborative Research Fund (Project No.: C7080-21GF) and General Research Fund (Project No.: 17209220).

Appendix

Table A
Summary of related review papers.

Research fields	Title	Summary	Resource
Human comfort	A review on factors related to patient comfort experience in hospitals	This study explores factors influencing patient comfort in hospitals and emphasizes the need for prioritizing patient well-being, including addressing indoor air pollution and implementing care guidelines.	[278]
Thermal	A review on predicted mean vote and adaptive thermal comfort models	This paper reviews Fanger's predicted mean vote and adaptive thermal comfort models in various buildings and climates, highlighting the need for an internationally recognized adaptive model for tropical climates.	[319]
Visual and design	A review study of cognitive design research on colors from a visual psychological perspective	This review explores the relationship between color psychology and cognitive design in different environments, emphasizing the impact of color on human emotions, stress levels, and visual comfort.	[259]
Acoustic	A state-of-the-art review of healthcare acoustics	Discusses the implementation of the latest acoustic technology in hospitals as mandated by the Sound & Vibration Design Guidelines, focusing on creating healthy environments for patients and staff.	[187]
Ventilation control	A systematic review of operating room ventilation	Reviews the complexity of ventilation systems in operating rooms, considering various factors like surgical instruments, room sizes, and staff, to provide a comfortable environment and prevent patient hypothermia.	[238]
Ventilation control	A systematic review of ventilation solutions for hospital wards: Addressing cross-infection and patient safety	This article reviews ventilation solutions in hospital wards, focusing on the role of air-cleaning techniques in reducing cross-infection risks and maintaining thermal comfort for patients and healthcare workers.	[213]
Design	Biophilia and Salutogenesis as restorative design approaches in healthcare architecture	This paper discusses the development of hospital designs that reconnect humans with nature, proposing a new design framework	[3]

(*continued on next page*)

Table A (*continued*)

Research fields	Title	Summary	Resource
		combining salutogenic and biophilic principles to reduce patient distress.	
Design on room types	Bringing the single versus multi-patient room debate to vulnerable patient populations: a systematic review of the impact of room types on hospitalized older people and people with neurological disorders	Investigates the suitability of single-patient rooms for older people and those with neurological disorders, revealing unclear benefits except for infection control.	[253]
IEQ	Building a satisfactory indoor environment for healthcare facility occupants: A literature review	Reviews methods to improve healthcare facility occupants' satisfaction focusing on visual, acoustic, thermal environments, and indoor air quality.	[254]
IEQ	Comfort and health of patients and staff, related to the physical environment of different departments in hospitals: a literature review	Explores the impact of hospital physical environments on the comfort and health of patients and staff, finding that staff satisfaction varies across departments.	[88]
Interior control	Control in healthcare interiors: the staff's perspective	Examines healthcare staff's perception of their workspaces, emphasizing the importance of controlling privacy, boundaries, and environmental factors for their satisfaction.	[39]
Physical environment design	Design approaches for promoting beneficial indoor environments in healthcare facilities: a review	Reviews design elements in healthcare facilities that positively impact patient and staff health and wellbeing, like single-bed rooms and natural daylight.	[244]
Physical environment design	Effects of the physical environment on children and families in hospital-based emergency departments: A systematic literature review	Reviews how emergency department physical environments affect children and families, focusing on control, positive distractions, and designing for safety and comfort.	[124]
Thermal and HVAC control methods	Energy efficiency and thermal comfort in hospital buildings: A review	Explores state-of-the-art technologies for energy efficiency in hospitals, emphasizing the importance of HVAC systems and the need for thermal comfort.	[115]
Acoustic and related control methods	Environmental noise in hospitals: a systematic review	Systematically reviews noise levels in hospitals and their impact on patients and healthcare professionals, highlighting the need for improved noise reduction measures.	[173]
Physical environment design	Healing environment: A review of the impact of physical environmental factors on users	Investigates the effects of physical environment on patient healing and staff well-being in healthcare facilities, emphasizing the need for evidence-based design.	[141]
Physical environment design	Healthy and sustainable hospital evaluation-A review of POE tools for hospital assessment in an evidence-based design framework	Compares tools for assessing hospital environments, focusing on health and sustainability criteria, and discusses the growing emphasis on the built environment's impact on health.	[57]
Physical environment design	Healthy work environments are critical for nurse job satisfaction: implications for Oman	This study examines the relationship between work environment and job satisfaction among nurses in Oman, finding that a favorable work environment greatly influences nurse satisfaction, more so than personal characteristics or hospital type.	[16]
Physical environment design	Hospital salutogenic public spaces: a conceptual framework of effective perceptional environment quality components on patients' satisfaction	This study aimed at the identification of perceptional environment properties in hospital public spaces that can affect salutogenic components and patients' overall satisfaction and suggested a conceptual framework.	[118]
IAQ	Indoor air quality in public utility environments—a review	This review discusses the impact of various factors on indoor air quality in public spaces like churches, museums, and hospitals, highlighting the link between poor indoor air quality and increased incidence of civilization diseases.	[261]
IAQ	Indoor air quality monitoring systems based on internet of things: A systematic review	This systematic review presents an analysis of indoor air quality monitoring systems using Internet of Things technology, focusing on design aspects, sensor types, and challenges in implementation.	[243]
Control methods	Integrated energy, daylighting, comfort and environmental performance analysis of window systems in patient rooms	This study investigates the effect of window system configurations on energy use, daylight access, visual comfort, and environmental performance in hospital patient rooms, using a multi-criteria evaluation approach.	[94]
Ventilation control and design	Natural and Mechanical Ventilation Concepts for Indoor Comfort and Well-Being with a Sustainable Design Perspective: A Systematic Review	This review explores the impact of natural and mechanical ventilation on indoor comfort and well-being, emphasizing the importance of a health-based approach to ventilation, especially after the COVID-19 outbreak.	[325]
Design	Role of healthcare-facilities layout design, healing architecture, on quality of services	Well-designed healthcare facility layouts are essential for the provision of high-quality healthcare services, which can reduce healthcare risks, minimize financial burden and improve patient satisfaction.	[221]
Green Roof	The Sustainable Prescription: Benefits of Green Roof Implementation for Urban Hospitals	This review offers insight into the environmental, societal, and economic benefits of green roofs installed on healthcare facilities.	[215]
Ventilation control and design	The ventilation of multiple-bed hospital wards in the tropics: A review	This paper reviews the ventilation of multi-bed hospital wards in tropical climates, discussing ward layouts, required indoor conditions, and emphasizing the need for specific studies and novel engineering controls for tropical ventilation.	[317]
Thermal	Thermal comfort applied in hospital environments: A literature review	Marking the 50th anniversary of the PMV model, this review examines studies on thermal comfort in hospital environments, identifying gaps in knowledge and emphasizing the need for more research on thermal comfort and productivity.	[225]
Thermal	Thermal comfort in hospital buildings – A literature review	This review critically examines existing studies on thermal comfort in hospitals, identifying key research themes, the importance of ventilation systems, and variations in thermal comfort needs based on health conditions and staff activities.	[324]
Thermal	Thermal comfort in hospitals – A literature review	This literature review focuses on thermal comfort in hospitals, suggesting the need for original studies on the relationship between thermal comfort conditions and productivity for hospital staff.	[161]

(*continued on next page*)

Table A (*continued*)

Research fields	Title	Summary	Resource
Physical environment design	Towards human-centered general hospitals: the potential of dementia-friendly design	This study investigates the effectiveness of dementia-friendly design in hospitals, using data from German Special Care Units, and highlights the potential of such designs in improving patients' self-care abilities and overall hospital design.	[164]

Table B
Nomenclature.

Abbreviations	
ACH	Air Changes per Hour
ACMV	Air-Conditioning and Mechanical Ventilation
ACV	Additional Curtain Ventilation
AI	Artificial Intelligence
BIM	Building Information Modelling
BMS	Building Management System
CFD	Computational Fluid Dynamics
CLR	Critical Literature Review
CSFR	Ceiling-Supplied Floor Return
CSCR	Ceiling-Supplied Ceiling Return
DTM	Dynamic Thermal Model
ECG	Electrocardiogram
EEG	Electroencephalography
ETFE	Ethylene-Tetra-Fluoro-Ethylene
FEM	Finite Element Method
GA	Genetic Algorithm
GBIC	Green Buildings Innovation Cluster
GSR	Galvanic Skin Response
HCAHPS	Hospital Consumer Assessment of Healthcare Providers and Systems
HEPA	High-Efficiency Particulate Air
HVAC	Heating, Ventilation, and Air Conditioning
IAQ	Indoor Air Quality
ICU	Intensive Care Unit
IEQ	Indoor Environmental Quality
LEED	Leadership in Energy and Environmental Design
MPC	Model Predictive Control
NPV	Natural Personal Ventilation
PBC	Planned Behavior Control
PCS	Personal Comfort System
PDE	Partial Differential Equation
PMV	Predicted Mean Vote
POV	Protected Occupation Ventilation
PSO	Particle Swarm Optimization
RANS	Reynolds-Averaged Navier-Stokes
RCP	Radiant Ceiling Panel
RSM	Response Surface Methodology
SEM	Structural Equation Modelling
SLR	Systematic Literature Review
TPB	Theory of Planned Behavior
UVGI	Ultraviolet Germicidal Irradiation
VR	Virtual Reality
WHO	World Health Organization
WOS	Web of Science
Symbols	
dBA	A-weighted Decibels
T	Temperature (K/ °C)
V	Air Velocity (m/s)

Data availability

The authors do not have permission to share data.

References

[1] N.M. Abd Rahman, L.C. Haw, A. Fazlizan, A. Hussin, M.S. Imran, Thermal comfort assessment of naturally ventilated public hospital wards in the tropics, Build. Environ. (2022) 207, https://doi.org/10.1016/j.buildenv.2021.108480.
[2] A. Rahman, N. Muhammad, L.C. Haw, A. Fazlizan, A. Hussin, M.S. Imran, Thermal comfort assessment of naturally ventilated public hospital wards in the tropics, Build. Environ. 207 (2022) 108480.
[3] M.S. Abdelaal, V. Soebarto, Biophilia and salutogenesis as restorative design approaches in healthcare architecture, Archit. Sci. Rev. 62 (3) (2019) 195–205, https://doi.org/10.1080/00038628.2019.1604313.
[4] G. Abhijeet Ganesh, S.L. Sinha, T.N. Verma, S.K. Dewangan, Energy consumption and thermal comfort assessment using CFD in a naturally ventilated indoor environment under different ventilations, Therm. Sci. Eng. Progr. 50 (2024) 102557, https://doi.org/10.1016/j.tsep.2024.102557.
[5] A. Ackley, O.I. Olanrewaju, O.N. Oyefusi, W.I. Enegbuma, T.S. Olaoye, A. E. Ehimatie, E. Ukpong, P. Akpan-Idiok, Indoor Environmental Quality (IEQ) in healthcare facilities: a systematic literature review and gap analysis, J. Build. Eng. 86 (2024) 108787, https://doi.org/10.1016/j.jobe.2024.108787.
[6] Z.A. Adamu, A. Price, The design and simulation of natural personalised ventilation (NPV) system for Multi-bed hospital wards, Buildings 5 (2) (2015) 381–404, https://doi.org/10.3390/buildings5020381.

[7] Z.A. Adamu, A.D.F. Price, M.J. Cook, Performance evaluation of natural ventilation strategies for hospital wards - a case study of Great Ormond Street Hospital, Build. Environ. 56 (2012) 211–222, https://doi.org/10.1016/j.buildenv.2012.03.011.

[8] A. Aganovic, M. Steffensen, G. Cao, CFD study of the air distribution and occupant draught sensation in a patient ward equipped with protected zone ventilation, Build. Environ. 162 (2019), https://doi.org/10.1016/j.buildenv.2019.106279.

[9] A. Aganovic, M. Steffensen, G. Cao, CFD study of the air distribution and occupant draught sensation in a patient ward equipped with protected zone ventilation, Build. Environ. 162 (2019) 106279, https://doi.org/10.1016/j.buildenv.2019.106279.

[10] N. Agarwal, C. Swaroop Meena, B.P. Raj, L. Saini, A. Kumar, N. Gopalakrishnan, A. Kumar, N.B. Balam, T. Alam, N.R. Kapoor, V. Aggarwal, Indoor air quality improvement in COVID-19 Pandemic: review, Sustain. Citi. Soc. 70 (2021) 102942, https://doi.org/10.1016/j.scs.2021.102942.

[11] A. Aguglia, G. Serafini, A. Escelsior, G. Canepa, M. Amore, G. Maina, Maximum temperature and solar radiation as predictors of bipolar patient admission in an emergency psychiatric ward, Int. J. Environ. Res. Public Health 16 (7) (2019), https://doi.org/10.3390/ijerph16071140.

[12] İ. Akova, E. Kiliç, H. Sümer, T. Keklikçi, Prevalence of sick building syndrome in hospital staff and its relationship with indoor environmental quality, Int. J. Environ. Health Res. 32 (6) (2022) 1204–1219, https://doi.org/10.1080/09603123.2020.1862067.

[13] F. Alam Khan, M. Asif, A. Ahmad, M. Alharbi, H. Aljuaid, Blockchain technology, improvement suggestions, security challenges on Smart grid and its application in healthcare for sustainable development, Sustain. Citi. Soc. 55 (2020) 102018, https://doi.org/10.1016/j.scs.2020.102018.

[14] L. Albala, T. Bober, G. Hale, B. Warfield, M.L. Collins, Z. Merritt, E. Steimetz, S. Nadler, Y. Lev, J. Hanifin, Effect on nurse and patient experience: overnight use of blue-depleted illumination, BMJ Open. Qual. 8 (3) (2019) e000692, https://doi.org/10.1136/bmjoq-2019-000692.

[15] W. AlBalkhy, D. Karmaoui, L. Ducoulombier, Z. Lafhaj, T. Linner, Digital twins in the built environment: definition, applications, and challenges, Automat. Construct. 162 (2024) 105368, https://doi.org/10.1016/j.autcon.2024.105368.

[16] A. Albashayreh, S.D. Al Sabei, O.M. Al-Rawajfah, H. Al-Awaisi, Healthy work environments are critical for nurse job satisfaction: implications for Oman, Int. Nurs. Rev. 66 (3) (2019) 389–395, https://doi.org/10.1111/inr.12529.

[17] M. Albornoz-Cabello, J.M. Pérez-Mármol, M.Á. Cardero-Durán, C.J. Barrios-Quinta, L. Espejo-Antúnez, Construction, factor structure, and internal consistency reliability of the hospital physical therapy perceived satisfaction questionnaire (H-Ptps), Int. J. Environ. Res. Public Health 17 (16) (2020) 1–14, https://doi.org/10.3390/ijerph17165857.

[18] Y. Al-Dmour, Beyond academia: investigating indoor environmental quality and its impact on postgraduate student satisfaction, Result. Eng. 24 (2024) 103190, https://doi.org/10.1016/j.rineng.2024.103190.

[19] S.S. Aljunid, M.Z.M. Taib, Z.A. Samah, Occupants' Satisfaction towards interior design quality at inpatient units of public hospitals in Malaysia: a sustainable Development Framework, Int. J. Environ. Sustain. Develop. 19 (3) (2020) 320–331, https://doi.org/10.1504/IJESD.2020.108167.

[20] B.S. Alotaibi, R. Codinhoto, D. Albadra, S. Lo, Combined multi-attribute inpatient thermal comfort requirements in hospitals: a designer's assessment method, J. Build. Eng. 42 (2021), https://doi.org/10.1016/j.jobe.2021.103039.

[21] B.S. Alotaibi, S. Lo, Thermal environment perceptions from a longitudinal study of indoor temperature profiles in inpatientwards, Buildings 10 (8) (2020), https://doi.org/10.3390/BUILDINGS10080136.

[22] B.S. Alotaibi, S. Lo, E. Southwood, D. Coley, Evaluating the suitability of standard thermal comfort approaches for hospital patients in air-conditioned environments in hot climates, Build. Environ. 169 (2020), https://doi.org/10.1016/j.buildenv.2019.106561.

[23] A.A. Alsarraf, M.M. Alobaidi, The effect of natural lighting on design of openings in hospital, J. Green Eng. 10 (10) (2020) 8233–8250.

[24] H.H. Alzoubi, S.M. Al-Rqaibat, The effect of Hospital design on indoor daylight quality in childrensection in King Abdullah University Hospital, Jordan, Sustain. Citi. Soc. 14 (1) (2015) 449–455, https://doi.org/10.1016/j.scs.2014.08.008.

[25] H.H. Alzoubi, A.S. Attia, Assessment of the acoustical standards in patient care units in Jordanian national hospitals in light of the international criteria: case of King Abdullah University Hospital, Alexand. Eng. J. 58 (4) (2019) 1205–1213, https://doi.org/10.1016/j.aej.2019.10.004.

[26] S. Alzubaidi, P.K. Soori, Energy effcient lighting system design for hospitals diagnostic and treatment room-a case study, J. Light Vis. Environ. 36 (1) (2012) 23–31, https://doi.org/10.2150/jlve.36.23.

[27] F. Amirbeiki Tafti, J. Han, M.K. Wekenborg, A.V. Moere, J.-H. Schierz, A. Heylighen, G. Marquardt, Beyond greenery: exploring influences of exposure to natural blue elements on patients using virtual reality, Build. Environ. 267 (2025) 112227, https://doi.org/10.1016/j.buildenv.2024.112227.

[28] P. Amoatey, I. Al-Harthy, M.A. Al-Mushaifari, K. Al-Jabri, A. Al-Mamun, Effect of ambient noise on indoor environments in a health care facility in Oman, Environ. Sci. Pollut. Res. 29 (10) (2022) 15081–15097, https://doi.org/10.1007/s11356-021-16875-y.

[29] P. Amoatey, I. Al-Harthy, M. Ali Al-Mushaifari, K. Al-Jabri, A. Al-Mamun, Effect of ambient noise on indoor environments in a health care facility in Oman, Environ. Sci. Pollut. Res. 29 (10) (2022) 15081–15097, https://doi.org/10.1007/s11356-021-16875-y.

[30] C.C. Andrade, A.S. Devlin, Stress reduction in the hospital room: applying Ulrich's theory of supportive design, J. Environ. Psychol. 41 (2015) 125–134, https://doi.org/10.1016/j.jenvp.2014.12.001.

[31] E.M. de Araújo Vieira, L.B.da Silva, E.L.de Souza, The influence of the workplace indoor environmental quality on the incidence of psychological and physical symptoms in Intensive Care units, Build. Environ. 109 (2016) 12–24.

[32] Ashrae, ANSI., Standard 55-2013: Thermal Environmental Conditions for Human Occupancy, American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc. Atlanta, 2013.

[33] N.A. Aziz, N.R. Nik Husain, M.R. Abdullah, N. Ismail, Z. Ismail, Indoor air quality in a Northeast Coast Malaysian Medical School, EnvironmentAsia 11 (2) (2018) 67–79, https://doi.org/10.14456/ea.2018.23.

[34] F. Azizpour, S. Moghimi, E. Salleh, S. Mat, C.H. Lim, K. Sopian, Thermal comfort assessment of large-scale hospitals in tropical climates: a case study of University Kebangsaan Malaysia Medical Centre (UKMMC), Energy Build. 64 (2013) 317–322, https://doi.org/10.1016/j.enbuild.2013.05.033.

[35] F. Azizpour, S. Moghimi, E. Salleh, S. Mat, C.H. Lim, K. Sopian, Thermal comfort assessment of large-scale hospitals in tropical climates: a case study of University Kebangsaan Malaysia Medical Centre (UKMMC), Energy Build. 64 (2013) 317–322, https://doi.org/10.1016/j.enbuild.2013.05.033.

[36] H. Azmoon, H. Dehghan, J. Akbari, S. Souri, The relationship between thermal comfort and light intensity with sleep quality and eye tiredness in shift work nurses, J. Environ. Public Health 2013 (2013) 639184, https://doi.org/10.1155/2013/639184.

[37] Y. Balali, A. Chong, A. Busch, S. O'Keefe, Energy modelling and control of building heating and cooling systems with data-driven and hybrid models—a review, Renew. Sustain. Energy Rev. 183 (2023) 113496, https://doi.org/10.1016/j.rser.2023.113496.

[38] C.A. Balaras, E. Dascalaki, A. Gaglia, HVAC and indoor thermal conditions in hospital operating rooms, Energy Build. 39 (4) (2007) 454–470, https://doi.org/10.1016/j.enbuild.2006.09.004.

[39] G. Ballice, Ç. Çetin, E.P.ç Özçelik, Z.T. Ultav, Control in healthcare interiors: the staff's perspective, Archit. Sci. Rev. 66 (4) (2023) 309–321, https://doi.org/10.1080/00038628.2023.2182269.

[40] P. Barrett, L. Barrett, The potential of positive places: senses, brain and spaces, Intell. Build. Int. 2 (3) (2010) 218–228, https://doi.org/10.3763/inbi.2010.0042.

[41] P. Barrett, F. Davies, Y. Zhang, L. Barrett, The impact of classroom design on pupils' Learning: final results of a holistic, multi-level analysis, Build. Environ. 89 (2015) 118–133, https://doi.org/10.1016/j.buildenv.2015.02.013.

[42] P. Barrett, Y. Zhang, J. Moffat, K. Kobbacy, A holistic, multi-level analysis identifying the impact of classroom design on pupils' Learning, Build. Environ. 59 (2013) 678–689, https://doi.org/10.1016/j.buildenv.2012.09.016.

[43] A. Barroso, B. Den Brinker, Boosting circadian rhythms with lighting: a model driven approach, Light. Res. Technol. 45 (2) (2013) 197–216, https://doi.org/10.1177/1477153512453667.

[44] M.V. Bayo, A.M. García, A. García, Noise levels in an urban hospital and workers' Subjective responses, Arch. Environ. Health. 50 (3) (1995) 247–251, https://doi.org/10.1080/00039896.1995.9940395.

[45] A.E.V.D. Berg, J. Maas, L.V.D. Hoven, K. Tanja-Dijkstra, Greening a geriatric ward reduces functional decline in elderly patients and is positively evaluated by hospital staff, J. Aging Environ. 35 (2) (2021) 125–144, https://doi.org/10.1080/26892618.2020.1805390.

[46] A.E.v.d. Berg, J. Maas, L. van den Hoven, K. Tanja-Dijkstra, Greening a geriatric ward reduces functional decline in elderly patients and is positively evaluated by hospital staff, J. Aging Environ. 35 (2) (2021) 125–144, https://doi.org/10.1080/26892618.2020.1805390.

[47] B. Berglund, T. Lindvall, D.H. Schwela, World Health Organization Occupational, and Environmental Health Team, Guidelines For Community Noise'. This document is the Outcome of the WHO-expert Task Force Meeting Held in London, United Kingdom, in April 1999. It bases On the Document Entitled 'Community noise' That Was Prepared For the World Health Organization and Published in 1995 By the Stockholm, University and Karolinska Institute, 1999.

[48] F.A. Berlanga, I. Olmedo, M.R. de Adana, J.M. Villafruela, J.F.S. José, F. Castro, Experimental assessment of different mixing air ventilation systems on ventilation performance and exposure to exhaled contaminants in hospital rooms, Energy Build. 177 (2018) 207–219, https://doi.org/10.1016/j.enbuild.2018.07.053.

[49] M.M. Bilec, R.J. Ries, K.L. Needy, M. Gokhan, A.F. Phelps, E. Enache-Pommer, M. J. Horman, S.E. Little, T.L. Powers, E. McGregor, C. Sheane, Analysis of the design process of Green children's hospitals: focus on process modeling and lessons learned, J. Green Build. 4 (1) (2009) 121–134, https://doi.org/10.3992/jgb.4.1.121.

[50] E. Bingham, D. Whitaker, C. Farnsworth, J. Smith, Evidence-based design in hospital renovation projects: design implementation for patient privacy and comfort, J. Architect. Engi. 28 (2) (2022), https://doi.org/10.1061/(ASCE)AE.1943-5568.0000530.

[51] D.-I. Bogatu, J. Shinoda, J.éJ.ín Aguilera, B.W. Olesen, F. Watanabe, Y. Kaneko, O.B. Kazanci, Human physiology for personal thermal comfort-based HVAC control – a review, Build. Environ. 240 (2023) 110418, https://doi.org/10.1016/j.buildenv.2023.110418.

[52] V.D. Bohbot, S. McKenzie, K. Konishi, C. Fouquet, V. Kurdi, R. Schachar, M. Boivin, P. Robaey, Virtual navigation strategies from childhood to senescence: evidence for changes across the life span, Front. Aging. Neurosci. 4 (2012) 10, https://doi.org/10.3389/fnagi.2012.00028.

[53] C. Boje, A. Guerriero, S. Kubicki, Y. Rezgui, Towards a semantic construction digital twin: directions for future research, Automat. Construct. 114 (2020) 103179, https://doi.org/10.1016/j.autcon.2020.103179.
[54] A. Booth, A. Sutton, M. Clowes, M.M.-S. James, Systematic Approaches to a Successful Literature Review (2021).
[55] Y. Bouzidi, Z. El Akili, A. Gademer, N. Tazi, A. Chahboun, How can we adapt thermal comfort for disabled patients? A case study of French healthcare buildings in summer, Energies 14 (15) (2021), https://doi.org/10.3390/en14154530.
[56] A. Brambilla, E. Brusamolin, A.A. Johnson, F. Scullica, S. Capolongo, Lessons from the first wave of COVID-19 in Italy: a collection of design strategies to face pandemic situations in healthcare facilities, J. Emerg. Manag. 21 (7) (2023) 167–184, https://doi.org/10.5055/JEM.0746.
[57] A. Brambilla, S. Capolongo, Healthy and sustainable Hospital evaluation-A review of POE tools for hospital assessment in an evidence-based design framework, Buildings 9 (4) (2019), https://doi.org/10.3390/buildings9040076.
[58] H. Cai, F. Fullam, L. Macallister, L.F. Fogg, J. Canar, I. Press, C. Weissman, O. Velasquez, Impact of inpatient unit design features on overall patient experience and perceived room-level call button response, Int. J. Environ. Res. Public Health 18 (18) (2021), https://doi.org/10.3390/ijerph18189747.
[59] N. Campion, C.L. Thiel, J. Focareta, M.M. Bilec, Understanding green Building design and healthcare outcomes: evidence-based design analysis of an oncology unit, J. Architect. Engi. 22 (3) (2016), https://doi.org/10.1061/(ASCE)AE.1943-5568.0000217.
[60] M. Canazei, W. Pohl, J. Weninger, H. Bliem, E.M. Weiss, C. Koch, A. Berger, B. Firulovic, C. Marth, Effects of adjustable dynamic bedroom lighting in a maternity ward, J. Environ. Psychol. 62 (2019) 59–66, https://doi.org/10.1016/j.jenvp.2019.01.010.
[61] I. Caner, N. Ilten, Evaluation of occupants' Thermal perception in a university hospital in Turkey, Proceed. Instit. Civi. Eng.: Eng. Sustain. 173 (8) (2020) 414–428, https://doi.org/10.1680/jensu.19.00059.
[62] I. Caner, N. Ilten, Evaluation of occupants' Thermal perception in a university hospital in Turkey, Proceedings. of. the. Institution. of. Civil. Engineers. -. Engineering. Sustainability. 173 (8) (2020) 414–428, https://doi.org/10.1680/jensu.19.00059.
[63] G.C. Cannistraro, M. Cannistraro, Hypothermia risk, monitoring and environment control in operating rooms, Int. J. Heat Technol. 34 (2) (2016) 165–171, https://doi.org/10.18280/ijht.340202.
[64] S.W. Casscells, E. Granger, T.V. Williams, T. Kurmel, L. May, L. Babeau, D. Boyd, D. Davis, S. Ayine, N. Thomas, TRICARE management activity healthcare facility evidence-based design survey, Mil. Med. 174 (3) (2009) 236–240.
[65] R. Cervinka, K. Röderer, I. Hämmerle, Evaluation of hospital gardens and implications for design: benefits from environmental psychology for architecture and landscape planning, J. Archit. Plann. Res. 31 (1) (2014) 43–56.
[66] K.G. Chang, H. Chien, The influences of landscape features on visitation of hospital green spaces—a choice experiment approach, Int. J. Environ. Res. Public Health 14 (7) (2017), https://doi.org/10.3390/ijerph14070724.
[67] C. Chen, Searching for intellectual turning points: progressive knowledge domain visualization, Proceed. Natl. Acad. Sci. 101 (suppl_1) (2004) 5303–5310, https://doi.org/10.1073/pnas.0307513100.
[68] K.W. Cheong, K.Y. Chong, Development and application of an indoor air quality audit to an air-conditioned building in Singapore, Build. Environ. 36 (2) (2001) 181–188, https://doi.org/10.1016/S0360-1323(99)00064-5.
[69] G. Chinazzo, R.K. Andersen, E. Azar, V.M. Barthelmes, C. Becchio, L. Belussi, C. Berger, S. Carlucci, S.P. Corgnati, S. Crosby, L. Danza, L.de Castro, M. Favero, S. Gauthier, R.T. Hellwig, Q. Jin, J. Kim, M.S. Khanie, D. Khovalyg, C. Lingua, A. Luna-Navarro, A. Mahdavi, C. Miller, I. Mino-Rodriguez, I. Pigliautile, A. L. Pisello, R.F. Rupp, A.-M. Sadick, F. Salamone, M. Schweiker, M. Syndicus, G. Spigliantini, N.G. Vasquez, D. Vakalis, M. Vellei, S. Wei, Quality Criteria for multi-domain studies in the indoor environment: critical review towards research guidelines and recommendations, Build. Environ. 226 (2022) 109719, https://doi.org/10.1016/j.buildenv.2022.109719.
[70] C.P. Chiodo, C. Keyser, D. Palms, C. Malone, E.M. Bluman, J.T. Smith, The impact of music therapy in the postoperative setting on overall patient satisfaction using validated outcome scores, J. Perianesthes. Nurs. 34 (2) (2019) 354–358, https://doi.org/10.1016/j.jopan.2018.04.006.
[71] M. Cho, Evaluating therapeutic Healthcare environmental criteria: architectural designers' Perspectives, Int. J. Environ. Res. Public Health 20 (2) (2023), https://doi.org/10.3390/ijerph20021540.
[72] D.H. Choi, S.H. Choi, D.H. Kang, Influence of surgical smoke on indoor air quality in hospital operating rooms, Aerosol. Air. Qual. Res. 17 (3) (2017) 821–830, https://doi.org/10.4209/aaqr.2016.05.0191.
[73] J.H. Choi, L.O. Beltran, H.S. Kim, Impacts of indoor daylight environments on patient average length of stay (ALOS) in a healthcare facility, Build. Environ. 50 (2012) 65–75, https://doi.org/10.1016/j.buildenv.2011.10.010.
[74] T. Clinton-Mcharg, C. Paul, R. Sanson-Fisher, H. Turon, M. Butler, R. Lindeman, Are the physical environments of treatment centres meeting recommendations for patient-centred care? Perceptions of haematological cancer patients, Int. J. Environ. Res. Public Health 18 (9) (2021), https://doi.org/10.3390/ijerph18094892.
[75] W. Czarniecki, M. Kopacz, W. Okołowicz, J. Gajewski, E. Grzedziński, Investigations of the microclimate in hospital wards, Energy Build. 16 (1–2) (1991) 727–733, https://doi.org/10.1016/0378-7788(91)90044-4.
[76] D. Dalirnaghadeh, S. Yilmazer, The effect of sound environment on spatial knowledge acquisition in a virtual outpatient polyclinic, Appl. Ergon. 100 (2022), https://doi.org/10.1016/j.apergo.2021.103672.
[77] I. Daniel Lawrence, S. Jayabal, P. Thirumal, Indoor air quality investigations in hospital patient room, Int. J. Biomed. Eng. Technol. 27 (1–2) (2018) 124–138, https://doi.org/10.1504/ijbet.2018.10014309.
[78] V. De Giuli, R. Zecchin, L. Salmaso, L. Corain, M. De Carli, Measured and perceived indoor environmental quality: padua Hospital case study, Build. Environ. 59 (2013) 211–226, https://doi.org/10.1016/j.buildenv.2012.08.021.
[79] S. Del Ferraro, S. Iavicoli, S. Russo, V. Molinaro, A field study on thermal comfort in an Italian hospital considering differences in gender and age, Appl. Ergon. 50 (2015) 177–184, https://doi.org/10.1016/j.apergo.2015.03.014.
[80] M.H. Depledge, R.J. Stone, W.J. Bird, Can natural and virtual environments be used to promote improved Human health and wellbeing? Environ. Sci. Technol. 45 (11) (2011) 4660–4665, https://doi.org/10.1021/es103907m.
[81] M.T.H. Derks, A.K. Mishra, M.G.L.C. Loomans, H.S.M. Kort, Understanding thermal comfort perception of nurses in a hospital ward work environment, Build. Environ. 140 (2018) 119–127, https://doi.org/10.1016/j.buildenv.2018.05.039.
[82] A.S. Devlin, C. Hetzel, M. Rathgeber, Does perceived control matter in the outpatient waiting room? Herd-Health Environ. Res. Des. J. (2023) https://doi.org/10.1177/19375867221143104.
[83] M. Dovjak, A. Kukec, Z. Kristl, M. Kosir, M. Bilban, M. Shukuya, A. Krainer, Integral control of health hazards in hospital environment, Indoor Built Environ. 22 (5) (2013) 776–795, https://doi.org/10.1177/1420326×12459867.
[84] M. Dovjak, M. Shukuya, A. Krainer, Individualisation of personal space in hospital environment, Int. J. Exergy 14 (2) (2014) 125–155, https://doi.org/10.1504/IJEX.2014.060279.
[85] M. Dovjak, M. Shukuya, A. Krainer, User-centred healing-oriented conditions in the design of hospital environments, Int. J. Environ. Res. Public Health 15 (10) (2018), https://doi.org/10.3390/ijerph15102140.
[86] M.A. Ebaid, A framework for implementing biophilic design in cancer healthcare spaces to enhance patients' experience', Int. J. Sustain. Build. Technol. Urban. Develop. 14 (2) (2023) 229–246, https://doi.org/10.22712/susb.20230018.
[87] A. Eijkelenboom, P.M. Bluyssen, Profiling outpatient staff based on their self-reported comfort and preferences of indoor environmental quality and social comfort in six hospitals, Build. Environ. 184 (2020), https://doi.org/10.1016/j.buildenv.2020.107220.
[88] A. Eijkelenboom, P.M. Bluyssen, Comfort and health of patients and staff, related to the physical environment of different departments in hospitals: a literature review, Intell. Build. Int. 14 (1) (2022) 95–113, https://doi.org/10.1080/17508975.2019.1613218.
[89] A. Eijkelenboom, D.H. Kim, P.M. Bluyssen, First results of self-reported health and comfort of staff in outpatient areas of hospitals in The Netherlands, Build. Environ. 177 (2020), https://doi.org/10.1016/j.buildenv.2020.106871.
[90] A. Eijkelenboom, M.A. Ortiz, P.M. Bluyssen, Preferences for indoor environmental and social comfort of outpatient staff during the COVID-19 Pandemic, an explanatory study, Int. J. Environ. Res. Public Health 18 (14) (2021), https://doi.org/10.3390/ijerph18147353.
[91] A. Eijkelenboom, P.M. Bluyssen, Profiling outpatient staff based on their self-reported comfort and preferences of indoor environmental quality and social comfort in six hospitals, Build. Environ. 184 (2020) 107220, https://doi.org/10.1016/j.buildenv.2020.107220.
[92] A. Eijkelenboom, D.H. Kim, P.M. Bluyssen, First results of self-reported health and comfort of staff in outpatient areas of hospitals in The Netherlands, Build. Environ. 177 (2020) 106871, https://doi.org/10.1016/j.buildenv.2020.106871.
[93] N. Eisazadeh, K. Allacker, F. De Troyer, Integrated energy, daylighting and visual comfort analysis of window systems in patient rooms, Sci. Technol. Built Environ. 27 (8) (2021) 1040–1055, https://doi.org/10.1080/23744731.2021.1912512.
[94] N. Eisazadeh, F.D. Troyer, K. Allacker, Integrated energy, daylighting, comfort and environmental performance analysis of window systems in patient rooms, Archit. Sci. Rev. 65 (5) (2022) 319–337, https://doi.org/10.1080/00038628.2022.2099806.
[95] A.A. Erdogan, M.Z. Yilmazoglu, Comparison of ventilation strategies in intensive care units for airborne infection control, Energy Source., Part A: Recov., Utiliz. Environ. Effect. 44 (3) (2022) 5829–5851, https://doi.org/10.1080/15567036.2022.2088900.
[96] M. Esrafilian-Najafabadi, F. Haghighat, Occupancy-based HVAC control systems in buildings: a State-of-the-art review, Build. Environ. 197 (2021) 107810, https://doi.org/10.1016/j.buildenv.2021.107810.
[97] M. Faghiholislam, H. Azemati, H. Keshmiri, S. Pourbagher, Extraction of effective cycles on reducing patients' Depression in the design of treatment spaces, Facilities 42 (1–2) (2024) 105–126, https://doi.org/10.1108/F-08-2022-0104.
[98] M. Fan, G. Cao, C. Pedersen, S. Lu, L.I. Stenstad, J.G. Skogås, Suitability evaluation on Laminar airflow and mixing airflow distribution strategies in operating rooms: a case study at St. Olavs Hospital, Build. Environ. 194 (2021), https://doi.org/10.1016/j.buildenv.2021.107677.
[99] X. Fan, X. Jie, F. Zou, D. Wang, H. Da, H. Li, H. Zhao, J. He, J. Liu, Y. Lu, Y. Liu, Association between outdoor air pollutants and risk of acute exacerbation of chronic obstructive pulmonary disease in Xi'an, China, Air Qual., Atmosph. Health (2024), https://doi.org/10.1007/s11869-024-01513-6.
[100] P.O. Fanger, Thermal comfort. Analysis and applications in environmental engineering. Thermal comfort. Analysis and applications in environmental engineering, 1970.
[101] F. Farhadi, M. Khakzand, M.A. Khanmohammadi, Z. Barzegar, Comparison of IAQ standards in healthcare facilities with the aim of providing acceptable standards in Iran, Pollution 9 (1) (2023) 344–367, https://doi.org/10.22059/poll.2022.342503.1463.

[102] A. Farina, P. Li, The acoustic complexity indices (ACIs), in: A. Farina, P. Li (Eds.), Methods in Ecoacoustics : The Acoustic Complexity Indices, Springer International Publishing, Cham, 2021, pp. 31–70.
[103] M. Fattahi, M. Sharbatdar, Machine-learning-based personal thermal comfort modeling for heat recovery using environmental parameters, Sustain. Energy Technolog. Assessm. 57 (2023), https://doi.org/10.1016/j.seta.2023.103294.
[104] T. Ferrante, T. Villani, Pre-occupancy evaluation in hospital rooms for efficient use of natural light—Improved proposals, Buildings 12 (12) (2022), https://doi.org/10.3390/buildings12122145.
[105] L.J. Fifield, K.J. Lomas, R. Giridharan, D. Allinson, Hospital wards and modular construction: summertime overheating and energy efficiency, Build. Environ. 141 (2018) 28–44, https://doi.org/10.1016/j.buildenv.2018.05.041.
[106] M. Figueiro, Sustainable lighting for healthcare facilities: more than just lumens per watt, J. Green Build. 3 (1) (2008) 74–89, https://doi.org/10.3992/jgb.3.1.74.
[107] M.G. Figueiro, D. Pedler, Red Light: a novel, non-pharmacological intervention to promote alertness in shift workers, J. Safe. Res. 74 (2020) 169–177, https://doi.org/10.1016/j.jsr.2020.06.003.
[108] W.J. Fisk, D. Black, G. Brunner, Changing ventilation rates in U.S. Offices: implications for health, work performance, energy, and associated economics, Build. Environ. 47 (2012) 368–372, https://doi.org/10.1016/j.buildenv.2011.07.001.
[109] N. Fransson, D. Västfjäll, J. Skoog, In search of the comfortable indoor environment: a comparison of the utility of objective and subjective indicators of indoor comfort, Build. Environ. 42 (5) (2007) 1886–1890, https://doi.org/10.1016/j.buildenv.2006.02.021.
[110] G.A. Ganesh, S.L. Sinha, T.N. Verma, Numerical simulation for optimization of the indoor environment of an occupied office building using double-panel and ventilation radiator, J. Build. Eng. 29 (2020) 101139, https://doi.org/10.1016/j.jobe.2019.101139.
[111] G.A. Ganesh, S.L. Sinha, T.N. Verma, S.K. Dewangan, Investigation of indoor environment quality and factors affecting Human comfort: a critical review, Build. Environ. 204 (2021) 108146, https://doi.org/10.1016/j.buildenv.2021.108146.
[112] G.A. Ganesh, S.L. Sinha, T.N. Verma, S.K. Dewangan, Numerical simulation for energy consumption and thermal comfort in a naturally ventilated indoor environment under different orientations of inlet diffuser, Build. Environ. 217 (2022) 109071, https://doi.org/10.1016/j.buildenv.2022.109071.
[113] C. Gao, S. Zhang, The restorative quality of patient ward environment: tests of six dominant design characteristics, Build. Environ. 180 (2020) 107039, https://doi.org/10.1016/j.buildenv.2020.107039.
[114] C. Gao, S. Zhang, Inpatient perceptions of design characteristics related to ward environments' Restorative quality, J. Build. Eng. 41 (2021) 102410, https://doi.org/10.1016/j.jobe.2021.102410.
[115] A.A. Gatea, M.F.M. Batcha, J. Taweekun, Energy efficiency and thermal comfort in hospital buildings: a review, Int. J. Integrat. Eng. 12 (3) (2020) 33–41, https://doi.org/10.30880/ijie.2020.12.03.005.
[116] A. Gebczynska-Janowicz, R. Janowicz, W. Targowski, R. Cudnik, K. Paszko, K. M. Zielińska-Dabkowska, Evaluation of medical staff satisfaction for workplace architecture in temporary COVID-19 Hospital: a case study in Gdańsk, Poland, Int. J. Environ. Res. Public Health 20 (1) (2023), https://doi.org/10.3390/ijerph20010639.
[117] S. Geng, H.W. Chau, S. Yan, W. Zhang, C. Zhang, Comparative analysis of hospital environments in Australia and China using the space syntax approach, Int. J. Build. Pathol. Adaptat. 39 (3) (2020) 525–546, https://doi.org/10.1108/IJBPA-04-2020-0031.
[118] F. Ghaffari, M. Shabak, N. Norouzi, S.N. Fallah, Hospital salutogenic public spaces: a conceptual framework of effective perceptional environment quality components on patients' Satisfaction', Int. J. Build. Pathol. Adaptat. (2021) https://doi.org/10.1108/IJBPA-05-2021-0071.
[119] L. Gharaibeh, S. Matarneh, B. Lantz, K. Eriksson, Quantifying the influence of BIM adoption: an in-depth methodology and practical case studies in construction, Result. Eng. 23 (2024) 102555, https://doi.org/10.1016/j.rineng.2024.102555.
[120] R. Giridharan, K.J. Lomas, C.A. Short, A.J. Fair, Performance of hospital spaces in summer: a case study of a 'Nucleus'-type hospital in the UK Midlands, Energy Build. 66 (2013) 315–328, https://doi.org/10.1016/j.enbuild.2013.07.001.
[121] M. Golbazi, C.B. Aktas, Leed certification and patient wellbeing in Green Healthcare facilities, J. Green Build. 15 (4) (2020) 3–18, https://doi.org/10.3992/jgb.15.4.3.
[122] M. Golbazi, C.B. Aktas, Leed certification and patient wellbeing in Green Healthcare facilities, J. Green Build. 15 (4) (2020) 3–18, https://doi.org/10.3992/jgb.15.4.3.
[123] P. Gong, Y. Cai, B. Chen, C. Zhang, S. Stravoravdis, S. Sharples, Q. Ban, Y. Yu, An artificial neural network-based model that can predict inpatients' Personal thermal sensation in rehabilitation wards, J. Build. Eng. 80 (2023) 108033, https://doi.org/10.1016/j.jobe.2023.108033.
[124] M. Gripko, A. Joseph, S. MohammadiGorji, Effects of the physical environment on children and families in hospital-based emergency departments: a systematic literature review, J. Environ. Psychol. 86 (2023), https://doi.org/10.1016/j.jenvp.2023.101970.
[125] H. Gu, W. Yan, E. Elahi, Y. Cao, Air pollution risks Human mental health: an implication of two-stages least squares estimation of interaction effects, Environ. Sci. Pollut. Res. 27 (2) (2020) 2036–2043, https://doi.org/10.1007/s11356-019-06612-x.
[126] L. Gualtieri, E. Rauch, R. Vidoni, Emerging research fields in safety and ergonomics in Industrial Collaborative Robotics: a systematic literature review, Robot. Comput. Integr. Manuf. 67 (2021) 101998, https://doi.org/10.1016/j.rcim.2020.101998.
[127] A.D. Gutierrez, H. Sezer, J.L. Ramirez, Required thermal comfort conditions inside hospital operating rooms (ORS): a numerical assessment, Front. Heat Mass Transfer 18 (2022), https://doi.org/10.5098/hmt.18.4.
[128] L. Hadjidemetriou, N. Stylianidis, D. Englezos, P. Papadopoulos, D. Eliades, S. Timotheou, M.M. Polycarpou, C. Panayiotou, A digital twin architecture for real-time and offline high granularity analysis in smart buildings, Sustain. Citi. Soc. 98 (2023) 104795, https://doi.org/10.1016/j.scs.2023.104795.
[129] M.M. Hammed, E.H.A. Afifi, T.S. Elhinnawy, Role of biophilic design in quality of hospital patients rooms, Int. J. Techn. Phys. Probl. Eng. 15 (4) (2023) 226–233.
[130] J. Han, H.-J. Kang, G. Hyun Kwon, A measurement for evaluating the environmental quality of advanced healthcare facilities: intelligent healthscape quality for medical staff, Build. Environ. 144 (2018) 532–541, https://doi.org/10.1016/j.buildenv.2018.08.023.
[131] M. Han, R. May, X. Zhang, X. Wang, S. Pan, Y. Da, Y. Jin, A novel reinforcement learning method for improving occupant comfort via window opening and closing, Sustain. Citi. Soc. 61 (2020) 102247, https://doi.org/10.1016/j.scs.2020.102247.
[132] C. Hedges, C. Hunt, P. Ball, Quiet time improves the patient experience, J. Nurs. Care. Qual. 34 (3) (2019) 197–202, https://doi.org/10.1097/NCQ.0000000000000363.
[133] S. Hignett, J. Lu, An investigation of the use of health building notes by UK healthcare building designers, Appl. Ergon. 40 (4) (2009) 608–616, https://doi.org/10.1016/j.apergo.2008.04.018.
[134] A.M. Hill, C.C. Crisp, A. Shatkin-Margolis, T. Tam, E. Yook, S. Kleeman, R. N. Pauls, The influence of postoperative environment on patient pain and satisfaction: a randomized trial, Am. J. Obstet. Gynecol. 223 (2) (2020), https://doi.org/10.1016/j.ajog.2020.05.006.
[135] S.H. Ho, L. Rosario, M.M. Rahman, Three-dimensional analysis for hospital operating room thermal comfort and contaminant removal, Appl. Therm. Eng. 29 (10) (2009) 2080–2092, https://doi.org/10.1016/j.applthermaleng.2008.10.016.
[136] K. Honarmand, C.J. Yarnell, C. Young-Ritchie, R. Maunder, F. Priestap, M. Abdalla, I.M. Ball, J. Basmaji, C.M. Bell, L. Jeffs, S. Shah, J. Chen, D. LeBlanc, J. Kayitesi, C. Eta-Ndu, S. Mehta, Personal, professional, and psychological impact of the COVID-19 Pandemic on hospital workers: a cross-sectional survey, PLOS ONE 17 (2) (2022), https://doi.org/10.1371/journal.pone.0263438.
[137] J. Horne, N. Dunne, N. Singh, M. Safiuddin, N. Esmaeili, M. Erenler, I. Ho, E. Luk, Building parameters linked with indoor transmission of SARS-CoV-2', Environ. Res. 238 (2023) 117156 https://doi.org/10.1016/j.envres.2023.117156.
[138] Y. Hua, F. Becker, T. Wurmser, J. Bliss-Holtz, C. Hedges, Effects of nursing unit spatial layout on nursing team communication patterns, quality of care, and patient safety, Herd-Health Environ. Res. Des. J. 6 (1) (2012) 8–38, https://doi.org/10.1177/193758671200600102.
[139] Y.-C. Huang, C.L. Chu, S.N. Chang Lee, S.J. Lan, C.H. Hsieh, Y.P. Hsieh, Building users' Perceptions of importance of indoor environmental quality in long-term care facilities, Build. Environ. 67 (2013) 224–230, https://doi.org/10.1016/j.buildenv.2013.05.004.
[140] Y.-C. Huang, C.-L. Chu, S.-N.C. Lee, S.-J. Lan, C.-H. Hsieh, Y.-P. Hsieh, Building users' Perceptions of importance of indoor environmental quality in long-term care facilities, Build. Environ. 67 (2013) 224–230, https://doi.org/10.1016/j.buildenv.2013.05.004.
[141] E.R.C.M. Huisman, E. Morales, J. van Hoof, H.S.M. Kort, Healing environment: a review of the impact of physical environmental factors on users, Build. Environ. 58 (2012) 70–80, https://doi.org/10.1016/j.buildenv.2012.06.016.
[142] T. Hulland, A. Su, M. Kingan, Noise in an inpatient hospital ward in New Zealand, Build. Acoust. 27 (4) (2020) 299–309, https://doi.org/10.1177/1351010×20916120.
[143] D.D. Huntsman, G. Bulaj, Healthy dwelling: design of biophilic interior environments fostering self-care practices for people living with migraines, chronic pain, and depression, Int. J. Environ. Res. Public Health 19 (4) (2022), https://doi.org/10.3390/ijerph19042248.
[144] R.-L. Hwang, T.-P. Lin, M.-J. Cheng, J.-H. Chien, Patient thermal comfort requirement for hospital environments in Taiwan, Build. Environ. 42 (8) (2007) 2980–2987, https://doi.org/10.1016/j.buildenv.2006.07.035.
[145] N. Iwata, S. Ichii, K. Egashira, Effects of bright artificial light on subjective mood of shift work nurses, Ind. Health 35 (1) (1997) 41–47, https://doi.org/10.2486/indhealth.35.41.
[146] N. Jain, E. Burman, S. Stamp, C. Shrubsole, R. Bunn, T. Oberman, E. Barrett, F. Aletta, J. Kang, P. Raynham, D. Mumovic, M. Davies, Building performance evaluation of a new hospital Building in the UK: balancing indoor environmental quality and energy performance, Atmosph. (Basel) 12 (1) (2021), https://doi.org/10.3390/atmos12010115.
[147] D. Jeong, H. Yi, J.H. Park, H.W. Park, K. Park, A vertical laminar airflow system to prevent aerosol transmission of SARS-CoV-2 in building space: computational fluid dynamics (CFD) and experimental approach, Indoor Built Environ. 31 (5) (2022) 1319–1338, https://doi.org/10.1177/1420326×211063422.
[148] J. Jeoung, S. Jung, T. Hong, J.-K. Choi, Blockchain-based IoT system for personalized indoor temperature control, Automat. Construct. 140 (2022) 104339, https://doi.org/10.1016/j.autcon.2022.104339.
[149] Z. Jia, S. Pan, H. Yu, Y. Liu, S. Wei, M. Qin, L. Chang, Y. Cui, Modeling occupant window behavior in hospitals—a case study in a maternity hospital in Beijing, China, Sustain. (Switzerl.) 15 (11) (2023), https://doi.org/10.3390/su15118606.
[150] S. Jiang, M. Wang, L. Ma, Gaps and requirements for applying automatic architectural design to building renovation, Automat. Construct. 147 (2023) 104742, https://doi.org/10.1016/j.autcon.2023.104742.

[151] A. Joseph, X. Quan, A.B. Keller, E. Taylor, U. Nanda, Y. Hua, Building a knowledge base for evidence-based healthcare facility design through a Post-occupancy evaluation toolkit, Intell. Build. Int. 6 (3) (2014) 155–169, https://doi.org/10.1080/17508975.2014.903163.
[152] N. Jovanovic, E. Miglietta, A. Podlesek, A. Malekzadeh, A. Lasalvia, J. Campbell, S. Priebe, Impact of the hospital built environment on treatment satisfaction of psychiatric In-patients, Psychol. Med. 52 (10) (2022) 1969–1980, https://doi.org/10.1017/S0033291720003815.
[153] D.F. Juang, C.H. Lee, T. Yang, M.C. Chang, Noise pollution and its effects on medical care workers and patients in hospitals, Int. J. Environ. Sci. Technol. 7 (4) (2010) 705–716, https://doi.org/10.1007/BF03326180.
[154] D. Jung, D.I. Kim, N. Kim, Bringing nature into hospital architecture: machine learning-based EEG analysis of the biophilia effect in virtual reality, J. Environ. Psychol. 89 (2023), https://doi.org/10.1016/j.jenvp.2023.102033.
[155] I. Kagan, S. Fridman, E. Shalom, S. Melnikov, The effect of working in an infection isolation room on hospital nurses' Job satisfaction, J. Nurs. Manag. 26 (2) (2018) 120–126, https://doi.org/10.1111/jonm.12516.
[156] M. Kameni Nematchoua, P. Ricciardi, S. Reiter, S. Asadi, C.M. Demers, Thermal comfort and comparison of some parameters coming from hospitals and shopping centers under natural ventilation: the case of Madagascar Island, J. Build. Eng. 13 (2017) 196–206, https://doi.org/10.1016/j.jobe.2017.07.014.
[157] S.K. Kang, S. Park, Effect of the ASPAN guideline on perioperative hypothermia among patients with upper extremity surgery under general anesthesia: a randomized controlled trial, J. Perianesth. Nurs. 35 (3) (2020) 298–306, https://doi.org/10.1016/j.jopan.2019.11.004.
[158] W. Khalid, S.A. Zaki, H.B. Rijal, F. Yakub, Investigation of comfort temperature and thermal adaptation for patients and visitors in Malaysian hospitals, Energy Build. 183 (2019) 484–499, https://doi.org/10.1016/j.enbuild.2018.11.019.
[159] M.A.I. Khan, C.J. Noakes, V.V. Toropov, Development of a numerical optimization approach to ventilation system design to control airborne contaminant dispersion and occupant comfort, Build. Simulat. 5 (1) (2012) 39–50, https://doi.org/10.1007/s12273-012-0058-7.
[160] M. Khan, M.J. Thaheem, M. Khan, A. Maqsoom, M. Zeeshan, Thermal comfort and ventilation conditions in healthcare facilities - part 1: an assessment of indoor environmental quality (IEQ), Environ. Eng. Manage. J. 19 (7) (2020) 917–933.
[161] J. Khodakarami, N. Nasrollahi, Thermal comfort in hospitals - a literature review, Renew. Sustain. Energy Rev. 16 (6) (2012) 4071–4077, https://doi.org/10.1016/j.rser.2012.03.054.
[162] J. Kim, R. de Dear, Nonlinear relationships between individual IEQ factors and overall workspace satisfaction, Build. Environ. 49 (2012) 33–40, https://doi.org/10.1016/j.buildenv.2011.09.022.
[163] S.K. Kim, Y. Hwang, Y.S. Lee, W. Corser, Occupant comfort and satisfaction in Green Healthcare Environments: a survey study focusing on healthcare staff, J. Sustain. Develop. 8 (1) (2015) 156–173, https://doi.org/10.5539/jsd.v8n1p156.
[164] J. Kirch, G. Marquardt, Towards Human-centred General hospitals: the potential of dementia-friendly design, Archit. Sci. Rev. 66 (5) (2023) 382–390, https://doi.org/10.1080/00038628.2021.1933889.
[165] X.F. Kong, C.L. Guo, Z. Lin, S.S. Duan, J.J. He, Y. Ren, J.L. Ren, Experimental study on the control effect of different ventilation systems on fine particles in a simulated hospital ward, Sustain. Citi. Soc. 73 (2021), https://doi.org/10.1016/j.scs.2021.103102.
[166] A. Konkani, B. Oakley, B. Penprase, Reducing hospital ICU noise: a behavior-based approach, J. Healthc. Eng. 5 (2) (2014) 229–246, https://doi.org/10.1260/2040-2295.5.2.229.
[167] B. Lam, E.M. Peijin Fan, K. Ooi, Z.-T. Ong, J.Y. Hong, W.-S. Gan, S.Y. Ang, Assessing the perceived indoor acoustic environment quality across building occupants in a tertiary-care public hospital in Singapore, Build. Environ. 222 (2022) 109403, https://doi.org/10.1016/j.buildenv.2022.109403.
[168] B. Lam, E.M.P. Fan, K. Ooi, Z.-T. Ong, J.Y. Hong, W.-S. Gan, S.Y. Ang, Assessing the perceived indoor acoustic environment quality across building occupants in a tertiary-care public hospital in Singapore, Build. Environ. 222 (2022) 109403, https://doi.org/10.1016/j.buildenv.2022.109403.
[169] L. Lan, W. Tushar, K. Otto, C. Yuen, K.L. Wood, Thermal comfort improvement of naturally ventilated patient wards in Singapore, Energy Build. 154 (2017) 499–512, https://doi.org/10.1016/j.enbuild.2017.07.080.
[170] S. Lee, Evaluating serviceability of healthcare servicescapes: service design perspective, Int. J. Des. 5 (2) (2011) 61–71.
[171] J.M. Levin, R.D. Winkelman, G.A. Smith, J. Tanenbaum, E.C. Benzel, T.E. Mroz, M.P. Steinmetz, The Association between the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) Survey and real-world clinical outcomes in Lumbar spine surgery, Spin. J. 17 (11) (2017) 1586–1593, https://doi.org/10.1016/j.spinee.2017.05.002.
[172] Y. Li, P. Liao, Y. Song, H. Chi, A systematic decision-support approach for healthcare facility layout design integrating resource flow and space adjacency optimization with simulation-based performance evaluation, J. Build. Eng. 77 (2023) 107465, https://doi.org/10.1016/j.jobe.2023.107465.
[173] E. de Lima Andrade, D.C. da Cunha e Silva, E.A. de Lima, R.A. de Oliveira, P.H. T. Zannin, A.C.G. Martins, Environmental noise in hospitals: a systematic review, Environ. Sci. Pollut. Res. 28 (16) (2021) 19629–19642, https://doi.org/10.1007/s11356-021-13211-2.
[174] A. Lin, M. Fang, C. Zhou, Analysis of ward lighting environment and design of comfortable ward lighting, Measure.:. Sens. (2024) 31, https://doi.org/10.1016/j.measen.2023.101020.
[175] E.H.C. Liu, S.-M. Tan, Patients' Perception of sound levels in the surgical suite, J. Clin. Anesth. 12 (4) (2000) 298–302, https://doi.org/10.1016/S0952-8180(00)00155-0.
[176] Y. Liu, Z. Wang, Z. Zhang, J. Hong, B. Lin, Investigation on the indoor environment quality of health care facilities in China, Build. Environ. 141 (2018) 273–287, https://doi.org/10.1016/j.buildenv.2018.05.054.
[177] Z. Liu, Z. Han, L. Hu, C. Hu, R. Rong, Influence of talking behavior of infected patients and the associated exposure risk in a ventilated negative-pressure ward, Build. Environ. 248 (2024), https://doi.org/10.1016/j.buildenv.2023.111065.
[178] M.G.L.C. Loomans, A.K. Mishra, M.T.H. Derks, J.J. Kraakman, H.S.M. Kort, Occupant response to transitions across indoor thermal environments in two different workspaces, Build. Environ. 144 (2018) 402–411, https://doi.org/10.1016/j.buildenv.2018.08.049.
[179] Y. Lu, D. Niu, S. Zhang, H. Chang, Z. Lin, Ventilation indices for evaluation of airborne infection risk control performance of air distribution, Build. Environ. 222 (2022) 109440, https://doi.org/10.1016/j.buildenv.2022.109440.
[180] Y.L. Lu, D. Niu, S. Zhang, H. Chang, Z. Lin, Ventilation indices for evaluation of airborne infection risk control performance of air distribution, Build. Environ. 222 (2022), https://doi.org/10.1016/j.buildenv.2022.109440.
[181] J. Lucas, T. Bulbul, C. Anumba, Gap analysis on the ability of guidelines and standards to support the performance of healthcare facilities', J. Perform. Construct. Facilit. 27 (6) (2013) 748–755, https://doi.org/10.1061/(ASCE)CF.1943-5509.0000364.
[182] H. Luo, L. Zhong, Ultraviolet germicidal irradiation (UVGI) for in-duct airborne bioaerosol disinfection: review and analysis of design factors, Build. Environ. 197 (2021) 107852, https://doi.org/10.1016/j.buildenv.2021.107852.
[183] C. Ma, O. Guerra-Santin, M. Mohammadi, Exploring the influence of indoor environment and spatial layout on changed behaviours of people with dementia in a nursing home, Build. Environ. 256 (2024) 111452, https://doi.org/10.1016/j.buildenv.2024.111452.
[184] S. Ma, W. Deng, J. Lu, T. Zhou, B. Liu, Investigation of thermal comfort and preferred temperatures for healthcare staff in hospitals in Ningbo, China, J. Build. Eng. 80 (2023), https://doi.org/10.1016/j.jobe.2023.108029.
[185] D.J. MacKenzie, L.G.U. Galbrun, Noise levels and Noise sources in acute care hospital wards, Build. Serv. Eng. Res. Technol. 28 (2) (2007) 117–131, https://doi.org/10.1177/0143624406074468.
[186] J. Mackrill, P. Jennings, R. Cain, Exploring positive Hospital ward soundscape interventions, Appl. Ergon. 45 (6) (2014) 1454–1460, https://doi.org/10.1016/j.apergo.2014.04.005.
[187] G. Madaras, A state-of-the-art review of healthcare acoustics, Sound Vibrat. 48 (8) (2014) 5–7.
[188] O.K. Magas, J.T. Gunter, J.L. Regens, Ambient air pollution and daily pediatric hospitalizations for asthma, Environ. Sci. Pollut. Res. 14 (1) (2007) 19–23, https://doi.org/10.1065/espr2006.08.333.
[189] M. Maghlakelidze, A. Mesa, A.O. Sawyer, M. Gath Morad, L. Ben-Alon, Biophilic design elements and natural materials in healthcare environments, J. Green Build. 19 (3) (2024) 1–39, https://doi.org/10.3992/jgb.19.3.3.
[190] F.J. Mahmood, A.Y. Tayib, Healing environment correlated with patients' Psychological comfort: post-occupancy evaluation of general hospitals, Indoor Built Environ. 30 (2) (2021) 180–194, https://doi.org/10.1177/1420326×19888005.
[191] F.J. Mahmood, A.Y. Tayib, The role of patients' Psychological comfort in optimizing indoor healing environments: a case study of the indoor environments of recently built hospitals in Sulaimani City, Kurdistan, Iraq', HERD: Health Environ. Res. Des. J. 13 (2) (2020) 68–82, https://doi.org/10.1177/1937586719894549.
[192] P. Maleetipwan-Mattsson, T. Laike, M. Johansson, Factors affecting optimal lighting use in shared hospital environments: a case-study, Build. Environ. 96 (2016) 260–269, https://doi.org/10.1016/j.buildenv.2015.11.026.
[193] A. Manyande, P. Salmon, Effects of pre-operative relaxation on post-operative analgesia: immediate increase and delayed reduction', Br. J. Health Psychol. 3 (1998) 215–224, https://doi.org/10.1111/j.2044-8287.1998.tb00568.x.
[194] L.J. McCunn, R. Gifford, Environmental design in acute care settings: a case study of a neurological rehabilitation unit, HERD 7 (1) (2013) 102–113, https://doi.org/10.1177/193758671300700109.
[195] L.J. McCunn, R. Gifford, Environmental design in acute care settings: a case study of a neurological rehabilitation unit, HERD-Health Environ. Res. Des. J. 7 (1) (2013) 102–113, https://doi.org/10.1177/193758671300700109.
[196] L.J. McCunn, S. Safranek, A. Wilkerson, R.G. Davis, Lighting control in patient rooms: understanding nurses' Perceptions of hospital Lighting using qualitative methods, HERD-Health Environ. Res. Des. J. 14 (2) (2021) 204–218, https://doi.org/10.1177/1937586720946669.
[197] L.J. McCunn, J. Wright, Hospital employees' Perceptions of circadian lighting: a pharmacy department case study, J. Faciliti. Manag. 17 (5) (2019) 428–441, https://doi.org/10.1108/JFM-04-2019-0016.
[198] S.P. McGrath, I.M. Perreard, M.D. Garland, K.A. Converse, T.A. Mackenzie, Improving patient safety and clinician workflow in the general care setting with enhanced surveillance monitoring, IEEE. J. Biomed. Health. Inform. 23 (2) (2019) 857–866, https://doi.org/10.1109/JBHI.2018.2834863.
[199] M.H. Mehraban, S.M. Sepasgozar, A. Ghomimoghadam, B. Zafari, AI-enhanced automation of building energy optimization using a hybrid stacked model and genetic algorithms: experiments with seven machine learning techniques and a deep neural network, Result. Eng. 26 (2025) 104994, https://doi.org/10.1016/j.rineng.2025.104994.

[200] A.K. Melikov, Advanced air distribution: improving health and comfort while reducing energy use, Indoor Air 26 (1) (2016) 112–124, https://doi.org/10.1111/ina.12206.
[201] C.C. Menassa, N. Taylor, J. Nelson, Optimizing hybrid ventilation in public spaces of complex buildings - a case study of the Wisconsin Institutes for Discovery, Build. Environ. 61 (2013) 57–68, https://doi.org/10.1016/j.buildenv.2012.12.009.
[202] E.A. Metwally, M. Refat Ismail, A.A. Farid, Development of an IoT occupant-centric metrics: a systematic literature review, Result. Eng. 23 (2024) 102614, https://doi.org/10.1016/j.rineng.2024.102614.
[203] S.D. Moeser, Cognitive mapping in a complex building, Environ. Behav. 20 (1) (1988) 21–49, https://doi.org/10.1177/0013916588201002.
[204] R. Mora, M. Meteyer, Thermal comfort in health-care settings', ASHRAE J. 61 (6) (2019) 10–19.
[205] A. Nejati, S. Rodiek, M. Shepley, Using visual simulation to evaluate restorative qualities of access to nature in hospital staff break areas, Landsc. Urban. Plan. 148 (2016) 132–138, https://doi.org/10.1016/j.landurbplan.2015.12.012.
[206] M.K. Nematchoua, P. Ricciardi, C. Buratti, Statistical analysis of indoor parameters an subjective responses of building occupants in a hot region of Indian Ocean; a case of Madagascar Island, Appl. Energy 208 (2017) 1562–1575, https://doi.org/10.1016/j.apenergy.2017.08.207.
[207] P.S. Nimlyat, Indoor Environmental Quality performance and occupants' Satisfaction [IEQPOS] as assessment criteria for Green Healthcare building rating, Build. Environ. 144 (2018) 598–610, https://doi.org/10.1016/j.buildenv.2018.09.003.
[208] P.S. Nimlyat, The impact of indoor environmental quality (IEQ) on patients' Health and comfort in Nigeria, Int. J. Build. Pathol. Adaptat. (2022), https://doi.org/10.1108/IJBPA-06-2021-0089.
[209] P.S. Nimlyat, M.Z. Kandar, Appraisal of indoor environmental quality (IEQ) in healthcare facilities: a literature review, Sustain. Citi. Soc. 17 (2015) 61–68, https://doi.org/10.1016/j.scs.2015.04.002.
[210] N.P.A. Nitidara, J. Sarwono, S. Suprijanto, F.X. Nugroho Soelami, The multisensory interaction between auditory, visual, and thermal to the overall comfort in public open space: a study in a tropical climate, Sustain. Citi. Soc. 78 (2022) 103622, https://doi.org/10.1016/j.scs.2021.103622.
[211] K. Nordström, D. Norbäck, R. Akselsson, Effect of air humidification on the sick building syndrome and perceived indoor air quality in hospitals: a four month longitudinal study, Occup. Environ. Med. 51 (10) (1994) 683–688, https://doi.org/10.1136/oem.51.10.683.
[212] K. Nordström, D. Norbäck, R. Akselsson, Effect of air humidification on the sick building syndrome and perceived indoor air-quality in hospitals - a 4 month longitudinal-study, Occup. Environ. Med. 51 (10) (1994) 683–688, https://doi.org/10.1136/oem.51.10.683.
[213] B. Nourozi, A. Wierzbicka, R. Yao, S. Sadrizadeh, A systematic review of ventilation solutions for hospital wards: addressing cross-infection and patient safety, Build. Environ. 247 (2024), https://doi.org/10.1016/j.buildenv.2023.110954.
[214] W. O'Brien, F. Tahmasebi, T. Hong, Occupant aspects of building energy codes and standards', ASHRAE J. 65 (3) (2023) 52–57.
[215] A.C. O'Hara, A.C. Miller, H. Spinks, A. Seifert, T. Mills, A.R. Tuininga, The sustainable prescription: benefits of green roof implementation for urban hospitals, Front. Sustain. Citi. 4 (2022), https://doi.org/10.3389/frsc.2022.798012.
[216] S. Oi-Zhen, C. Weng-Wai, T. Yu-Tian, Quality of healing environment in healthcare facilities, Jurnal Teknologi 74 (2) (2015) 101–108, https://doi.org/10.11113/jt.v74.4529.
[217] S. Okcu, E. Ryherd, H. Pelton, C. Zimring, Characterizing impulsiveness of hospital sound environments, Noise Control Eng. J. 60 (3) (2012) 246–257, https://doi.org/10.3397/1.3701002.
[218] Olson, K., C. Sinsky, S.T. Rinne, T. Long, R. Vender, S. Mukherjee, M. Bennick, and M. Linzer. 2019. 'Cross-sectional survey of workplace stressors associated with physician burnout measured by the mini-Z and the Maslach burnout inventory'. Stress Health 35(2):157–75. doi:10.1002/smi.2849.
[219] M.J. Page, J.E. McKenzie, P.M. Bossuyt, I. Boutron, T.C. Hoffmann, C.D. Mulrow, L. Shamseer, J.M. Tetzlaff, E.A. Akl, S.E. Brennan, The PRISMA 2020 statement: an updated guideline for reporting systematic reviews, BMJ 372 (2021).
[220] J. Pai, A study in hospital noise–a case from Taiwan, Int. J. Occupat. Safet. Ergon. 13 (1) (2007) 83–90, https://doi.org/10.1080/10803548.2007.11076711.
[221] Y. Parsia, P.F.M. Tamyez, Role of healthcare-facilities layout design, healing Architecture, on quality of services, Int. J. Civi. Eng. Technol. 9 (4) (2018) 598–601.
[222] E.S. Patterson, E.B.N. Sanders, C.M. Sommerich, S.A. Lavender, J. Li, K.D. Evans, Meeting patient expectations during hospitalization: a grounded theoretical analysis of patient-centered room elements, HERD-Health Environ. Res. Des. J. 10 (5) (2017) 95–110, https://doi.org/10.1177/1937586717696700.
[223] H.K. Pelton, E. Ryherd, M. Martin, Acoustical design of a burn acute care unit for enhanced patient comfort, Noise Control Eng. J. 57 (1) (2009) 32–41, https://doi.org/10.3397/1.3075073.
[224] T. Peng, Y. Zhang, X. Jiang, Y. Yang, Z. Fang, Z. Zheng, Investigation of pregnant women thermal comfort in the waiting area of the hospital in South China, Guangzhou, J. Build. Eng. 44 (2021), https://doi.org/10.1016/j.jobe.2021.103254.
[225] P.F.D.C. Pereira, E.E. Broday, A.A.P. Xavier, Thermal comfort applied in Hospital environments: a literature review, Appl. Sci. (Switzerl.) 10 (20) (2020) 1–22, https://doi.org/10.3390/app10207030.
[226] M. Pirsaheb, H. Yarmohammadi, M. Dabirian, D. Abdollahzadeh, Y. Sohrabi, A study of public and local noise level in wards of a State hospital in Kermanshah, J. Eng. Appl. Sci. 11 (5) (2016) 1069–1073, https://doi.org/10.3923/jeasci.2016.1069.1073.
[227] A. Pourshaghaghy, M. Omidvari, Examination of thermal comfort in a hospital using PMV-PPD model, Appl. Ergon. 43 (6) (2012) 1089–1095, https://doi.org/10.1016/j.apergo.2012.03.010.
[228] Y. Qi, Y. Yan, S.S. Lau, Y. Tao, Evidence-based design for waiting space environment of pediatric clinics—Three hospitals in Shenzhen as case studies, Int. J. Environ. Res. Public Health 18 (22) (2021), https://doi.org/10.3390/ijerph182211804.
[229] Y. Qian, J. Leng, Q. Chun, H. Wang, K. Zhou, A year-long field investigation on the spatio-temporal variations of occupant's thermal comfort in Chinese traditional courtyard dwellings, Build. Environ. 228 (2023) 109836, https://doi.org/10.1016/j.buildenv.2022.109836.
[230] A. Rossetti, J. Loewy, W. Chang-Lit, N.H. van Dokkum, E. Baumann, G. Bouissou, J. Mondanaro, T. O'Connor, G. Asch-Ortiz, H. Mitaka, Effects of live music on the perception of noise in the SICU/PICU: a patient, caregiver, and medical staff environmental study, Int. J. Environ. Res. Public Health 20 (4) (2023), https://doi.org/10.3390/ijerph20043499.
[231] A. Rossetti, J. Loewy, W. Chang-Lit, N.H.van Dokkum, E. Baumann, G. Bouissou, J. Mondanaro, T. O'Connor, G. Asch-Ortiz, H. Mitaka, Effects of live music on the perception of noise in the SICU/PICU: a patient, caregiver, and medical staff environmental study, Int. J. Environ. Res. Public Health 20 (4) (2023) 3499, https://doi.org/10.3390/ijerph20043499.
[232] S. Roy, P. Satvaya, The effects of lamp types and surface reflectance combinations on the subjective perception of a simulated lit hospital ward environment, Facilities 40 (11–12) (2022) 697–718, https://doi.org/10.1108/F-01-2022-0013.
[233] S. Ruggiero, F. Tariello, G.P. Vanoli, Evaluation of comfort models considering the peculiarities of hospitalization: bedding, clothing and reduced activity of patients, Buildings 12 (3) (2022), https://doi.org/10.3390/buildings12030343.
[234] T. Rus, G. Cruciat, G. Nemeti, R. Mare, D. Muresan, Thermal comfort in maternity wards: summer vs. Winter conditions, J. Build. Eng. 51 (2022), https://doi.org/10.1016/j.jobe.2022.104356.
[235] T. Rus, G. Nemeti, F.V. Domnita, I.G. Goidescu, D. Muresan, Indoor thermal environment evaluation of postpartum patients in a tertiary level maternity in Romania during summer, Sci. Technol. Built Environ. 27 (7) (2021) 948–959, https://doi.org/10.1080/23744731.2021.1906084.
[236] T. Rus, G. Cruciat, G. Nemeti, R. Mare, D. Muresan, Thermal comfort in maternity wards: summer vs. Winter conditions, J. Build. Eng. 51 (2022) 104356, https://doi.org/10.1016/j.jobe.2022.104356.
[237] T. Rus, G. Nemeti, F. Vasile Domnita, I. Gabriel Goidescu, D. Muresan, Indoor thermal environment evaluation of postpartum patients in a tertiary level maternity in Romania during summer, Sci. Technol. Built Environ. 27 (7) (2021) 948–959, https://doi.org/10.1080/23744731.2021.1906084.
[238] S. Sadrizadeh, A. Aganovic, A. Bogdan, C. Wang, A. Afshari, A. Hartmann, C. Croitoru, A. Khan, M. Kriegel, M. Lind, Z. Liu, A. Melikov, J. Mo, H. Rotheudt, R. Yao, Y. Zhang, O. Abouali, H. Langvatn, O. Sköldenberg, G. Cao, A systematic review of operating room ventilation, J. Build. Eng. 40 (2021), https://doi.org/10.1016/j.jobe.2021.102693.
[239] A.K. Sahu, T.N. Verma, S.L. Sinha, Numerical simulation of air flow in multiple beds intensive care unit of hospital, Int. J. Automot. Mech. Eng. 16 (2) (2019) 6796–6807, https://doi.org/10.15282/ijame.16.2.2019.24.0511.
[240] A.K. Sahu, S.L. Sinha, T.N. Verma, Numerical simulation method to predict air flow and contaminant control in a multiple bed intensive care unit of hospital, in: A.A. Rizvanov, B.K. Singh, P. Ganasala (Eds.), Advances in Biomedical Engineering and Technology, Springer, Singapore, 2021, pp. 9–19.
[241] I. Said, S.Z. Salleh, M.S.A. Bakar, I. Mohamad, Caregivers' Evaluation on hospitalized children's preferences concerning garden and ward, J. Asia. Architect. Build. Eng. 4 (2) (2005) 331–338, https://doi.org/10.3130/jaabe.4.331.
[242] P. Saikia, D. Rakshit, R. Narayanaswamy, F. Wang, Energy performance and indoor airflow analysis of a healthcare ward designed with resource conservation objectives, J. Build. Eng. 44 (2021), https://doi.org/10.1016/j.jobe.2021.103296.
[243] J. Saini, M. Dutta, G. Marques, Indoor air quality monitoring systems based on internet of things: a systematic review, Int. J. Environ. Res. Public Health 17 (14) (2020) 1–22, https://doi.org/10.3390/ijerph17144942.
[244] H. Salonen, M. Lahtinen, S. Lappalainen, N. Nevala, L.D. Knibbs, L. Morawska, K. Reijula, Design approaches for promoting beneficial indoor environments in healthcare facilities: a review, Intell. Build. Int. 5 (1) (2013) 26–50, https://doi.org/10.1080/17508975.2013.764839.
[245] H.H. Sameh, E.M. Omar, N.M. Ezz El-Dein, Criteria for evaluating the saving and production of energy in hospitals "nursing units, Int. J. Eng. Res. Technol. 12 (5) (2019) 670–678.
[246] C. Sanna, L. Marras, A. Desogus, B. Marras, N. Montero, G. Bertolino, M. Schintu, V. Coroneo, Evaluation of Rhodotorula Spp. Contamination in hospital environments, Environ. Monit. Assess. 193 (3) (2021), https://doi.org/10.1007/s10661-021-08908-3.
[247] A. Santurtún, R. Almendra, P. Fdez-Arroyabe, A. Sanchez-Lorenzo, D. Royé, M. T. Zarrabeitia, P. Santana, Predictive value of three thermal comfort indices in low temperatures on cardiovascular morbidity in the Iberian Peninsula, Sci. Total Environ. 729 (2020), https://doi.org/10.1016/j.scitotenv.2020.138969.
[248] M.K. Satheesan, K.W. Mui, L.T. Wong, A numerical study of ventilation strategies for infection risk mitigation in general inpatient wards, Build. Simulat. 13 (4) (2020) 887–896, https://doi.org/10.1007/s12273-020-0623-4.

[249] S. Sattayakorn, M. Ichinose, R. Sasaki, Clarifying thermal comfort of healthcare occupants in tropical region: a case of indoor environment in Thai hospitals, Energy Build. 149 (2017) 45–57, https://doi.org/10.1016/j.enbuild.2017.05.025.
[250] K.M. Schledermann, T. Bjørner, A.S. West, T.S. Hansen, Evaluation of staff's perception of a circadian lighting system implemented in a hospital, Build. Environ. 242 (2023), https://doi.org/10.1016/j.buildenv.2023.110488.
[251] S. Secchi, N. Setola, L. Marzi, V. Amodeo, Analysis of the acoustic comfort in hospital: the case of maternity rooms, Buildings 12 (8) (2022), https://doi.org/10.3390/buildings12081117.
[252] K.T. Seng, L.C. Shen, N.S. Ling, S.P. Chin, T.J. Xin, T.K. En, A.A. Jama, A. Hariri, N.F. Nasir, Comparative study on the indoor air quality in critical areas of hospitals in Malaysia, Nat. Environ. Pollut. Technol. 22 (2) (2023) 921–927, https://doi.org/10.46488/NEPT.2023.v22i02.035.
[253] M.M. Shannon, R. Lipson-Smith, M. Elf, J. Olver, S. Kramer, J. Bernhardt, Bringing the single versus multi-patient room debate to vulnerable patient populations: a systematic review of the impact of room types on hospitalized older people and people with neurological disorders, Intell. Build. Int. 12 (3) (2020) 180–198, https://doi.org/10.1080/17508975.2018.1548339.
[254] X. Shen, H. Zhang, Y. Li, K. Qu, L. Zhao, G. Kong, W. Jia, Building a satisfactory indoor environment for healthcare facility occupants: a literature review, Build. Environ. 228 (2023) 109861, https://doi.org/10.1016/j.buildenv.2022.109861.
[255] Z.N. Shi, H. Qian, X.H. Zheng, Z.F. Lv, Y.G. Li, L. Liu, P.V. Nielsen, Seasonal variation of window opening behaviors in two naturally ventilated hospital wards, Build. Environ. 130 (2018) 85–93, https://doi.org/10.1016/j.buildenv.2017.12.019.
[256] H.-kyung Shin, K.-woo Kim, Sound absorbing ceiling to reduce heavy weight floor impact Sound, Build. Environ. 180 (2020) 107058, https://doi.org/10.1016/j.buildenv.2020.107058.
[257] S.A. Shumaker, J.E. Reizenstein, Environmental factors affecting inpatient stress in acute care hospitals, Environ. Stress (1982) 179–223.
[258] B.V.F. Silva, J.B. Holm-Nielsen, S. Sadrizadeh, M.P.R. Teles, M. Kiani-Moghaddam, A. Arabkoohsar, Sustainable, green, or smart? Pathways for energy-efficient healthcare buildings', Sustain. Citi. Soc. 100 (2024) 105013 https://doi.org/10.1016/j.scs.2023.105013.
[259] M. Singh, R.M. Singari, M. Bholey, A review study of cognitive design research on colors from a visual psychological perspective, Int. J. Exper. Res. Rev. 30 (2023) 75–86, https://doi.org/10.52756/ijerr.2023.v30.009.
[260] J. Skoog, N. Fransson, L. Jagemar, Thermal environment in Swedish hospitals: summer and winter measurements, Energy Build. 37 (8) (2005) 872–877, https://doi.org/10.1016/j.enbuild.2004.11.003.
[261] M. Śmiełowska, M. Marć, B. Zabiegała, Indoor air quality in public utility environments—A review, Environ. Sci. Pollut. Res. 24 (12) (2017) 11166–11176, https://doi.org/10.1007/s11356-017-8567-7.
[262] R.M. Smith, A. Rae, Patient comfort and radiant ceiling heating in a hospital ward, Build. Environ. 12 (3) (1977) 143–146, https://doi.org/10.1016/0360-1323(77)90011-7.
[263] E.-S. Song, W.H. Kim, B.H. Lee, D.W. Han, J.H. Lee, B. Kim, Assessment of color perception and preference with eye-tracking analysis in a dental treatment environment, Int. J. Environ. Res. Public Health 18 (15) (2021), https://doi.org/10.3390/ijerph18157981.
[264] E.-S. Song, W.-H. Kim, B.-H. Lee, D.-W. Han, J.-H. Lee, B. Kim, Assessment of color perception and preference with eye-tracking analysis in a dental treatment environment, Int. J. Environ. Res. Public Health 18 (15) (2021) 7981, https://doi.org/10.3390/ijerph18157981.
[265] T. Sookchaiya, V. Monyakul, S. Thepa, Assessment of the thermal environment effects on Human comfort and health for the development of novel air conditioning system in tropical regions, Energy Build. 42 (10) (2010) 1692–1702, https://doi.org/10.1016/j.enbuild.2010.04.012.
[266] J.F. Stichler, Healthy work environments for the ageing nursing workforce, J. Nurs. Manag. 21 (7) (2013) 956–963, https://doi.org/10.1111/jonm.12174.
[267] H. Stopps, B. Huchuk, M.F. Touchie, W. O'Brien, Is anyone home? A critical review of occupant-centric smart HVAC controls implementations in residential buildings, Build. Environ. 187 (2021) 107369, https://doi.org/10.1016/j.buildenv.2020.107369.
[268] Y. Sun, S. Kojima, K. Nakaohkubo, J. Zhao, S. Ni, Analysis and evaluation of indoor environment, occupant satisfaction, and energy consumption in General Hospital in China, Buildings 13 (7) (2023), https://doi.org/10.3390/buildings13071675.
[269] V. Surawattanasakul, W. Sirikul, R. Sapbamrer, K. Wangsan, J. Panumasvivat, P. Assavanopakun, S. Muangkaew, Respiratory symptoms and skin sick building syndrome among office workers at University Hospital, Chiang Mai, Thailand: associations with Indoor Air Quality, AIRMED Project, Int. J. Environ. Res. Public Health 19 (17) (2022), https://doi.org/10.3390/ijerph191710850.
[270] V. Surawattanasakul, W. Sirikul, R. Sapbamrer, K. Wangsan, J. Panumasvivat, P. Assavanopakun, S. Muangkaew, Respiratory symptoms and skin sick building syndrome among office workers at University Hospital, Chiang Mai, Thailand: associations with Indoor Air Quality, AIRMED Project, Int. J. Environ. Res. Public Health 19 (17) (2022) 10850, https://doi.org/10.3390/ijerph191710850.
[271] H. Tan, K.Y. Wong, M.H.D. Othman, H.Y. Kek, B.B. Nyakuma, W.S. Ho, H. Hashim, R.A. Wahab, D.D.C.V. Sheng, N.H.A. Wahab, A.S. Yatim, Why do ventilation strategies matter in controlling infectious airborne particles? A comprehensive numerical analysis in isolation ward, Build. Environ. 231 (2023) 110048, https://doi.org/10.1016/j.buildenv.2023.110048.
[272] H. Tang, J. Ding, C. Li, J. Li, A field study on indoor environment quality of Chinese inpatient buildings in a hot and humid region', Build. Environ. 151 (2019) 156–167, https://doi.org/10.1016/j.buildenv.2019.01.046.
[273] H. Tang, J. Ding, Z. Lin, On-site measurement of indoor environment quality in a Chinese healthcare facility with a semi-closed hospital street, Build. Environ. 173 (2020), https://doi.org/10.1016/j.buildenv.2019.106637.
[274] H. Tang, Y. Ding, B.C. Singer, Post-occupancy evaluation of indoor environmental quality in ten nonresidential buildings in Chongqing, China, J. Build. Eng. 32 (2020), https://doi.org/10.1016/j.jobe.2020.101649.
[275] K.R.-H. Teoh, J. Hassard, T. Cox, Doctors' Working conditions, wellbeing and hospital quality of care: a multilevel analysis', Saf. Sci. 135 (2021) 105115 https://doi.org/10.1016/j.ssci.2020.105115.
[276] A. Teran-Somohano, A.E. Smith, A sequential space syntax approach for healthcare facility layout design, Comput. Ind. Eng. 177 (2023), https://doi.org/10.1016/j.cie.2023.109038.
[277] C.L. Thiel, K.L. Needy, R. Ries, D. Hupp, M.M. Bilec, Building design and performance: a comparative longitudinal assessment of a children's hospital, Build. Environ. 78 (2014) 130–136, https://doi.org/10.1016/j.buildenv.2014.04.001.
[278] Y. Tian, A review on factors related to patient comfort experience in hospitals, J. Health., Populat. Nutrit. 42 (1) (2023), https://doi.org/10.1186/s41043-023-00465-4.
[279] H. Toodekharman, M. Abravesh, S. Heidari, Visual comfort assessment of hospital patient rooms with climate responsive facades, J. Daylight. 10 (1) (2023) 17–30, https://doi.org/10.15627/JD.2023.2.
[280] D. Tranfield, D. Denyer, P. Smart, Towards a methodology for developing evidence-informed management knowledge by means of systematic review, Brit. J. Manage. 14 (3) (2003) 207–222, https://doi.org/10.1111/1467-8551.00375.
[281] Ö.S.O. Trängeberg, M.W. Stomberg, Listening to music during regional anesthesia: patients' Experiences and the effect on mood, J. Perianesth. Nurs. 28 (5) (2013) 291–297, https://doi.org/10.1016/j.jopan.2013.01.008.
[282] P.T. Tuna, S. Kursun, I. Kara, Effect of active and passive heating methods used in different areas of perioperative processes on thermal comfort and anxiety: a randomized control trial, J. Basic Clin. Health Sci. 6 (1) (2022) 225–237, https://doi.org/10.30621/jbachs.950156.
[283] R.S. Ulrich, Effects of interior design on wellness: theory and recent scientific research, J. Health Care Inter. Des. 3 (1991) 97–109.
[284] E.E. Ungar, Vibration criteria for healthcare facility floors, Sound Vibrat. 41 (9) (2007) 26–27.
[285] P. Uścinowicz, A. Bogdan, Directions of modification of the model of perception of the thermal environment by patients of selected hospital wards, Energies 15 (11) (2022), https://doi.org/10.3390/en15113965.
[286] P. Uścinowicz, A. Bogdan, M. Szyłak-Szydłowski, M. Młynarczyk, D. Ćwiklińska, Subjective assessment of indoor air quality and thermal environment in patient rooms: a survey study of Polish hospitals, Build. Environ. 228 (2023), https://doi.org/10.1016/j.buildenv.2022.109840.
[287] P. Uścinowicz, A. Bogdan, M.ław Szyłak-Szydłowski, M. Młynarczyk, D. Ćwiklińska, Subjective assessment of indoor air quality and thermal environment in patient rooms: a survey study of Polish hospitals, Build. Environ. 228 (2023) 109840, https://doi.org/10.1016/j.buildenv.2022.109840.
[288] P. Uścinowicz, M. Chludzińska, A. Bogdan, Thermal environment conditions in Polish operating rooms, Build. Environ. 94 (2015) 296–304, https://doi.org/10.1016/j.buildenv.2015.08.003.
[289] P. Uścinowicz, M. Chludzińska, A. Bogdan, Thermal environment conditions in Polish operating rooms, Build. Environ. 94 (2015) 296–304, https://doi.org/10.1016/j.buildenv.2015.08.003.
[290] F.O. Uzuegbunam, F.N. Aja, E.O. Ibem, Influence of building attributes on thermal comfort in naturally ventilated hospital wards in the hot-humid tropics of Southeast Nigeria, Int. J. Build. Pathol. Adaptat. (2023), https://doi.org/10.1108/IJBPA-12-2022-0189.
[291] F.O. Uzuegbunam, F.N. Aja, E.O. Ibem, Influence of building attributes on thermal comfort in naturally ventilated hospital wards in the hot-humid tropics of Southeast Nigeria, Int. J. Build. Pathol. Adaptat. (2023), https://doi.org/10.1108/IJBPA-12-2022-0189 ahead-of-print(ahead-of-print).
[292] P. Valdiserri, S. Cesari, M. Coccagna, P. Romio, S. Mazzacane, Experimental data and simulations of performance and thermal comfort in a patient room equipped with radiant ceiling panels, Buildings 10 (12) (2020) 1–18, https://doi.org/10.3390/buildings10120235.
[293] R. Van Gaever, V.A. Jacobs, M. Diltoer, L. Peeters, S. Vanlanduit, Thermal comfort of the surgical staff in the operating room, Build. Environ. 81 (2014) 37–41, https://doi.org/10.1016/j.buildenv.2014.05.036.
[294] T. Van Renterghem, Towards explaining the positive effect of vegetation on the perception of environmental noise, Urban For. Urban Green. 40 (2019) 133–144, https://doi.org/10.1016/j.ufug.2018.03.007.
[295] J. Verheyen, N. Theys, L. Allonsius, F. Descamps, Thermal comfort of patients: objective and subjective measurements in patient rooms of a Belgian healthcare facility, Build. Environ. 46 (5) (2011) 1195–1204, https://doi.org/10.1016/j.buildenv.2010.12.014.
[296] T.N. Verma, S.L. Sinha, Numerical simulation of contaminant control in multi-patient intensive care unit of hospital using computational fluid dynamics, J. Med. Imaging Health Inform. 5 (5) (2015) 1088–1092, https://doi.org/10.1166/jmihi.2015.1500.
[297] A. Wagdy, A. Sherif, H. Sabry, R. Arafa, I. Mashaly, Daylighting simulation for the configuration of external sun-breakers on south oriented windows of hospital patient rooms under a clear desert sky, Sol. Energy 149 (2017) 164–175, https://doi.org/10.1016/j.solener.2017.04.009.

[298] C. Wang, F. Zhang, J. Wang, J.K. Doyle, P.A. Hancock, C.M. Mak, S. Liu, How indoor environmental quality affects occupants' Cognitive functions: a systematic review, Build. Environ. 193 (2021) 107647, https://doi.org/10.1016/j.buildenv.2021.107647.
[299] G. Wang, A. Badal, X. Jia, J.S. Maltz, K. Mueller, K.J. Myers, C. Niu, M. Vannier, P. Yan, Z. Yu, R. Zeng, Development of Metaverse for Intelligent Healthcare', Nat. Mach. Intell. 4 (11) (2022) 922–929, https://doi.org/10.1038/s42256-022-00549-6.
[300] Y. Wang, K.K.L. Wong, H. Du, J. Qing, J. Tu, Design configuration for a higher efficiency air conditioning system in large space building, Energy Build. 72 (2014) 167–176, https://doi.org/10.1016/j.enbuild.2013.12.041.
[301] WELL. n.d.'WELL Standard'. https://v2.wellcertified.com/en/wellv2/concepts.
[302] S. Willems, D. Saelens, A. Heylighen, Discrepancies between predicted and actual indoor environmental (Dis)comfort: the role of hospitalized patients' Adaptation strategies, Build. Res. Inform. 50 (7) (2022) 792–809, https://doi.org/10.1080/09613218.2022.2038060.
[303] S. Willems, D. Saelens, A. Heylighen, 'Patient well-being, adaptation of and to indoor conditions, and hospital room design: two mixed methods case studies, Build. Res. Inform. 50 (1–2) (2022) 105–133, https://doi.org/10.1080/09613218.2021.2004386.
[304] S. Willems, D. Saelens, A. Heylighen, Patient well-being, adaptation of and to indoor conditions, and hospital room design: two mixed methods case studies, Build. Res. Inform. 50 (1–2) (2022) 105–133, https://doi.org/10.1080/09613218.2021.2004386.
[305] Q. Wu, N. Li, X. Cai, Y. He, Y. Du, Impact of indoor environmental quality (IEQ) factors on occupants' Environmental perception and satisfaction in hospital wards, Build. Environ. 245 (2023), https://doi.org/10.1016/j.buildenv.2023.110918.
[306] Y. Wu, Q. Meng, L. Li, J. Mu, Interaction between sound and thermal influences on patient comfort in the hospitals of China's Northern heating region, Appl. Sci. (Switzerl.) 9 (24) (2019), https://doi.org/10.3390/app9245551.
[307] Y. Wu, Q. Meng, L. Li, J. Mu, Interaction between sound and thermal influences on patient comfort in the hospitals of China's Northern heating region, Appl. Sci. 9 (24) (2019) 5551, https://doi.org/10.3390/app9245551.
[308] Z. Xie, Y. Tian, Y. Xu, F. Zhong, S. Li, X. Zhu, Q. Yuan, Molecular design of epoxy resin and the driving forces in adhesion with cementitious materials, Appl. Surf. Sci. 689 (2025) 162498, https://doi.org/10.1016/j.apsusc.2025.162498.
[309] X.D. Xuan, Study of indoor environmental quality and occupant overall comfort and productivity in LEED- and Non-LEED-certified healthcare settings, Indoor Built Environ. 27 (4) (2018) 544–560, https://doi.org/10.1177/1420326×16684007.
[310] X. Xuan, Effectiveness of indoor environment quality in LEED-certified healthcare settings, Indoor Built Environ. 25 (5) (2016) 786–798.
[311] B. Yan, W. Yang, F. He, W. Zeng, Occupant behavior impact in buildings and the artificial intelligence-based techniques and data-driven approach solutions, Renew. Sustain. Energy Rev. 184 (2023) 113372, https://doi.org/10.1016/j.rser.2023.113372.
[312] D. Yan, T. Hong, B. Dong, A. Mahdavi, S. D'Oca, I. Gaetani, X. Feng, IEA EBC Annex 66: definition and simulation of occupant behavior in buildings, Energy Build. 156 (2017) 258–270, https://doi.org/10.1016/j.enbuild.2017.09.084.
[313] H.X. Yang, X. Luo, B. Chen, H.P. Xie, L. Huang, Q.Q. Shi, Y. Zhang, M. Ni, A numerical investigation on indoor air ventilation design and aerosol transportation: a case study in a hospital hall, Simulat.-Transact. Soc. Model. Simulat. Int. (2023), https://doi.org/10.1177/00375497231194348.
[314] Q. Yang, J.G. Cruz-Garza, S. Kalantari, Brain–Computer interfaces as an architectural design tool: feasibility and usability study, Automat. Construct. 154 (2023) 105011, https://doi.org/10.1016/j.autcon.2023.105011.
[315] Y. Yang, H. Yang, Q. Li, L. Zhang, Z. Dong, Simulation study on airflow organization and environment in reconstructed Fangcang Shelter Hospital based on CFD, Buildings 13 (5) (2023), https://doi.org/10.3390/buildings13051269.
[316] M.A. Yar, F. Kazemi, The role of dish gardens on the physical and neuropsychological improvement of hospitalized children, Urban For. Urban Green. 53 (2020), https://doi.org/10.1016/j.ufug.2020.126713.
[317] Y.H. Yau, D. Chandrasegaran, A. Badarudin, The ventilation of multiple-bed hospital wards in the Tropics: a review, Build. Environ. 46 (5) (2011) 1125–1132, https://doi.org/10.1016/j.buildenv.2010.11.013.
[318] Y.H. Yau, B.T. Chew, Thermal comfort study of hospital workers in Malaysia, Indoor Air 19 (6) (2009) 500–510, https://doi.org/10.1111/j.1600-0668.2009.00617.x.
[319] Y.H. Yau, B.T. Chew, A review on predicted mean vote and adaptive thermal comfort models, Build. Serv. Eng. Res. Technol. 35 (1) (2014) 23–35, https://doi.org/10.1177/0143624412465200.
[320] Y.H. Yau, K.S. Phuah, Indoor air quality study in four Malaysian hospitals for Centralized and Non-Centralized ACMV Systems, Air Qual., Atmosph. Health 16 (2) (2023) 375–390, https://doi.org/10.1007/s11869-022-01280-2.
[321] J. Yin, J. Yuan, N. Arfaei, P.J. Catalano, J.G. Allen, J.D. Spengler, Effects of biophilic indoor environment on stress and anxiety recovery: a between-subjects experiment in virtual reality, Environ. Int. 136 (2020), https://doi.org/10.1016/j.envint.2019.105427.
[322] O. Yongson, I.A. Badruddin, Z.A. Zainal, P.A. Aswatha Narayana, Airflow analysis in an air conditioning room, Build. Environ. 42 (3) (2007) 1531–1537, https://doi.org/10.1016/j.buildenv.2006.01.002.
[323] S. Yoon, J. Koo, Situ model fusion for building digital twinning, Build. Environ. 243 (2023) 110652, https://doi.org/10.1016/j.buildenv.2023.110652.
[324] F. Yuan, R. Yao, S. Sadrizadeh, B. Li, G. Cao, S. Zhang, S. Zhou, H. Liu, A. Bogdan, C. Croitoru, A. Melikov, C.A. Short, B. Li Professor, Thermal comfort in Hospital buildings – a literature review, J. Build. Eng. 45 (2022), https://doi.org/10.1016/j.jobe.2021.103463.
[325] L. Zaniboni, R. Albatici, Natural and mechanical ventilation concepts for indoor comfort and well-being with a sustainable design perspective: a systematic review, Buildings 12 (11) (2022), https://doi.org/10.3390/buildings12111983.
[326] L. Zaniboni, G. Pernigotto, J. Toftum, A. Gasparella, B.W. Olesen, Subjective and objective assessment of thermal comfort in physiotherapy centers, Build. Environ. 176 (2020), https://doi.org/10.1016/j.buildenv.2020.106808.
[327] L. Zaniboni, G. Pernigotto, J. Toftum, A. Gasparella, B.W. Olesen, Thermal comfort in physiotherapy centers: evaluation of the neutral temperature and interaction with the other comfort domains, Build. Environ. 206 (2021), https://doi.org/10.1016/j.buildenv.2021.108289.
[328] J. Zeisel, N.M. Silverstein, J. Hyde, S. Levkoff, M. Powell Lawton, W. Holmes, Environmental correlates to behavioral health outcomes in Alzheimer's Special care units, Gerontologist 43 (5) (2003) 697–711, https://doi.org/10.1093/geront/43.5.697.
[329] Z.J. Zhai, A.L. Osborne, Simulation-based feasibility study of improved air conditioning systems for hospital operating room, Front. Architect. Res. 2 (4) (2013) 468–475, https://doi.org/10.1016/j.foar.2013.09.003.
[330] Z. Zhan, W. Xu, L. Xu, X. Qi, W. Song, C. Wang, Z. Huang, BIM-based Green hospital building performance pre-evaluation: a case study, Sustainability. 14 (4) (2022), https://doi.org/10.3390/su14042066.
[331] H. Zhang, X. Xie, S. Hong, H. Lv, Impact of metabolism and the clothing thermal resistance on inpatient thermal comfort, Energy Built Environ. 2 (2) (2021) 223–232, https://doi.org/10.1016/j.enbenv.2020.07.002.
[332] J. Zhang, Y. Zhao, S. Wen, D. Tu, J. Liu, S. Liu, Assessment of COVID-19 infection risk, thermal comfort, and energy efficiency in negative pressure isolation wards with varied ventilation modes, Energy Build. 308 (2024) 114002, https://doi.org/10.1016/j.enbuild.2024.114002.
[333] M. Zhang, R. Xu, H. Wu, J. Pan, X. Luo, Human–robot collaboration for on-site construction', Automat. Construct. 150 (2023) 104812 https://doi.org/10.1016/j.autcon.2023.104812.
[334] S. Zhang, K. Song, Q. Ban, P. Gong, R. Li, Z. Peng, Indoor air quality and smoking control in healthcare environments in Northern China, Sustain. (Switzerl.) 15 (5) (2023), https://doi.org/10.3390/su15054041.
[335] S. Zhang, J. Zheng, Y. Wu, Field study of air environment perceptions and influencing factors in waiting spaces of general hospitals in winter cities, Build. Environ. 183 (2020), https://doi.org/10.1016/j.buildenv.2020.107203.
[336] S. Zhang, J. Zheng, Y. Wu, Field study of air environment perceptions and influencing factors in waiting spaces of general hospitals in winter cities, Build. Environ. 183 (2020) 107203, https://doi.org/10.1016/j.buildenv.2020.107203.
[337] S. Zhang, Z. Lin, Effective moisture temperature: ventilation performance index accounting for effects of air temperature and relative humidity on thermal comfort, Build. Environ. 243 (2023) 110625, https://doi.org/10.1016/j.buildenv.2023.110625.
[338] S. Zhang, D. Niu, Y. Lu, Z. Lin, Contaminant removal and Contaminant dispersion of air distribution for overall and local airborne infection risk controls, Sci. Total Environ. 833 (2022) 155173, https://doi.org/10.1016/j.scitotenv.2022.155173.
[339] Y.Q. Zhang, X. Liu, Q.L. Meng, B. Li, L. Caneparo, Physical environment research of the Family ward for a healthy residential environment, Front. Public Health 10 (2022), https://doi.org/10.3389/fpubh.2022.1015718.
[340] Y. Zhang, P. Tzortzopoulos, M. Kagioglou, Healing built-environment effects on health outcomes: environment–occupant–health framework, Build. Res. Inform. 47 (6) (2019) 747–766, https://doi.org/10.1080/09613218.2017.1411130.
[341] Y. Zhao, M. Mourshed, Design indicators for better accommodation environments in hospitals: inpatients' perceptions', Intell. Build. Int. 4 (4) (2012) 199–215, https://doi.org/10.1080/17508975.2012.701186.
[342] Y. Zhao, M. Mourshed, Patients' Perspectives on the design of hospital outpatient areas, Buildings 7 (4) (2017), https://doi.org/10.3390/buildings7040117.
[343] X. Zou, Y. Zhou, Spatial cognition of the visually impaired: a case study in a familiar environment, Int. J. Environ. Res. Public Health 20 (3) (2023), https://doi.org/10.3390/ijerph20031753.